



WHO INFLUENZA CENTRE LONDON

**Report prepared for the WHO annual
consultation on the composition of influenza
vaccine for the Northern Hemisphere 2013/14**

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Northern Hemisphere Vaccine Recommendation Meeting for 2013-14 influenza season.

Summary of NIMR Results.

Final Report for WHO meeting: 18-20 February 2013.

Influenza Activity in the European Region, October 2012 – February 2013

Weekly reporting on influenza activity in the Region started in week 40/2012. Activity started early compared to the 2011-2012 influenza season with levels of transmission increasing slowly since week 46/2012, and detections in week 05/2013 passing that seen at the peak (week 08/2012) for the 2011-2012 season (**Figure 1**). Approximately 30,000 influenza detections have been reported across the whole WHO Euro Region with type A detections (70%) predominating over type B (30%), but these relative proportions are slightly different for western (EU/EAA) countries (68% & 32%) and eastern countries (82% & 18%). These patterns may change as type B activity often increases later in the influenza season, but influenza B detections as a percentage of total influenza detections across the Region is running at 30% compared to 10% in the previous season (**Figure 2**). Within type A, the H1pdm09 subtype (71%) is predominating over H3 viruses (29%) by a 2.5-fold margin across the Region (**Figures 1 & 2**): this represents a reversal of the situation in 2011-2012 (**Figure 2**). In terms of the intensity of influenza activity and the degree of geographic spread within countries there is evidence of west to east spread across the Region (**Figure 3**). However, within individual countries the relative proportions of influenza type A/B and subtype H1/H3 detections shows considerable variation (**Figure 4**). Overall, influenza activity across the Region has been greater than that seen in the 2011-2012 season, as exemplified in the UK through the HPA/Q-Surveillance National Syndromic Surveillance System based on a population sample of 20×10^6 in England, Wales and Northern Ireland: Influenza Like Illness (ILI) showed peak activity in weeks 52/2012-01/2013 but remained well within normal seasonal activity (**Figure 5**).

Antigenic and genetic characteristics of influenza viruses recovered from clinical specimens collected after 31 August 2012

Influenza positive specimens received at NIMR

Samples, viruses or clinical specimens, with collection dates after 31 August 2012 have been received from 32 countries in Europe, Africa, the Middle East and the Far East. The majority (69%) were type A viruses with A(H1N1)pdm09 viruses predominating over A(H3N2) viruses by a ratio of 1.5:1 (**Table 1**). Of the type B viruses (31% of all specimens received) B-Yamagata lineage viruses predominated over those of the Victoria lineage by a ratio of 3.3:1. These proportions of influenza types/subtypes/lineages are comparable to those reported within the WHO Euro Region (**Figures 1 & 2**). **Table 2** shows the isolation rates for samples received that were from specimens collected after 31 August 2012. **Table 3** summarises the number of viruses of the A(H1N1), A(H3N2) subtypes, and the two B-lineages that showed ≤ 2 -, 4- and ≥ 8 -fold reductions in HI titres with post-infection ferret antisera compared with the titres against their representative homologous vaccine/reference viruses. For A(H3N2) viruses the results of plaque-reduction based neutralisation assays are also summarised (**Table 3**).

Influenza A(H1N1)pdm09 virus analysis.

Table 4 summarises antigenic results for the A(H1N1)pdm09 viruses received.

The vast majority of the H1N1 viruses were antigenically similar to the vaccine virus (**Table 4**) with six viruses showing a 4-fold reduction and two ≥ 8 -fold reductions in HI titre compared with the titre of the vaccine virus (A/California/7/2009) with its homologous post-infection ferret antiserum. The results are shown in **Tables 5 to 10** and the HA1 amino acid substitutions/polymorphisms associated with 4- and ≥ 8 -fold reductions in HI titre are indicated where known. Forty-four of the 98 viruses, with specimen collection dates since 31 August 2012, analysed in HI assays had been subjected to HA, NA and M gene sequence analysis at the time of compilation of this report.

Figure 6 shows a phylogenetic tree of the HA genes of representative H1N1 viruses. The HA genes of H1N1 viruses cluster into eight genetic groups with A/California/7/2009 representing group 1. The other seven groups, all containing S203T substitutions compared to A/California/7/2009, are defined by the following amino acid substitutions in HA1:

- (2) **N31D, S162N** (resulting in the gain of a glycosylation site) & **A186T**, e.g. A/Czech Republic/32/2011;
- (3) **A134T & S183P** often with additional substitutions, e.g. A/Hong Kong/3934/2011;
- (4) **N125D**, e.g. A/Christchurch/16/2010;
- (5) **D97N, R205K, I216V & V249L**, e.g. A/Astrakhan/1/2011, sometimes with additional substitutions;

- (6) **D97N & S185T**, e.g. A/St Petersburg/27/2011, often with additional substitutions;
- (7) **S185T & A197T** often with **S143G**, e.g. A/St Petersburg/100/2011, and/or other substitutions;
- (8) **A186T & V272A**, e.g. A/Norway/2552/2012.

Viruses recovered from specimens collected since 31 August 2012 fell predominantly within genetic groups 6 and 7, with a few falling in genetic group 8, as indicated in **Tables 5 to 10**. In addition an outlier 'West African' genetic group containing viruses from Cote d'Ivoire, Ghana and Senegal, lacking the S203T substitution, was identified. The virus A/South Australia/162/2012 is also an outlier, carrying six unusual amino acid substitutions in HA1.

Amino acid alignment of the HAs for a selected group of viruses is shown in **Figure 7**: the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 8 illustrates the location of amino acid residues, defining genetic groups 6, 7 and 8, and the 'West African' group, on the structure of the H1 HA.

Figure 9 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 10** shows an amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent but three viruses, indicated on the NA phylogeny (**Figure 9**) by dashed lines (---), appear to be HA/NA reassortants.

Figure 25 shows a phylogenetic tree of representative H1N1 M genes. All showed resistance to adamantanes due to S31N substitution in the M2 ion-channel protein. Amino acid substitutions in M1 and M2 are highlighted.

Antiviral analysis.

All viruses recovered from specimens with collection dates after 31 August 2012 were sensitive (normal inhibition) to the neuraminidase inhibitors zanamivir and oseltamivir (**Table 30**). Two viruses with collection dates in August 2012, A/Netherlands/492/2012 and A/Netherlands/507/2012 (**Table 6**), carried NA H275Y substitutions and one from July 2012 (A/Salta/1341/2012) carried NA N295S – these three were resistant to oseltamivir (highly reduced inhibition). All viruses for which M-gene sequences were determined encoded the M2 S31N substitution which confers adamantane resistance.

Conclusion.

The vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus.

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses have been received from Europe, Africa (Algeria, Ghana, Senegal, Tunisia), the Middle-East (Turkey) and from Hong Kong SAR. As described previously, these viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. Those viruses with sufficient titre in HA assays using guinea pig red blood cells in the presence of 20nM oseltamivir, to circumvent the NA-mediated binding of H3N2 viruses to the red blood cells, were analysed by HI assay. Approximately 93% of viruses propagated retained sufficient HA titre in the presence of oseltamivir to allow HI analysis (**Table 2**). Virus neutralisation assays were used to complement HI with approximately one third of the H3N2 viruses received being analysed in the plaque reduction neutralisation assay (PRNA).

The sample collection details and a summary of the results are shown in **Table 11**. The results of HI assays are shown in **Tables 12 to 20**. The results from the PRNA on MDCK-SIAT1 cells are shown in **Tables 21 and 22**. Virus gene sequences included on the phylogenetic trees are highlighted.

In the HI assays two different antisera raised against the egg-propagated vaccine virus, A/Victoria/361/2011, were used. Overall, just under 5% of test viruses propagated in MDCK-SIAT1 cells reacted with these antisera within 4-fold of the titre of the homologous egg-propagated virus. A similar low level of reactivity was seen with antiserum raised against the egg-propagated virus A/Hawaii/22/2012 (~7.5% of test viruses propagated in MDCK-SIAT1 cells reacted with these antisera within 4-fold of the titre of the homologous egg-propagated virus). Better reactivity was seen with antiserum raised against the egg-propagated virus A/Texas/50/2012 with 45% of test viruses reacting with titres within 4-fold of the titre of the homologous virus. Notably, only 7.5% showed reactivity within 2-fold of the homologous titre for egg-propagated A/Texas/50/2012.

Test viruses propagated in MDCK-SIAT1 cells showed better reactivity with antisera raised against the cell-propagated reference strains: A/Alabama/5/2010, A/Stockholm/18/2011, A/Berlin/93/2011, A/Athens/112/2012 and a cell-propagated counterpart of A/Victoria/361/2011. With two antisera raised against the latter A/Victoria/361/2011 virus, just over 95% of test viruses showed HI titres within 4-fold of the titre of homologous cell-propagated virus. Similar results were seen for the other cell-propagated reference viruses: 95% of viruses reacted within 4-fold of the titre of the homologous virus with antiserum raised against A/Alabama/5/2010; for antiserum raised against A/Stockholm/18/2011, 96% of test viruses reacted within 4-fold of the titre of the homologous virus; with antiserum raised against A/Athens/112/2012, 99%; and all viruses reacted within 4-fold of the titre of the homologous virus with antiserum raised against A/Berlin/93/2011.

The results of PRNA are shown in **Tables 21 and 22**. Shown on the tables are the genetic groups to which the HA gene of the reference viruses and the test viruses, where known, are assigned, and those with substitutions at amino acid positions 128 and 142 in HA1 are noted. The plaque morphology of the viruses is also described. The results of PRNA were consistent with the results from HI assays done in the presence of oseltamivir. In PRNA none of the test viruses reacted within 4-fold of the titre of the homologous virus with antisera raised against the egg-propagated vaccine virus, A/Victoria/361/2011. However, all viruses

reacted well with antiserum raised against the cell-propagated counterpart to the A/Victoria/361/2011 vaccine virus having titres within 4-fold of the titre of the homologous virus. Similarly, all test viruses gave titres within 4-fold of the reference viruses with antisera raised against cell-propagated A/Stockholm/18/2011 and A/Athens/112/2012. As seen in the HI assays, test viruses showed somewhat better reactivity with antiserum raised against the egg-propagated reference virus A/Texas/50/2012 than with antiserum raised against the egg-propagated vaccine virus A/Victoria/361/2011, with 70% of test viruses propagated in tissue culture showing neutralisation titres within 4-fold of that of the homologous virus A/Texas/50/2012. However, as in the HI assays, a minority of test viruses, 12.5% of those tested, showed reactivity within 2-fold of the titre of the homologous egg-propagated virus, A/Texas/50/2012.

Several egg-propagated viruses, including high growth reassortants have been examined by HI assay (**Tables 18 & 19**). Each was antigenically similar to the cell-propagated reference viruses (A/Alabama/5/2010, A/Stockholm/18/2011, A/Berlin/93/2011, A/Victoria/361/2011 and A/Athens/112/2012). In contrast, they showed variable reactivity with antisera raised against egg-propagated viruses, notably the high growth reassortants carrying the A/Hawaii/22/2012 or A/Texas/50/2012 HA and NA genes shown on **Table 19** showed titres 8-fold reduced with antiserum raised against the egg-propagated vaccine virus A/Victoria/361/2011 compared with the homologous titre of the vaccine virus.

Phylogenetic analysis of the HA genes of representative H3N2 viruses is shown in **Figure 11**. The HA genes fell into the A/Victoria/208 genetic clade and predominantly into genetic groups 3A, 3B and 3C, and groups 5 and 6. Amino acid substitutions that mark these genetic groups are:

(3A) **N144D** (resulting in the loss of a potential glycosylation site which is predicted not to be utilised in the context of the background amino acid sequence), **N145S**, **V223I** & **HA2 D158N**, e.g. A/Stockholm/18/2011;

(3B) **N145S**, **A198S**, **V223I**, **N312S** & **HA2 D158N**, e.g. A/Athens/112/2012; some also carry the substitution **N144D** (resulting in the loss of a potential glycosylation site which, as pointed out above, is predicted not to be utilised in this sequence context);

(3C) **S45N** (resulting in the gain of a potential glycosylation site), **T48I**, **S145N**, **A198S**, **V223I** & **N312S**, e.g. A/Victoria/361/2011, with some viruses also carrying the substitutions **Q33R** and **N278K**, e.g. A/Berlin/93/2011, A/Texas/50/2012 and A/Hawaii/22/2012. A major sub-group carries the reversion **N145S** and a further subset of this group carry the additional substitutions **T128A** (which results in the loss of a potential glycosylation site) and **R142G**, both in the 140 loop of HA1 that corresponds to antigenic site A;

(5) **K2E**, **N8D** (resulting in the loss of a potential glycosylation site), **D53N**, **Y94H**, **I230V** & **E280A**, e.g. A/Acores/SU43/2012;

(6) **D53N**, **Y94H**, **S199A**, **I230V** & **E280A**, e.g. A/Iowa/19/2010.

Figure 12 shows an HA amino acid alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 11**). The current vaccine virus, A/Victoria/361/2011,

is shown in red, glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group to which each of the sequences belongs is also shown.

Figure 13 shows the location on the 2005 H3 HA structure of amino acid substitutions that define genetic groups 3A, 3B and 3C, and groups 5 and 6. Several of the substitutions that have accumulated are exposed and could influence the antigenicity of the virus.

Figure 14 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 15** shows a NA amino acid alignment for a representative selection of these viruses. As above, A/Victoria/361/2011 is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown.

The HA and NA phylogenetic trees are generally congruent but it is noteworthy that the viruses of the predominating genetic groups all carry the dual substitutions **S367N** and **K369T** in their NA which results in the addition of a glycosylation site at residue **367**. Other glycosylation sites have been lost at residues **402**, in the majority of recently collected viruses, and at residue **329** in genetic group 7. **Figure 16** localises the predominant variable sites on the structure of an N2 NA along with amino residues that influence the ability of viruses to agglutinate red blood cells through the NA.

Antiviral resistance.

All 176 viruses collected after 2012-08-31 and analysed were sensitive to both zanamivir and oseltamivir neuraminidase inhibitors (**Table 39**). A single virus, A/Cairo/136/2012, was predicted to be sensitive to the adamantane antiviral drugs based on the sequence of the M2-coding region of RNA segment 7 encoding **N31S** substitution (**Figure 25**).

Conclusion.

The vast majority of recent viruses analysed show reduced reactivity with antisera raised against the egg-propagated vaccine virus A/Victoria/361/2011 in both virus neutralisation and HI assays, with the latter being carried out in the presence of oseltamivir. However, test viruses reacted well with antisera raised against reference viruses that are genetically similar to A/Victoria/361/2011 but propagated in cells. There is no evidence of antigenic drift from the cell-propagated reference viruses but the test viruses are antigenically distinguishable from egg-propagated viruses.

Influenza B viruses.

Influenza B viruses represented 31% of samples received (**Table 1**). B/Yamagata lineage viruses predominated over those of the B/Victoria lineage by over 3:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from countries in Europe, Africa (Algeria), the Middle East (Jordan) and from Hong Kong SAR. The collection details and summary results are shown in **Table 23**.

HI results of the propagated viruses are shown in **Tables 24 to 29**. Virus gene sequences included in the phylogenetic trees are highlighted. Over 90% of cell-propagated viruses displayed low reactivity with antisera raised against the egg-propagated previous vaccine virus B/Brisbane/60/2008 and egg-propagated B/Malta/636714/2011. All viruses showed reactivity within 4-fold of the titre of the homologous virus with antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells. In **Tables 24 to 29** these antisera were raised against B/Paris/1762/2008, B/Hong Kong/514/2009 and B/Odessa/3886/2010, and these viruses are considered as surrogate cell-propagated antigens representing the egg-propagated prototype strain. Three viruses that showed increased reactivity with antiserum raised against the egg-propagated viruses showed mutation or polymorphism in the HA gene sequence that affect glycosylation at amino acid position **197** of HA1, a glycosylation site that is commonly lost on egg passage.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 17** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 18**. The HA genes of the majority of recently collected viruses fall into the B/Brisbane/60 clade subgroup now designated clade 1A, and others fall into the subgroup designated clade 1B. **Figure 19** shows a phylogenetic tree for the NA gene and **Figure 20** shows a corresponding amino acid alignment for a selection of NA glycoproteins. Highlighted in the trees are three intra-clade reassortants between the HA gene from clade 1A or 1B and the NA gene of clade 3, exemplified by B/Denmark/23/2012 and B/Norway/2090/2012.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa (Ghana), the Middle East (Turkey) and from Hong Kong SAR. The collection details and results are summarised in **Table 30**.

The results of HI analysis of the propagated viruses of the B/Yamagata lineage are shown in **Tables 31 to 38**; virus sequences included in the phylogenetic trees are highlighted and, where known, the clade into which the HA falls is shown. Three different ferret antisera raised against the current, egg-propagated, vaccine virus B/Wisconsin/1/2010 were used to analyse the reactivity of test viruses. In the majority of assays (**Tables 31 to 37**) only 6% (4 of 66) of test viruses showed reactivity within 4-fold of the titre of B/Wisconsin/1/2010. In **Table 38**, although the antiserum had been used in most of the other tests, the homologous titre was markedly reduced and 40% (18 of 45) showed reactivity within 2-fold of the homologous titre for B/Wisconsin/1/2010. Post-infection ferret antisera raised against cell-propagated reference viruses carrying HA genes falling within genetic clade 3 showed that

test viruses reacted within 4-fold of the titre of the homologous virus with varying efficiency: with antiserum raised against B/Serbia/1984/2011, 54% of viruses reacted within 4-fold of the titre of the homologous virus; with antiserum raised against B/Stockholm/12/2011, 74%; and B/Novosibirsk/1/2012, 78%. Similarly, antisera raised against cell-propagated viruses carrying HA genes falling within genetic clade 2 also showed that test viruses reacted within 4-fold of the titre of the homologous virus with varying efficiency: with antisera raised against B/Estonia/55669/2011, 65% of test viruses reacted within 4-fold of the titre of the homologous virus; with antiserum raised against B/Hong Kong/3577/2012, 85%; and of 63 viruses tested with antiserum raised against the cell-propagated B/Massachusetts/02/2012 all 63 reacted within 4-fold of the titre of the homologous virus.

It was clear that many, but not all, viruses carrying clade 2 HA genes could be differentiated antigenically from those carrying clade 3 HA genes by certain antisera, but not all antisera were able to antigenically differentiate between viruses falling in the alternate clades.

Figure 21 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid alignment of a selection of the HA glycoproteins is shown in **Figure 22**. The HA genes of just over half of the recently collected viruses fall into the B/Brisbane/3/2007 clade (Clade 2), which includes the reference viruses B/Estonia/55669/2011, B/Massachusetts/02/2012 and B/Hong Kong/3577/2012. The remaining HA genes fall into the B/Wisconsin/1/2010 clade (Clade 3), which includes the reference viruses B/Stockholm/12/2011, B/Serbia/1894/2011 and B/Novosibirsk/1/2012. **Figure 23** shows a phylogenetic tree for NA genes and **Figure 24** shows a corresponding amino acid alignment for selected NA glycoproteins.

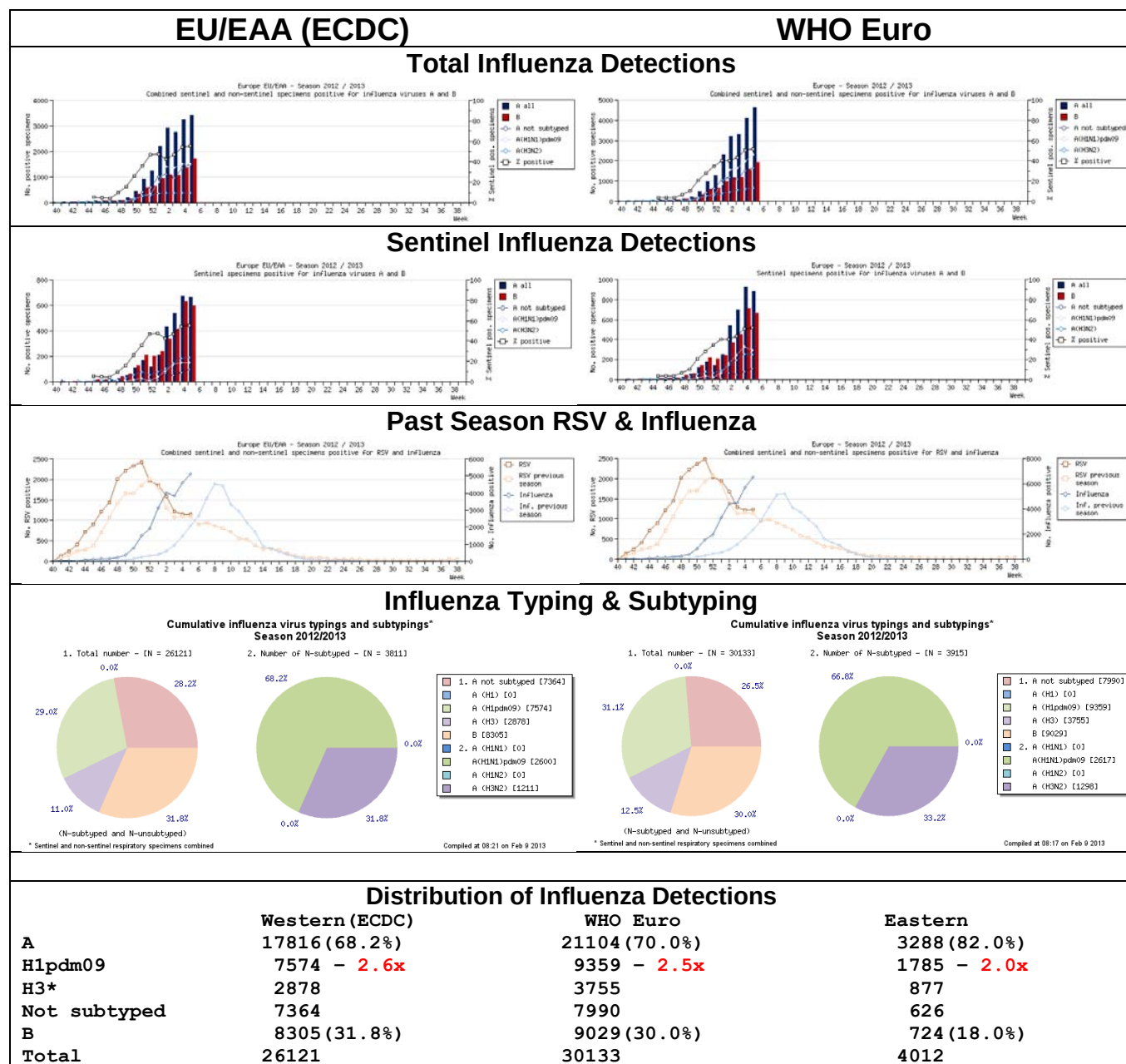
Antiviral Analysis.

No evidence for reduced sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was seen for influenza B viruses after 31 August 2012 (**Table 39**); further analyses are underway.

Conclusion.

The number of influenza B viruses analysed has increased and viruses of the B/Yamagata lineage have predominated. For viruses of the B/Victoria lineage most were antigenically similar to viruses closely related to B/Brisbane/60/2008. For the B/Yamagata lineage viruses many viruses showed reduced activity with antiserum raised against the egg-propagated vaccine virus, A/Wisconsin/1/2010. Test viruses generally showed better reactivity either with an antiserum raised against the cell-propagated virus, B/Stockholm/12/2011, which, like A/Wisconsin/1/2010, has an HA gene belonging to clade 3, or with antisera raised against the cell-propagated virus, B/Estonia/55669/2011, which carries an HA gene belonging to clade 2.

Figure 1. Influenza Activity in Europe: 2012-13 Season (weeks 40-05)



* Ratios for H1pdm09:H3 viruses are shown (x)

Figure 2. Comparison of 2011-12 & 2012-13 Influenza Seasons in Europe

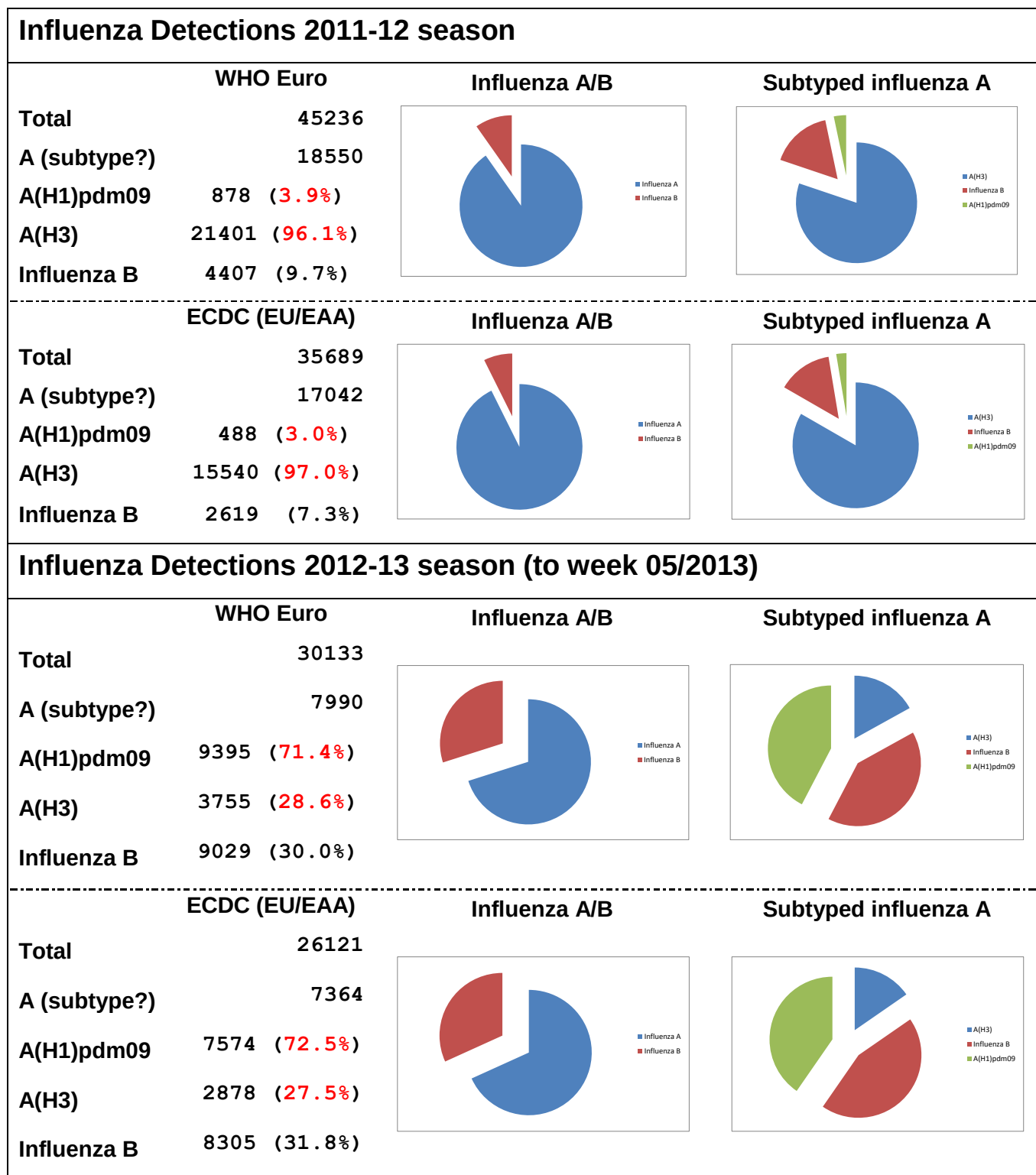


Figure 3. Development of the 2012-13 Influenza Season in Europe

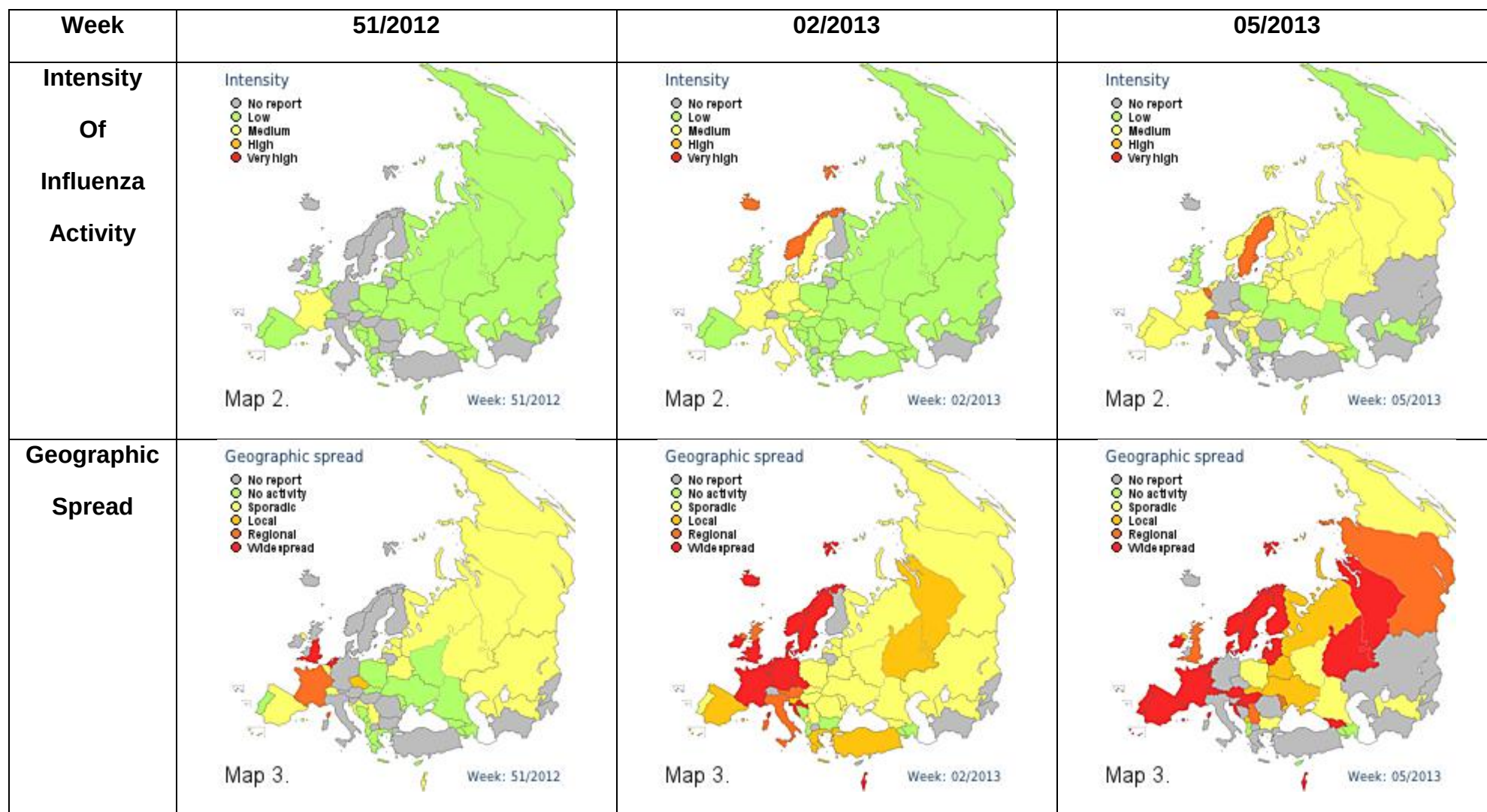


Figure 4. Patterns of Influenza Type/Subtype Circulation in in WHO Euro Region Countries

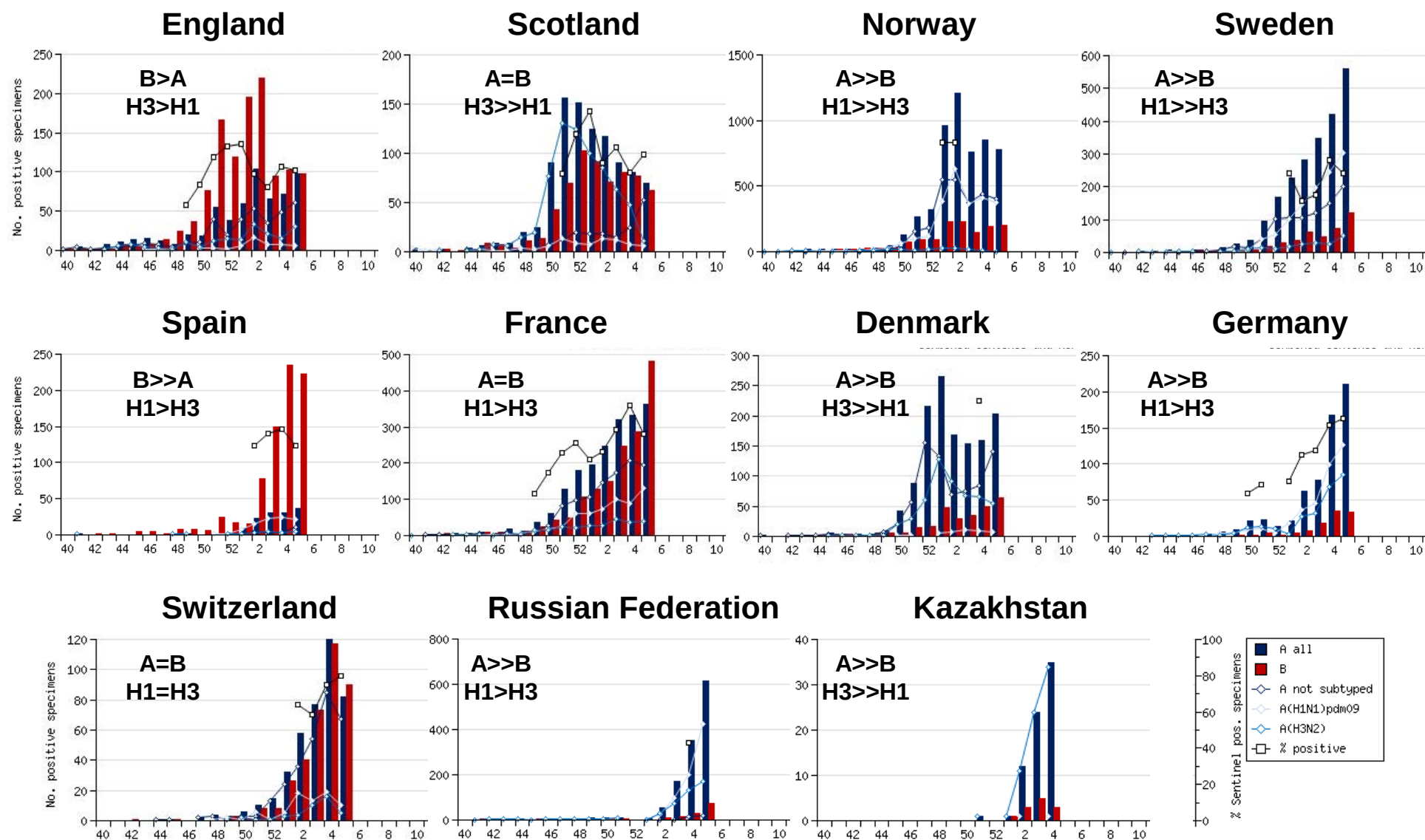
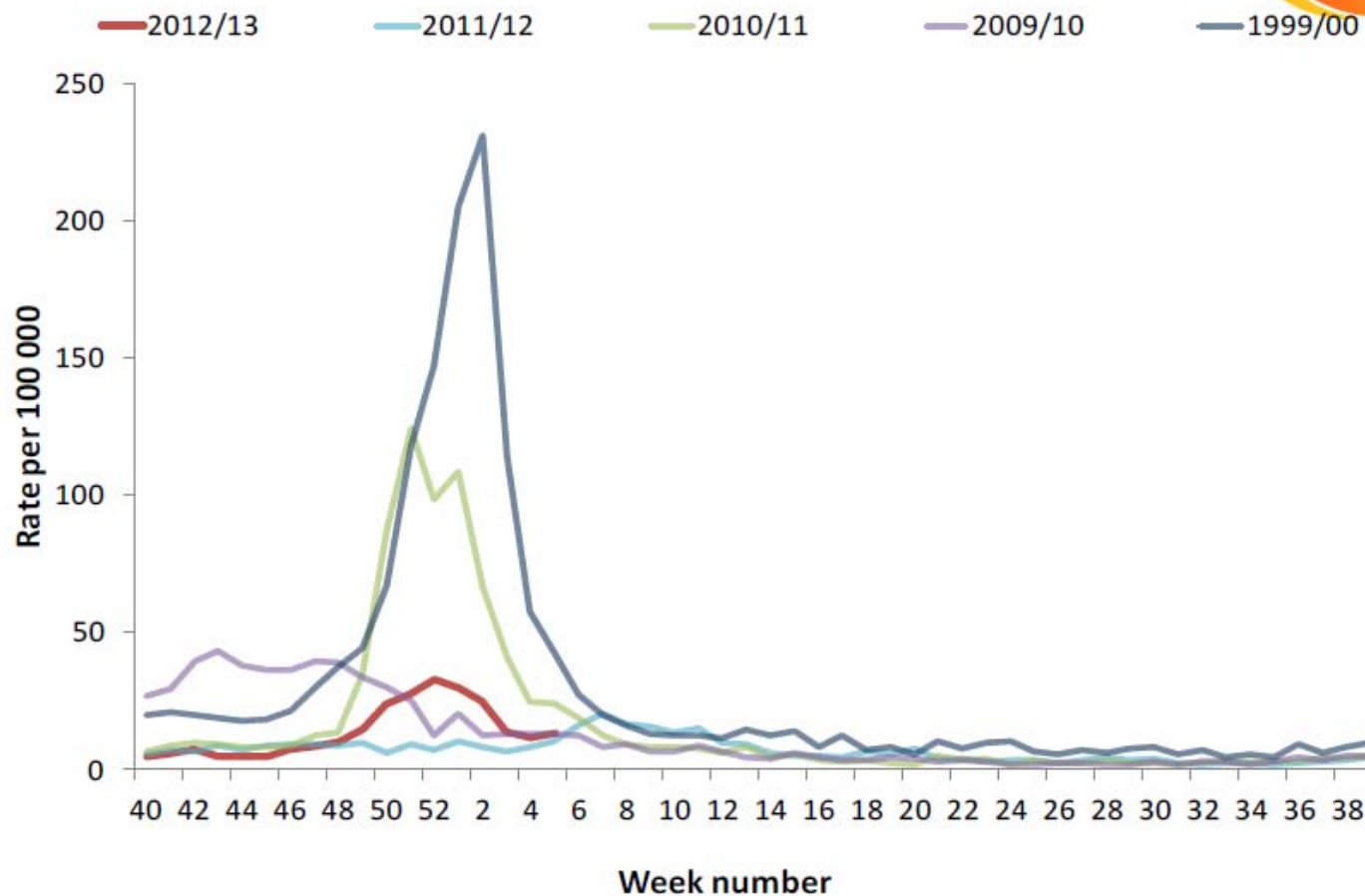


Figure 5. Influenza-Like Illness estimated from the UK HPA/Q-surveillance syndromic surveillance system

RCGP: Influenza-like illness current and recent seasons



Note: Weekly Q-Surveillance national thresholds for ILI incidence rates are: <20 per 100,000 (baseline activity); 20-70 (normal seasonal activity); 70-130 (above average activity); >130 (exceptional activity).

Table 1. Summary of clinical samples and isolates received, with collection dates since 2012-09-01, by country

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated ¹	Number received	Number propagated ¹
SEPTEMBER											
Denmark	2				2	2					
France	1									1	1
Ghana	1									1	1
Norway	2									2	2
Senegal	1				1	1					
Spain	1				1	0					
OCTOBER											
Algeria	1		1	in process						1	in process
France	6				3	3		2	in process	1	in process
Germany	2				1	1				1	1
Ghana	4		3	3						1	1
Israel	1				1	1					
Norway	14		6	0	2	2		1	1	5	4
Romania	1				1	1					
Senegal	16		2	0	14	14					
Spain	2	1								1	0
Sweden	2		1	1	1	1					
United Kingdom	8		1	0	2	2		2	2	3	3
NOVEMBER											
Algeria	4				2	2		1	in process	1	in process
Austria	1				1	1					
Belgium	3		1	1				1	1	1	1
Denmark	7		1	1	1	1		2	2	3	3
Egypt	12	10			2	2					
Finland	1							1	1		
France	11		5	5	1	1				5	in process
Germany	8				5	2				3	3
Ghana	4		4	in process							
Hong Kong	3							2	in process	1	in process
Ireland	6							1	in process	5	5
Israel	1		1	in process							
Italy	5		2	2	1	1				2	2
Jordan	2		1	in process				1	1		
Netherlands	1									1	in process
Norway	27		13	in process	1	1		2	in process	11	in process
Portugal	2				1	1				1	1
Senegal	2		2	1							
Spain	10				1	1	2	1	1	6	5
Sweden	7		1	1	5	5				1	1
Switzerland	1		1	1							
Turkey	2				1	in process	1				
Ukraine	2	2									
United Kingdom	3				2	2		1	1		
DECEMBER											
Austria	8	4	2	0	2	2					
Belgium	41	1	13	in process	4	3				23	in process
Denmark	2				1	1				1	1
Egypt	72	61			11	11					
France	33		7	in process	15	15		6	6	5	5
Germany	23		6	6	12	12		1	1	4	4
Ghana	8		4	in process	3	in process				1	in process
Greece	1				1	0					
Hong Kong	3		2	in process	1	in process					
Ireland	12		1	in process	3	in process	3			5	in process
Israel	1		1	1							
Italy	14				3	3				11	in process
Jordan	11	6	1	1				4	4		
Latvia	2		1	1						1	1
Netherlands	3		1	in process	2	in process					
Norway	29		28	18	1	1					
Portugal	9		1	0	1	1		1	1	6	6
Senegal	3		3	1							
Spain	9						1	6	in process	2	in process
Slovenia	5		3	3				1	0	1	0
Switzerland	17		9	in process	3	in process				5	in process
Tunisia	2						2				
Turkey	3				3	in process					
Ukraine	4		1	1	1	1	2				
United Kingdom	11		2	2	4	4		1	1	4	4
JANUARY											
Belgium	19	2	7	in process	1	1				9	in process
Denmark	3		1	1	2	2					
Estonia	21	2	9	in process	2	in process	7			1	1
Finland	7		3	3	3	3		1	1		
France	1		1	1							
Germany	9		2	2	2	2		1	1	4	4
Greece	7	1	2	1	3	1	1				
Hong Kong	6		1	in process	2	in process		1	in process	2	in process
Ireland	4				1	1		1	1	2	2
Israel	13		7	in process	5	5				1	1
Italy	17		13	in process	1	1	2	1	1		
Latvia	7		3	3	3	3				1	1
Malta	24		24	in process							
Netherlands	2		1	in process						1	in process
Norway	4		4	3							
Portugal	9		5	3	2	2				2	2
Romania	7		4	in process			3				
Russia	1				1	in process					
Slovenia	7		3	0	2	0		2	0		
Spain	5		5	5							
Tunisia	2		1	1	1	0					
Turkey	11		10	in process	1	in process					
Ukraine	3		2	2						1	1
32 Countries											
	705	90	240	75	156	123	24	45	27	150	67
			34.0%		22.1%			6.4%		21.3%	
			486 (69%)				219 (31%)				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Table 2. Isolation rates for samples collected after 2012-09-01

	Clinical sample			Virus isolate		
	number processed	number propagated ¹	percent recovered	number processed	number propagated ¹	percent recovered
H1	71	36	51%	74	61	82%
H3	19	14	74%	119	114	96%
B Victoria	3	3	100%	30	28	93%
B Yamagata	12	8	67%	85	83	98%

1. Propagated to sufficient titre to perform HI assay

Table 3. Summary of isolates with collection dates after 2012-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays

	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ¹	96	88 (92%)	6 (6%)	2 (2%)
H3 ²	121	91 (75%)	24 (20%)	6 (5%)
H3 ^b	36	36 (100%)		
B/Victoria ³	31	26 (84%)	4 (13%)	1 (3%)
B/Yamagata ⁴	91	38 (42%)	25 (27%)	28 (31%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titre for ferret antiserum raised against vaccine virus A/California/7/2009
2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011

b. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011 in plaque-reduction neutralisation assays

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)
4. Compared to homologous titre for ferret antiserum raised against prototype virus B/Stockholm/12/2011

Table 4. Summary of influenza A (H1N1)pdm09 samples analysed, collected between 2012-09-01 and 2013-01-31

MONTH	H1N1pdm09					
	Country	Number received	Number propagated ¹	Fold reduction in HI titre		
				≤2 fold ²	4 fold ²	≥8 fold ²
OCTOBER						
Algeria	1	in process				
Ghana	3	3	3			
Sweden	1	1	1			
NOVEMBER						
Belgium	1	1	1			
Denmark	1	1	1			
France	5	5	5			
Ghana	4	in process				
Israel	1	in process				
Italy	2	2	2			
Jordan	1	in process				
Norway	13	in process	1		1	
Senegal	2	1				
Sweden	1	1	1			
Switzerland	1	1	1			
DECEMBER						
Belgium	13	in process	4			
France	7	in process	3	2		
Germany	6	6	6			
Ghana	4	in process				
Hong Kong	2	in process				
Ireland	1	in process				
Israel	1	1			1	
Jordan	1	1	1			
Latvia	1	1	1			
Netherlands	1	in process				
Norway	28	18	18			
Senegal	3	1	1			
Slovenia	3	3	2	1		
Switzerland	9	in process	1	1		
Ukraine	1	1	1			
United Kingdom	2	2	2			
JANUARY						
Belgium	7	in process				
Denmark	1	1	1			
Estonia	9	in process	1			
Finland	3	3	2	1		
France	1	1		1		
Germany	2	2	2			
Greece	2	1	1			
Hong Kong	1	in process				
Israel	7	in process	2			
Italy	13	in process	5			
Latvia	3	3	3			
Malta	24	in process				
Netherlands	1	in process				
Norway	4	3	3			
Portugal	5	3	3			
Romania	4	in process				
Spain	5	5	5			
Tunisia	1	1	1			
Turkey	10	in process	1			
Ukraine	2	2	2			
	225	75	88	6	2	
			92%	6%	2%	

1. Propagated to sufficient titre to perform HI assays

2. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

All viruses with >2 fold reduction in HI titre contained HA substitutions, as a result of cell propagation, that affect receptor binding/antigenicity

Table 5. Antigenic analyses of influenza A(H1N1)pdm09 viruses 2012-10-02

			Haemagglutination inhibition titre							
			Post infection ferret sera							
Viruses	Collection date	Passage history	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/C'church 16-Oct F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11
Genetic Group						Group 4	Group 3	Group 5	Group 6	Group 7
REFERENCE VIRUSES										
A/California/7/2009	2009-04-09	E1/E2	1280	1280	1280	640	640	640	640	1280
A/Bayern/69/2009	2009-07-01	MDCK4/MDCK1	80	320	160	80	40	80	80	80
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT1/MDCK2	640	1280	1280	320	160	320	320	160
A/Christchurch/16/2010	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560
A/Hong Kong/3934/2011	2011-03-29	MDCK2/MDCK2	640	160	320	640	1280	1280	640	1280
A/Astrakhan/1/2011	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	2560	2560	1280	5120
A/St. Petersburg/27/2011	2011-02-14	E1/E2	5120	1280	5120	2560	2560	5120	5120	5120
A/St. Petersburg/100/2011	2011-03-14	E1/E2	2560	1280	2560	1280	2560	2560	2560	5120
TEST VIRUSES										
A/Corrientes/66335/2012	2012-08-03	SIAT2/MDCK1	2560	1280	2560	2560	2560	2560	5120	5120
A/Formosa/V2361/2012	2012-08-03	SIAT2/MDCK2	5120	2560	5120	5120	5120	5120	5120	5120
A/Tucumán/11956/2012	2012-08-07	MDCK2/MDCK1	2560	640	1280	2560	2560	2560	2560	5120
A/Misiones/6563/2012	2012-08-07	SIAT2/MDCK1	1280	640	1280	2560	2560	2560	2560	5120
A/Misiones/6592/2012	2012-08-08	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120
A/Tucumán/11972/2012	2012-08-10	SIAT2/MDCK1	2560	640	2560	2560	5120	5120	5120	5120
A/Corrientes/67120/2012	2012-08-15	SIAT2/MDCK1	5120	1280	2560	2560	5120	5120	5120	5120
A/Santa Fe/915526/2012	2012-08-17	SIAT2/MDCK1	5120	1280	2560	2560	5120	5120	5120	5120
A/Catamarca/R4297/2012	2012-08-17	SIAT2/MDCK1	2560	1280	2560	2560	5120	5120	2560	5120
A/Santa Fe/1667/2012	2012-08-21	SIAT2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120
A/Misiones/7214/2012	2012-08-22	MDCK2 /MDCK1	5120	1280	2560	2560	5120	5120	5120	5120
A/Tucumán/A382/2012	2012-08-23	SIAT2/MDCK1	2560	640	1280	1280	2560	2560	2560	5120
A/Tucumán/A480/2012	2012-08-31	MDCK1/MDCK1	5120	1280	2560	2560	5120	5120	5120	5120
Sequences in phylogenetic trees			Vaccine							

Table 6. Antigenic analyses of influenza A(H1N1)pdm09 viruses 2012-11-06

Haemagglutination inhibition titre ¹											
Post infection ferret sera											
Viruses	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/C'church 16-Oct F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic Group						Group 4	Group 3	Group 5	Group 6	Group 7	Group 6
REFERENCE VIRUSES											
A/California/7/2009	2009-04-09	E1/E2	640	640	640	320	640	320	320	640	640
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	40	160	80	40	<	40	40	<	40
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT1/MDCK2	320	1280	640	160	80	80	160	160	320
A/Christchurch/16/2010	42010-07-12	E2/E2	640	1280	1280	5120	1280	1280	640	2560	2560
A/Hong Kong/3934/2011	32011-03-29	MDCK2/MDCK3	160	80	160	320	640	640	320	640	640
A/Astrakhan/1/2011	52011-02-28	MDCK1/MDCK5	640	320	640	640	1280	640	640	2560	1280
A/St. Petersburg/27/2011	62011-02-14	E1/E2	640	640	640	640	1280	1280	640	2560	2560
A/St. Petersburg/100/2011	72011-03-14	E1/E2	1280	640	1280	640	1280	1280	1280	5120	2560
A/Hong Kong/5659/2012	62012-05-21	MDCK4	640	320	640	640	1280	1280	1280	2560	2560
TEST VIRUSES											
A/Netherlands/492/2012	62012-08-14	x/MDCK2/MDCK1	2560	640	1280	1280	2560	2560	1280	5120	5120
A/Stockholm/27/2012	72012-08-15	MDCK3/MDCK1	80	320	160	80	80	80	160	160	160
A/Netherlands/507/2012	62012-08-17	x/MDCK2/MDCK1	640	320	320	160	160	160	160	320	320
A/Stockholm/32/2012	62012-10-25	MDCK2/MDCK1	1280	640	1280	1280	1280	1280	1280	2560	5120
A/Stockholm/34/2012	72012-11-06	MDCK1/MDCK1	320	320	320	320	640	640	640	1280	1280
Sequences in phylogenetic trees			Vaccine								
1. <= <40											

G155X

N156X

Table 7. Antigenic analyses of influenza A(H1N1)pdm09 viruses 2013-01-22

			Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
Viruses		Collection date	Passage History	A/Cal 7/09 F05/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
	Genetic group						Group 4	Group 3	Group 5	Group 6	Group 7	Group 6
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	640	640	640	320	1280	640
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	80	40	<	40	40	40	40
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK2	640	1280	640	320	80	160	320	160	640
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	1280	1280	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	1280	640	640	640	1280	1280	1280	2560	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Hong Kong/5659/2012	6	2012-05-21	MDCK4/MDCK1	320	160	320	160	640	640	320	640	1280
TEST VIRUSES												
A/Slovenia/1905/2012	6	2012-12-05	MDCK1/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560
A/Slovenia/1908/2012	6	2012-12-06	MDCKx/MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560
A/Norway/2493/2012	6	2012-12-08	MDCK2	640	320	640	320	640	640	640	1280	1280
A/Norway/2531/2012	6	2012-12-11	MDCK2	640	320	640	640	1280	640	640	2560	2560
A/Norway/2552/2012	8	2012-12-17	MDCK2	1280	320	640	640	1280	640	1280	2560	2560
A/Lyon/2757/2012	7	2012-12-17	MDCK2/MDCK1	320	160	160	160	320	320	320	640	640
A/Norway/46/2013	6	2012-12-18	MDCK2	1280	320	640	640	1280	1280	1280	2560	2560
A/Norway/2667/2012	7	2012-12-19	MDCK2	640	320	320	320	640	640	640	1280	640
A/Norway/2620/2012	6	2012-12-20	MDCK2	1280	640	1280	1280	1280	1280	1280	5120	2560
A/England/658/2012	6	2012-12-20	MDCK1/MDCK1	1280	640	1280	640	1280	1280	1280	5120	2560
A/Lyon/2783/2012	6	2012-12-20	MDCK2/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560
A/Lyon-CHU/52.271/2013	7	2012-12-27	MDCK2/MDCK1	1280	640	640	1280	1280	1280	1280	5120	5120
A/Lyon/2/2013	7	2012-12-28	MDCK2/MDCK1	320	320	320	320	640	640	320	1280	1280
A/Norway/120/2013	8	2013-01-02	MDCK2	1280	640	640	1280	1280	1280	1280	5120	2560
A/Limoges/26/2013	6	2013-01-03	MDCK1/MDCK1	320	320	320	640	320	320	320	640	640

Sequences in phylogenetic trees

Vaccine

1. < = <40

Table 8. Antigenic analyses of influenza A(H1N1)pdm09 viruses 2013-01-25

			Haemagglutination inhibition titre								
			Post infection ferret sera								
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic group						Group 4	Group 3	Group 5	Group 6	Group 7	Group 6
REFERENCE VIRUSES											
A/California/7/2009	2009-04-09	E1/E2	2560	1280	640	640	1280	640	640	2560	640
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	80	320	80	40	40	40	80	40	40
A/Lviv/N6/2009	2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	160	160	320	160	640
A/Christchurch/16/2010	2010-07-12	E2/E2	2560	2560	1280	5120	2560	1280	2560	5120	320
A/Hong Kong/3934/2011	2011-03-29	MDCK2/MDCK3	640	320	640	640	1280	640	1280	2560	2560
A/Astrakhan/1/2011	2011-02-28	MDCK1/MDCK5	1280	640	640	1280	1280	1280	1280	2560	2560
A/St. Petersburg/27/2011	2011-02-14	E1/E2	2560	1280	1280	1280	2560	1280	2560	5120	5120
A/St. Petersburg/100/2011	2011-03-14	E1/E2	2560	640	1280	1280	2560	1280	2560	5120	5120
A/Hong Kong/5659/2012	2012-05-21	MDCK4/MDCK1	640	160	320	320	640	640	640	1280	1280
TEST VIRUSES											
A/Norway/2505/2012	2012-12-06	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	2560
A/Norway/2607/2012	2012-12-06	MDCK2	2560	2560	1280	2560	2560	1280	2560	5120	5120
A/Norway/2494/2012	2012-12-08	MDCK2	2560	1280	1280	1280	2560	1280	2560	5120	5120
A/Lyon/2665/2012	2012-12-10	MDCK2/MDCK2	1280	640	640	640	1280	1280	1280	2560	2560
A/Norway/45/2013	2012-12-14	MDCK3	1280	640	1280	1280	2560	1280	2560	2560	2560
A/Berlin/167/2012	2012-12-14	C2/MDCK1	2560	640	1280	2560	2560	2560	2560	5120	5120
A/Slovenia/1940/2012	2012-12-14	MDCK2/SIAT2	640	320	640	640	1280	640	1280	2560	2560
A/Norway/48/2013	2012-12-15	MDCK3	1280	320	640	640	2560	2560	1280	5120	2560
A/Thüringen/37/2012	2012-12-17	C2/MDCK1	2560	640	1280	1280	2560	1280	2560	5120	2560
A/Thüringen/38/2012	2012-12-17	C2/MDCK1	2560	1280	2560	2560	2560	2560	2560	5120	5120
A/Thüringen/39/2012	2012-12-17	C2/MDCK1	2560	1280	2560	2560	2560	2560	2560	5120	5120
A/Baden Württemberg/173/2012	2012-12-17	C1/MDCK1	2560	1280	2560	2560	2560	2560	2560	5120	5120
A/Norway/62/2013	2012-12-25	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	2560
A/Norway/130/2013	2012-12-27	MDCK3	1280	320	1280	640	1280	1280	2560	2560	2560
A/Norway/129/2013	2012-12-28	MDCK3	1280	640	1280	640	2560	2560	1280	5120	2560
A/Norway/51/2013	2012-12-30	MDCK3	1280	320	1280	1280	1280	1280	1280	5120	2560
A/Norway/56/2013	2012-12-30	MDCK2	2560	1280	1280	1280	2560	2560	2560	5120	1280
A/Norway/29/2013	2012-12-31	MDCK3	1280	1280	1280	2560	1280	1280	1280	2560	5120
A/Norway/35/2013	2013-01-02	MDCK3	1280	640	1280	1280	2560	1280	2560	2560	2560
A/Norway/108/2013	2013-01-03	MDCK3	1280	640	1280	1280	1280	2560	1280	2560	2560
A/Sachsen/1/2013	2013-01-03	C2/MDCK1	2560	1280	2560	2560	2560	2560	2560	5120	2560
A/Tunisia/159/2013	2013-01-04	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560

Sequences in phylogenetic trees

Vaccine

Table 9. Antigenic analyses of pandemic influenza A(H1N1) viruses 2013-01-30

			Haemagglutination inhibition titre ¹								
			Post infection ferret sera								
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic group						Group 4	Group 3	Group 5	Group 6	Group 7	Group 6
REFERENCE VIRUSES											
A/California/7/2009	2009-04-09	E1/E2	1280	640	1280	640	640	640	640	1280	1280
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	80	160	80	40	<	40	40	<	40
A/Lviv/N6/2009	2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	160	160	320
A/Christchurch/16/2010	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560	2560
A/Hong Kong/3934/2011	2011-03-29	MDCK2/MDCK3	320	320	640	320	1280	640	640	1280	1280
A/Astrakhan/1/2011	2011-02-28	MDCK1/MDCK5	640	320	640	640	1280	640	1280	2560	2560
A/St. Petersburg/27/2011	2011-02-14	E1/E2	1280	640	640	640	640	1280	640	2560	2560
A/St. Petersburg/100/2011	2011-03-14	E1/E2	1280	640	1280	640	1280	1280	1280	5120	2560
A/Hong Kong/5659/2012	2012-05-21	MDCK4/MDCK1	160	80	160	160	320	320	320	640	1280
TEST VIRUSES											
A/Ghana/FS-0895/2012	2012-07-13	SIAT3/MDCK1	640	320	640	320	1280	640	640	2560	1280
A/Ghana/DILI-0517/2012	2012-08-21	SIAT3/MDCK1	640	320	640	320	640	640	640	1280	1280
A/Ghana/DILI-0518/2012	2012-08-21	SIAT3/MDCK1	320	640	640	160	320	320	320	1280	640
A/Ghana/DILI-0880/2012	2012-10-19	SIAT3/MDCK1	640	160	640	640	1280	1280	1280	2560	1280
A/Ghana/DILI-0899/2012	2012-10-25	SIAT2/MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	5120
A/Ghana/DILI-0902/2012	2012-10-25	SIAT2/MDCK1	640	320	640	320	1280	640	1280	2560	1280
A/Paris/1667/2012	2012-11-01	MDCK2/MDCK1	640	640	640	640	1280	640	1280	2560	1280
A/Dakar/22/2012	2012-11-03	MDCK2	2560	640	1280	1280	2560	2560	2560	5120	2560
A/Paris/1773/2012	2012-11-20	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280
A/Strasbourg/1866/2012	2012-11-21	MDCK2/MDCK1	1280	640	640	1280	1280	1280	1280	5120	2560
A/Paris/1836/2012	2012-11-28	MDCK2/MDCK1	640	320	320	320	640	320	640	1280	1280
A/Paris/1878/2012	2012-11-29	MDCK2/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560
A/Berlin/166/2012	2012-12-11	C1/MDCK2	1280	1280	1280	1280	2560	1280	2560	2560	5120
A/Jordan/30505/2012	2012-12-12	C1/MDCK1	640	640	640	640	640	1280	1280	2560	2560
A/England/661/2012	2012-12-21	SIAT1/MDCK3	1280	640	640	640	1280	1280	1280	2560	2560
A/Athens GR/1/2013	2013-01-02	MDCK2	640	320	640	640	1280	1280	1280	2560	1280
A/Niedersachsen/1/2013	2013-01-07	C1/MDCK2	1280	640	640	1280	1280	1280	1280	2560	2560
A/Ukraine/3/2013	2013-01-07	MDCK2/MDCK1	640	160	640	320	640	640	640	1280	1280
A/Lisboa/EuroEva42/2013	2013-01-08	MDCK2	640	320	640	320	640	640	1280	1280	1280
A/Ukraine/11/2013	2013-01-08	MDCK2/MDCK1	640	320	640	320	1280	640	1280	2560	2560

G155X

Sequences in phylogenetic trees

1. < = <40

Table 10. Antigenic analyses of pandemic influenza A(H1N1) viruses 2013-02-05

Haemagglutination inhibition titre											
Post infection ferret sera											
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6
Genetic group											
REFERENCE VIRUSES											
A/California/7/2009	2009-04-09	E1/E2	1280	640	1280	640	640	640	640	640	640
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	80	160	80	40	40	40	40	40	40
A/Lviv/N6/2009	2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	160	160	320
A/Christchurch/16/2010	42010-07-12	E2/E2	2560	1280	2560	5120	2560	1280	1280	2560	2560
A/Hong Kong/3934/2011	32011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	320	640	640
A/Astrakhan/1/2011	52011-02-28	MDCK1/MDCK5	1280	320	1280	640	1280	1280	640	2560	2560
A/St. Petersburg/27/2011	62011-02-14	E1/E2	1280	640	1280	640	1280	1280	1280	2560	2560
A/St. Petersburg/100/2011	72011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Hong Kong/5659/2012	62012-05-21	MDCK4/MDCK1	320	160	320	320	640	640	320	1280	1280
TEST VIRUSES											
A/Chiang Rai/312/2012	62012-02-10	C2/C1/MDCK1	640	640	640	640	1280	1280	1280	2560	2560
A/Bangladesh/2021/2012	62012-07-21	E4/E1	320	320	160	160	160	160	80	160	320
A/Tennessee/09/2012	62012-10-31	C2/MDCK1	640	640	640	640	1280	1280	640	2560	1280
A/Norway/2289/2012		LLCMK2/MDCK1/MDCK1	1280	640	1280	1280	1280	1280	2560	2560	2560
A/Belgium/G860/2012	72012-11-08	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560
A/Norway/2254/2012		MDCK2/MDCK2	160	640	320	80	80	80	160	80	320
A/Pavia/175/2012		Cx/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560
A/Denmark/76/2012		MDCK3/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560
A/Parma/158/2012		MDCK2/MDCK1	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Switzerland/7502824/2012	72012-11-29	MDCK2	1280	640	640	1280	1280	1280	2560	5120	2560
A/Ukraine/523/2012	62012-12-02	MDCK2/MDCK2	1280	320	640	640	2560	1280	1280	2560	2560
A/Belgium/G906/2012		MDCK2	1280	640	1280	1280	2560	2560	2560	2560	2560
A/Switzerland/7546304/2012	72012-12-07	MDCK2	1280	640	640	640	1280	1280	1280	2560	2560
A/Belgium/G916/2012		MDCK2	2560	1280	1280	1280	2560	2560	2560	5120	5120
A/Belgium/G917/2012	72012-12-07	MDCK2	640	320	640	640	1280	640	640	2560	2560
A/Dakar/20/2012	outlier W Africa	MDCK3	2560	640	1280	1280	2560	5120	2560	5120	5120
A/Belgium/G932/2012		MDCK2	1280	640	640	640	1280	1280	1280	2560	2560
A/Switzerland/7587022/2012		MDCK3	320	640	320	160	80	80	160	160	320
A/Norway/2597/2012		MDCK4	2560	640	1280	640	2560	2560	1280	5120	2560
A/Latvia/12-39425p/2012		MDCK2/MDCK1	1280	640	1280	1280	1280	1280	1280	5120	2560
A/Israel/11/2012		C1/MDCK1	160	640	320	80	80	160	160	80	160
A/Pavia/1/2013		Cx/MDCK1	1280	640	1280	1280	1280	1280	2560	2560	2560
A/Israel/18/2013		C2/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560
A/Latvia/1-30371p/2013		MDCK1/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560
A/Roma/01/2013		Cx/MDCK1	1280	640	1280	1280	2560	1280	2560	2560	2560
A/Navarra/19/2013		E2/E1	1280	1280	1280	1280	1280	1280	1280	2560	2560
A/Estonia/74192/2013		MDCK2/MDCK1	640	320	640	640	1280	640	640	2560	2560
A/Denmark/08/2013		MDCK1/MDCK1	1280	320	640	640	2560	2560	1280	2560	2560
A/Roma/05/2013		Cx/MDCK2	640	320	640	320	1280	1280	1280	2560	2560
A/Roma/06/2013		Cx/MDCK2	1280	320	1280	1280	1280	1280	1280	2560	2560
A/Roma/07/2013		Cx/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560
A/Acores/HAH480803/2012	62013-01-07	MDCK3	2560	640	1280	1280	2560	2560	2560	5120	5120
A/Acores/HAH7518/2012	62013-01-07	MDCK2	640	640	640	1280	1280	1280	1280	2560	2560
A/Israel/28/2013		C1/MDCK1	1280	640	1280	640	2560	2560	1280	2560	2560
A/Latvia/1-36685/2013		MDCK1/MDCK1	640	320	640	640	1280	640	640	2560	2560
A/Navarra/111/2013		SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Madrid/61/2013		SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Baleares/100/2013		SIAT1/MDCK1	640	320	640	640	1280	1280	640	1280	1280
A/Navarra/108/2013		SIAT1/MDCK1	1280	640	1280	640	1280	2560	1280	2560	2560
A/Turkey/600/2013		C0/MDCK1	2560	640	1280	1280	2560	2560	2560	5120	5120
A/Latvia/1-40822p/2013		MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	5120
A/Finland/30820/13	62013-01-21	MDCK1/MDCK1	1280	640	1280	1280	1280	1280	1280	5120	5120
A/Finland/314/2013		MDCK1/MDCK1	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Finland/302/2013	62013-01-24	MDCK3/MDCK1	320	320	320	320	160	320	320	320	640
Sequences in phylogenetic trees											

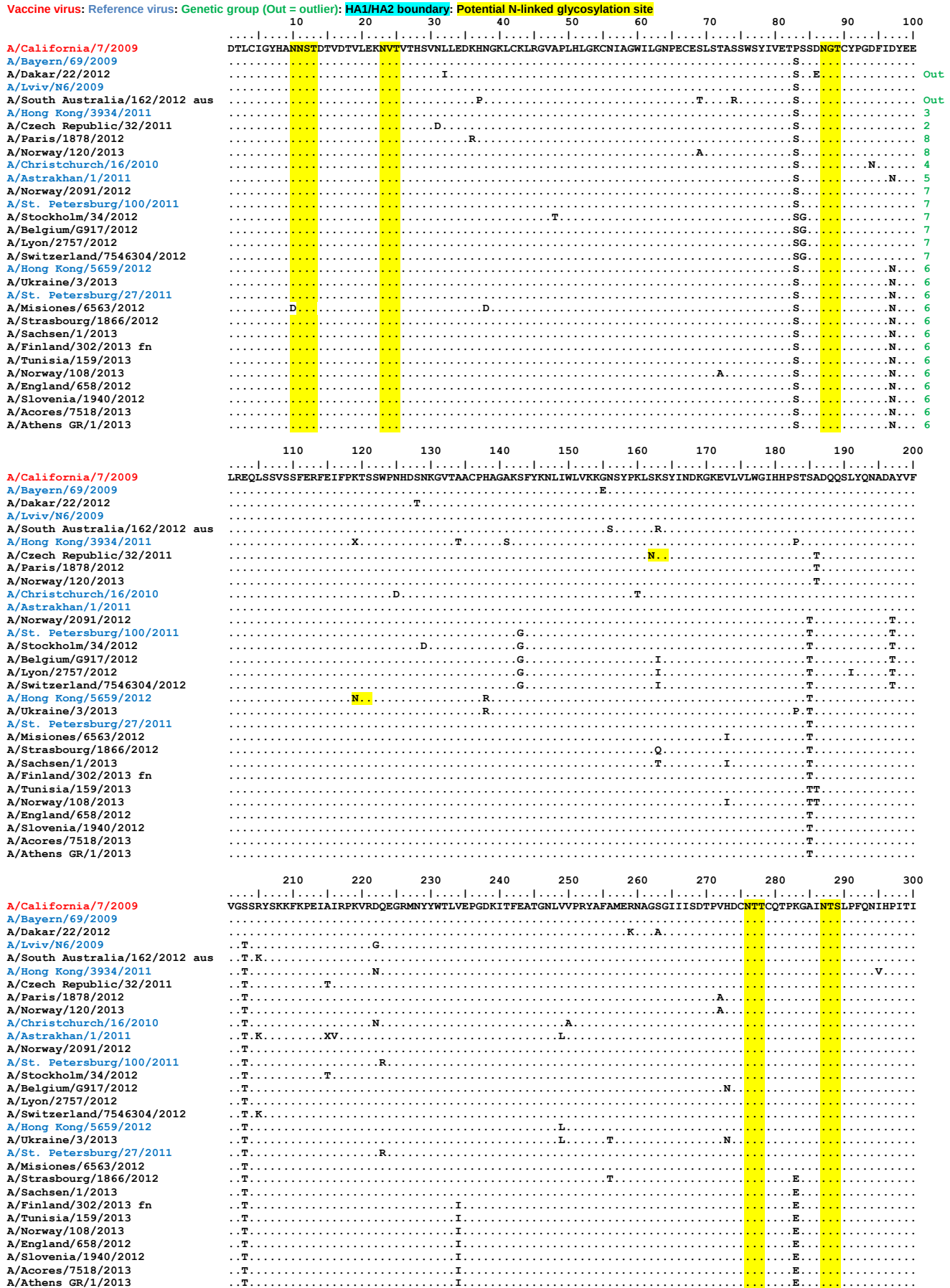
Vaccine virus
Reference viruses

Jan 2013

proposed serology antigens



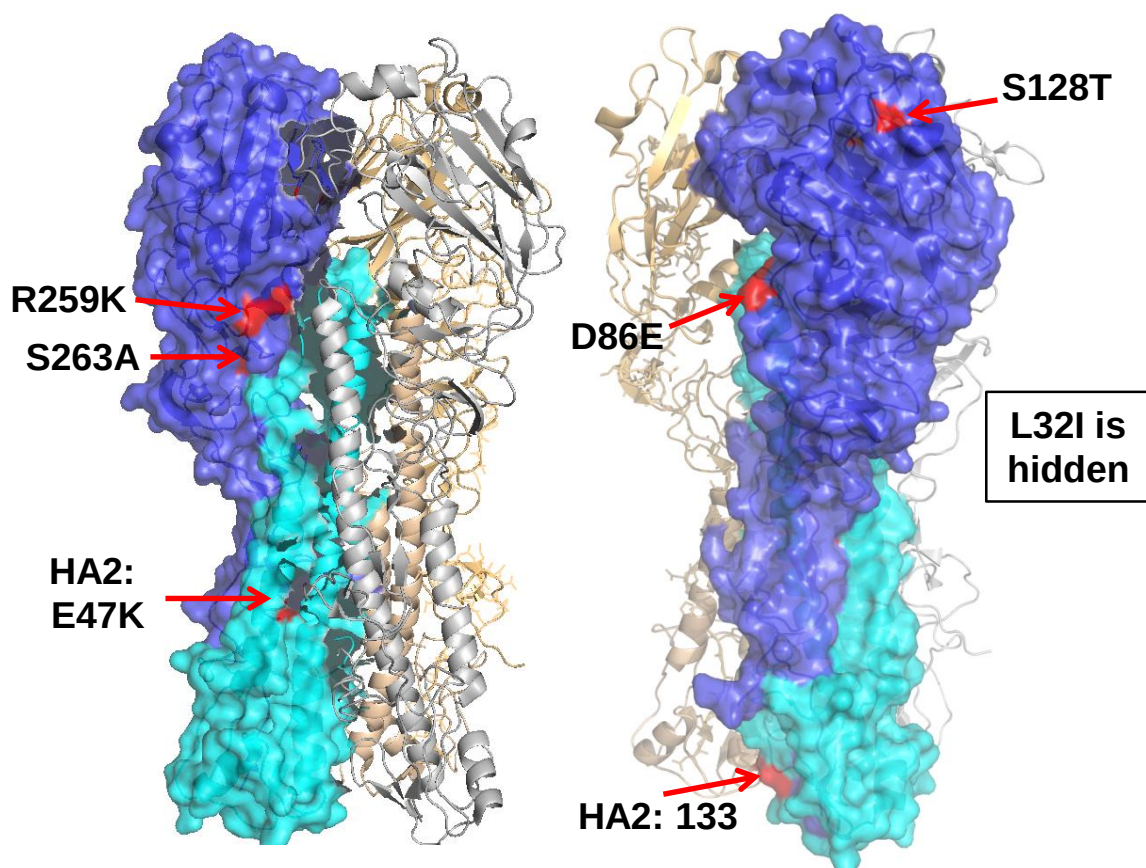
Figure 7. HA protein alignment for A(H1N1)pdm09 viruses.



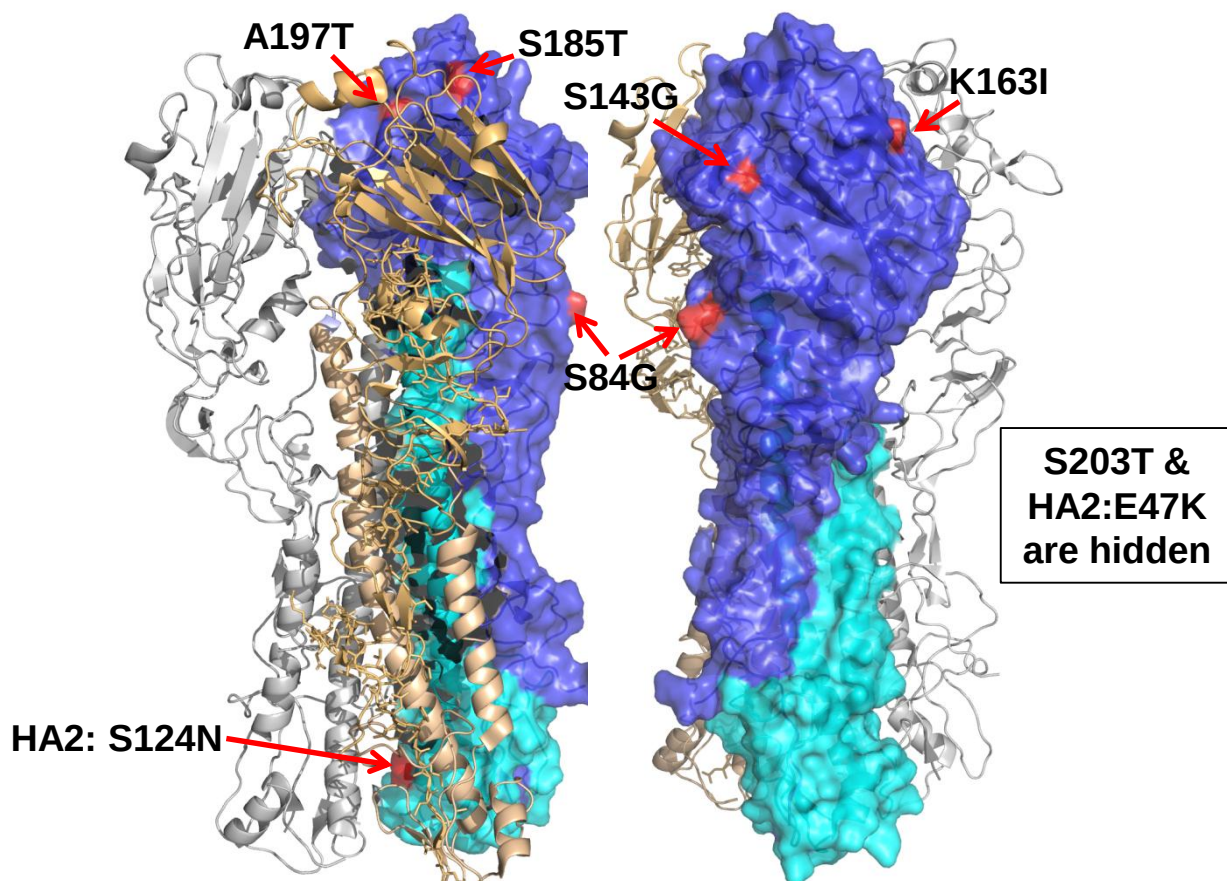
	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVSTKLR	LATGLRNPISIQSR	GLFGALAGFIEGGWT	GMVDGWYGYHHQNEQ	SGGYAADLKSTQNA	IDEITNKNVSVIEKMNT	QFTAVGKEFNHL			
A/Bayern/69/2009		V								
A/Dakar/22/2012		V					K			
A/Lviv/N6/2009		V								
A/South Australia/162/2012 aus		V								
A/Hong Kong/3934/2011		V								
A/Czech Republic/32/2011		V					K			
A/Paris/1878/2012		V					K			
A/Norway/120/2013		V				Q	K			
A/Christchurch/16/2010		V					K			
A/Astrakhan/1/2011		V					K			
A/Norway/2091/2012		V					K			
A/St. Petersburg/100/2011		V					K			
A/Stockholm/34/2012		V					K			
A/Belgium/G917/2012		V					K			
A/Lyon/2757/2012		V					K			
A/Switzerland/7546304/2012		V					K			
A/Hong Kong/5659/2012		V					K			
A/Ukraine/3/2013		V					K			
A/St. Petersburg/27/2011		V					K			
A/Misiones/6563/2012		V					K			
A/Strasbourg/1866/2012		V		X			K			
A/Sachsen/1/2013		V					K			
A/Finland/302/2013 fn		V					K			
A/Tunisia/159/2013		V					K			
A/Norway/108/2013		V					K			
A/England/658/2012		V					K			
A/Slovenia/1940/2012		V					V.K			
A/Acores/7518/2013		V					K			
A/Athens GR/1/2013		V					K			
	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRLENLKKVDDG	FLDIWTYNALLVLE	NERTLDYHDSNVKNLY	EKVRSQLKNNAKEI	GNCGCFEYHKCDNT	CMESVKN	NGTYDPKYSEEAK	LNREEI		
A/Bayern/69/2009										
A/Dakar/22/2012						V				
A/Lviv/N6/2009										
A/South Australia/162/2012 aus	I									
A/Hong Kong/3934/2011								I		
A/Czech Republic/32/2011				N						
A/Paris/1878/2012							DK			
A/Norway/120/2013							DK			
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/Norway/2091/2012				N		T				K
A/St. Petersburg/100/2011				N						
A/Stockholm/34/2012				N						
A/Belgium/G917/2012				N						
A/Lyon/2757/2012				X						
A/Switzerland/7546304/2012				N						
A/Hong Kong/5659/2012				N						
A/Ukraine/3/2013				N						
A/St. Petersburg/27/2011				N						
A/Misiones/6563/2012				N						
A/Strasbourg/1866/2012				N						K
A/Sachsen/1/2013				N						K
A/Finland/302/2013 fn				N						K
A/Tunisia/159/2013				N						K
A/Norway/108/2013				N						K
A/England/658/2012				N						K
A/Slovenia/1940/2012	V			N						K
A/Acores/7518/2013				N						K
A/Athens GR/1/2013				N						K
	510	520	530	540						
A/California/7/2009	DGVKLESTRIYQ	ILAIYSTVASSLV	LVSLGAISFWMC	NGSLQCRICI						
A/Bayern/69/2009										
A/Dakar/22/2012		A								
A/Lviv/N6/2009										
A/South Australia/162/2012 aus					V					
A/Hong Kong/3934/2011										
A/Czech Republic/32/2011										
A/Paris/1878/2012	V	A								
A/Norway/120/2013		A								
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/Norway/2091/2012										
A/St. Petersburg/100/2011										
A/Stockholm/34/2012										
A/Belgium/G917/2012	V	A								
A/Lyon/2757/2012		A								
A/Switzerland/7546304/2012		A								
A/Hong Kong/5659/2012										
A/Ukraine/3/2013										
A/St. Petersburg/27/2011										
A/Misiones/6563/2012										
A/Strasbourg/1866/2012										
A/Sachsen/1/2013										
A/Finland/302/2013 fn										
A/Tunisia/159/2013										
A/Norway/108/2013										
A/England/658/2012										
A/Slovenia/1940/2012										
A/Acores/7518/2013										
A/Athens GR/1/2013										

Vaccine virus: Reference virus: Genetic group (Out = outlier): HA1/HA2 boundary: Potential N-linked glycosylation site

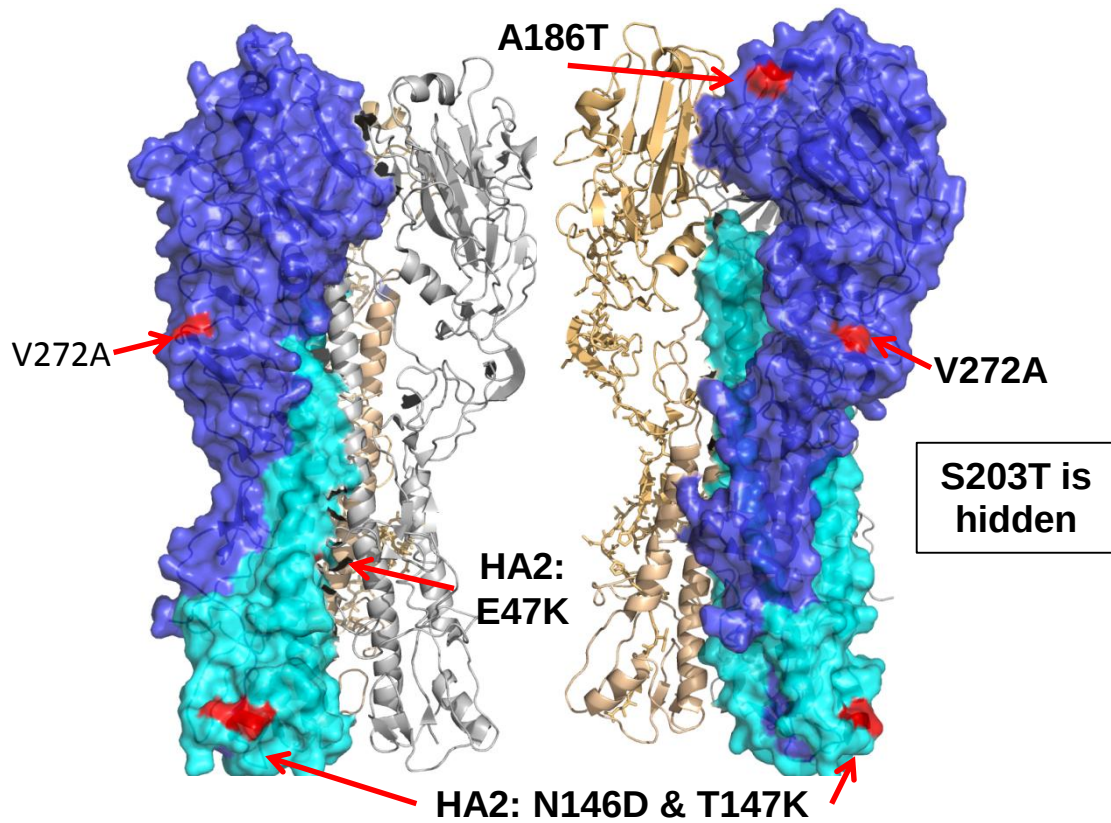
Figure 8. H1-HA locations of genetic group defining amino acid substitutions



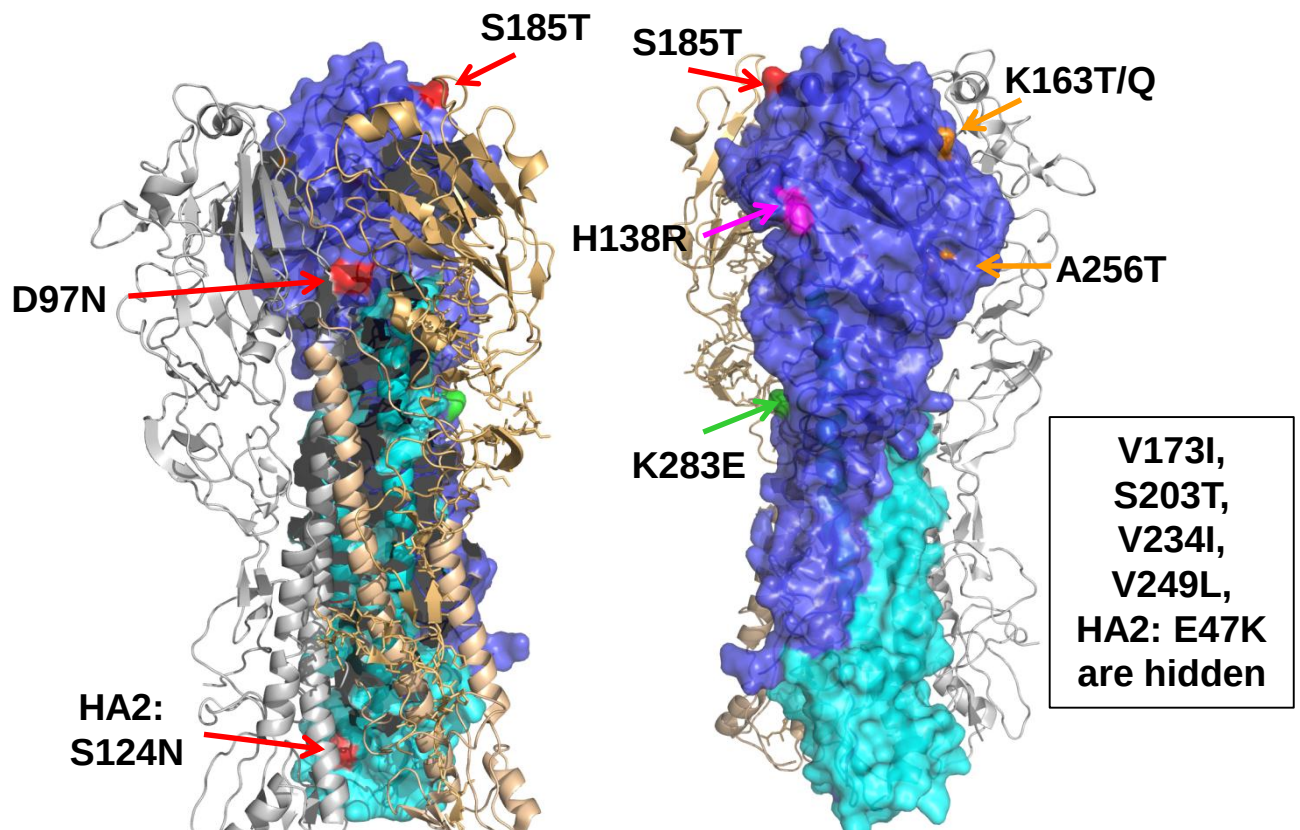
H1 HA – W. Africa group



H1 HA – group 7



H1 HA – group 8



H1 HA – group 6

Figure 9. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

Vaccine virus
Reference viruses

Collection date

Oct 2012

Nov 2012

Dec 2012

Jan 2013

See HI Tables
for antigenic
characterisation

proposed serology
antigens

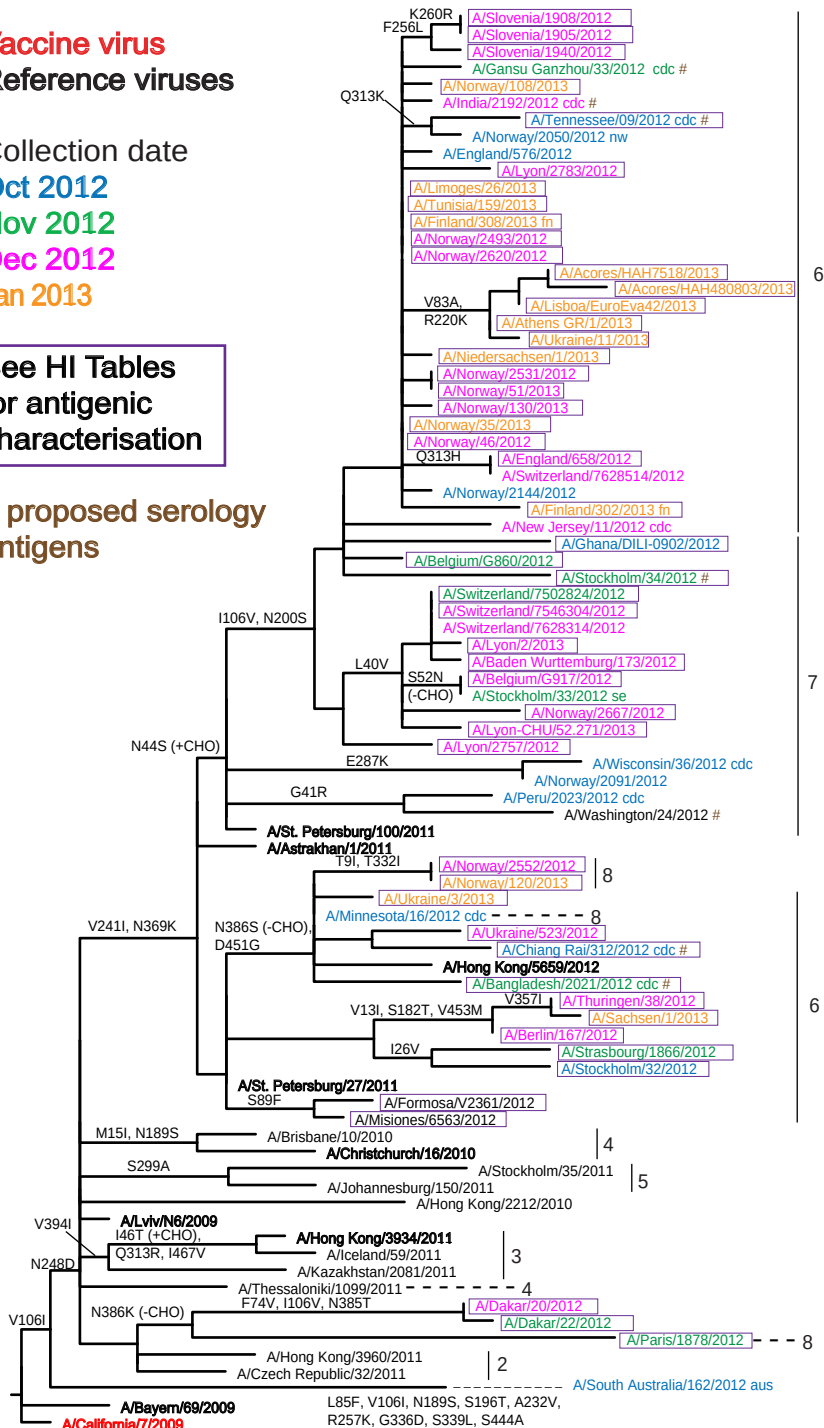


Figure 10. NA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: **Reference virus:** **Genetic group (Out = outlier):** **Potential N-linked glycosylation site**

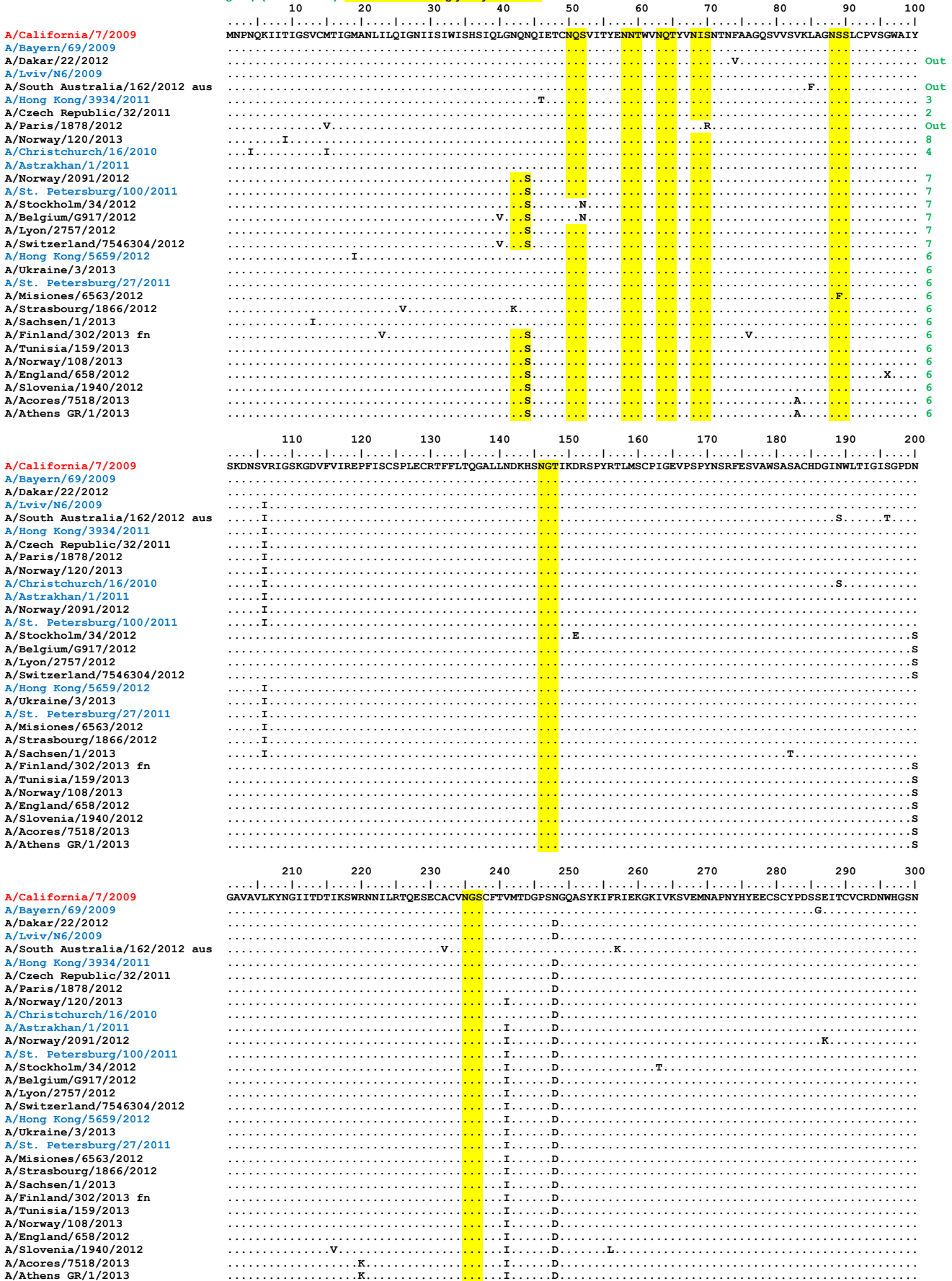


Table 11. Summary of influenza A (H3N2) samples analysed, collected between 2012-09-01 and 2013-01-31

MONTH	H3N2				
	Number received	Number propagated ¹	Fold reduction in HI titre		
			≤2 fold ²	4 fold ²	≥8 fold ²
Country					
SEPTEMBER					
Denmark	2	2	2		
Senegal	1	1	1		
OCTOBER					
France	3	3	2	1	
Germany	1	1	1		
Israel	1	1			1
Norway	2	2	2		
Romania	1	1	1		
Senegal	14	14	14		
Sweden	1	1	1		
United Kingdom	2	2	1	1	
NOVEMBER					
Algeria	2	2	2		
Austria	1	1	1		
Denmark	1	1	1		
Egypt	2	2	2		
France	1	1	1		
Germany	5	2	1	1	
Italy	1	1	1		
Norway	1	1	1		
Portugal	1	1	1		
Spain	1	1	1		
Sweden	5	5	5		
Turkey	1	in process			
United Kingdom	2	2	2		
DECEMBER					
Austria	2	2	2		
Belgium	4	3	3		
Denmark	1	1	1		
Egypt	11	11	11		
France	15	15	9	5	1
Germany	12	12	1	10	1
Ghana	3	in process			
Hong Kong	1	in process			
Ireland	3	in process	1		
Italy	3	3	1	1	1
Netherlands	2	in process			
Norway	1	1	1		
Portugal	1	1		1	
Switzerland	3	in process		2	
Turkey	3	in process	2		
Ukraine	1	1	1		
United Kingdom	4	4	3	1	
JANUARY					
Belgium	1	1	1		
Denmark	2	2	2		
Estonia	2	in process			
Finland	3	3	3		
Germany	2	2	1		1
Greece	3	1	1		
Hong Kong	2	in process			
Ireland	1	1	1		
Israel	5	5	3	1	1
Italy	1	1	1		
Latvia	3	3	3		
Portugal	2	2	2		
Russia	1	in process			
Turkey	1	in process			
	151	123	98	24	6

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011

Table 12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2012-09-18

Viruses		Collection Date	Passage History	Haemagglutination inhibition titre ¹												
				Post infection ferret sera												
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Ohio 2/12	A/SAust 30/12	A/Mary 2/12	
				F35/11	F7/10	F27/10	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12	F27/12	F28/12	F29/12	
Genetic group				group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3B	group 3A	group 3B			
REFERENCE VIRUSES																
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	80	80	80	
A/Victoria/208/2009		2009-06-02	E3/E2	1280	5120	1280	2560	2560	5120	2560	2560	2560	1280	1280	5120	
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	40	80	160	320	40	40	40	
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT5	<	<	40	160	40	80	160	80	320	80	80	80	
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	1280	2560	1280	2560	320	640	1280	
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	320	160	640	5120	640	320	320	160	640	1280	
A/Berlin/93/2011	3C	2011-12-07	NVD3/S4	80	80	160	320	320	320	640	320	640	160	160	160	
A/Victoria/361/2011	3C	2011-10-24	M2/S2	80	80	160	320	320	320	320	320	640	160	160	160	
A/Athens/112/2012	3B	2012-02-01	SIAT5	40	40	80	160	80	160	320	320	640	160	160	160	
A/Ohio/2/2012	3B	2012-03-08	NVD3/S4	160	80	640	160	320	160	320	320	640	640	160	160	
A/South Australia/30/2012	3A	2012-05-10	E2/E1	160	40	320	320	320	320	320	320	640	320	160	80	
A/Maryland/2/2012	3B	2012-04-01	E5/E1	160	640	320	640	640	1280	640	320	640	320	1280	2560	
TEST VIRUSES																
A/England/565/2012		2012-05-31	SIAT2/SIAT3	<	<	40	80	40	80	160	160	320	80	80	80	
A/Attécoube/GR741/2012	3A	2012-07-06	Cx/SIAT1	160	320	320	1280	640	640	1280	2560	2560	640	640	80	
A/Buenos Aires/1004423/2012	3C	2012-07-24	SIAT2/SIAT1	160	160	320	640	640	640	1280	1280	1280	640	320	80	
A/Buenos Aires/10435982/2012	3C	2012-07-27	SIAT2/SIAT1	160	160	320	640	320	320	1280	1280	1280	320	320	80	
A/England/568/2012	3C	2012-08-18	SIAT1/SIAT3	<	<	40	80	40	80	160	160	160	80	80	80	
A/England/569/2012	3C	2012-08-18	SIAT2/SIAT2	<	40	<	160	40	80	160	160	320	80	80	80	
								Vaccine	Vaccine							

Vaccine

Vaccine

1. < = <40

Table 13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2012-12-27

Haemagglutination inhibition titre ¹														
Viruses	Collection Date	Passage History	Post infection ferret sera											
			A/Perth	A/Vic	A/Ala	A/Stock	A/Iowa	A/Vic	A/Berlin	A/Vic	A/Athens	A/Texas	A/Hawaii	
			16/09	208/09	5/10	18/11	19/10	361/11	93/11	361/11	112/12	50/12	22/12	
			F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F14/12	F16/12	F36/12	F37/12	
Genetic group			group 5		group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C		
REFERENCE VIRUSES														
A/Perth/16/2009	2009-07-04	E3/E2	1280	80	320	160	160	160	320	640	640	320	160	
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	1280	2560	5120	2560	5120	2560	2560	5120	5120	
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	40	80	80	80	80	160	320	160	320	80
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT5	40	40	40	160	80	80	320	320	160	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	1280	1280	1280	2560	640
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	320	160	640	2560	640	640	160	2560	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT5	160	80	160	320	160	320	640	640	320	640	320
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	80	80	160	320	160	160	640	640	320	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT8	40	80	80	160	80	160	320	320	640	640	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	640	640	1280	640	1280	1280	1280	1280	2560	1280
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	640	640	1280	640	1280	1280	1280	2560	5120
TEST VIRUSES														
A/Stockholm/12-16700/2012	3C	unknown	MDCK2/SIAT1	40	80	80	320	160	160	640	640	640	640	320
A/Stockholm/30/2012	5	2012-07-12	MDCK1/SIAT1	40	80	160	160	160	160	320	640	320	640	160
A/Stockholm/29/2012	3C	2012-07-27	MDCK1/SIAT1	40	80	80	160	80	160	320	320	320	640	320
A/Denmark/68/2012	3C	2012-09-25	Cx/SIAT1	40	80	80	320	160	160	640	640	320	640	320
A/Denmark/67/2012	3C	2012-09-27	MDCK3/SIAT1	40	80	80	320	160	160	640	640	320	640	320
A/Stockholm/31/2012	5	2012-10-17	MDCK2/SIAT1	80	80	320	320	320	320	640	640	640	1280	320
A/England/575/2012	3C	2012-10-23	SIAT1/SIAT1	40	80	80	320	160	160	320	160	160	640	320
A/Mecklenburg Vorpommern/4/2012	5	2012-10-30	C2/SIAT2	40	40	80	160	80	80	320	320	320	320	160
A/England/587/2012	3C	2012-10-31	SIAT1/SIAT1	<	80	40	160	40	80	320	320	160	320	160
A/England/586/2012	3C	2012-11-02	SIAT1/SIAT1	<	80	80	160	80	80	320	320	160	320	160
A/Niedersachsen/29/2012	3C	2012-11-05	C1/SIAT1	40	40	80	160	80	80	640	320	640	640	320
A/Stockholm/38/2012	5	2012-11-05	MDCK1/SIAT1	80	80	160	320	320	160	640	640	640	640	320
A/Denmark/71/2012	3C	2012-11-08	MDCK4/SIAT1	40	80	80	320	160	160	640	640	320	640	320
A/Stockholm/39/2012	3A	2012-11-08	MDCK1/SIAT1	40	80	80	320	160	80	320	320	320	640	160
A/England/593/2012	3C	2012-11-12	SIAT1/SIAT1	<	80	40	160	80	80	320	320	160	320	160
A/Stockholm/40/2012	5	2012-11-14	MDCK1/SIAT1	80	80	160	160	160	160	320	640	320	640	160
A/Stockholm/37/2012	5	2012-11-15	MDCK0/SIAT1	80	160	160	320	320	160	640	640	320	1280	320
A/Stockholm/36/2012	3C	2012-11-16	MDCK1/SIAT1	40	80	80	320	160	160	640	640	320	640	320
A/Denmark/78/2012	3C	2012-12-05	MDCK2/SIAT1	40	80	40	160	80	80	320	320	320	320	160
Sequences in phylogenetic trees								Vaccine		Vaccine				

1. < = <40

Table 14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-01-15

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret sera										
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12
				F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F14/12	F16/12	F36/12	F37/12
				group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C		
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	160
A/Victoria/208/2009		2009-06-02	E3/E1	1280	5120	1280	2560	5120	2560	5120	2560	2560	5120	5120
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT3	40	<	160	80	80	160	160	320	320	320	80
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT5	80	80	80	320	160	320	320	320	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	640	1280	640	1280	1280	1280	2560	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	160	320	320	80	640	1280	640	320	160	2560	640
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT5	160	160	320	320	320	320	1280	1280	640	1280	640
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	160	160	320	320	320	160	640	1280	640	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	160	160	320	160	160	640	640	640	640	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	640	640	1280	640	640	1280	1280	1280	5120	1280
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	640	640	1280	640	1280	1280	1280	2560	2560
TEST VIRUSES														
A/Austria/705367/2012	5	2012-11-06	SIAT1/SIAT1	160	160	320	320	320	160	640	640	320	1280	320
A/Austria/709778/2012	3C	2012-12-06	SIAT1/SIAT1	80	160	320	320	320	160	640	640	640	1280	640
A/Norway/2496/2012	3C	2012-12-10	SIAT2	40	80	160	320	160	160	640	640	640	640	320
A/Austria/710638/2012	5	2012-12-11	SIAT1/SIAT1	160	160	640	640	640	320	1280	1280	1280	1280	640
Sequences in phylogenetic trees									Vaccine		Vaccine			

1. < = <40

Table 15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-01-22

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret sera										
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12
				F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12
group 5 group 3A group 6 group 3C group 3C group 3C group 3B group 3C group 3C														
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	320	160	160	320	320	640	640	160
A/Victoria/208/2009		2009-06-02	E3/E2	640	5120	2560	1280	2560	2560	2560	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT3	40	<	160	80	160	80	160	160	320	320	80
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT5	80	80	40	320	160	160	320	320	640	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	2560	640	2560	1280	2560	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	320	640	160	640	2560	640	320	320	5120	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT3	320	160	80	160	160	160	640	640	640	1280	320
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT3	160	160	80	320	320	160	640	640	1280	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT5	80	160	80	320	160	320	640	320	1280	1280	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	640	640	1280	1280	1280	2560	640	1280	5120	2560
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	640	640	1280	640	1280	640	2560	5120	5120
TEST VIRUSES														
A/Poitiers/2407/2012	3C	2012-10-24	MDCK2/SIAT1	<	40	<	80	40	40	160	160	160	320	80
A/Niedersachsen/31/2012		2012-11-21	C2/SIAT1	40	80	160	160	160	160	640	160	640	640	320
A/England/625/2012	3C	2012-12-04	MDCK1/SIAT1	<	40	80	320	160	160	640	320	640	640	320
A/Lyon/2623/2012		2012-12-05	MDCK3/SIAT1	<	40	80	160	160	80	320	160	320	320	160
A/Berlin/165/2012	3C	2012-12-05	C2/SIAT1	<	40	40	160	80	80	320	80	320	640	320
A/Poitiers/2745/2012	3C	2012-12-06	MDCK2/SIAT1	<	40	<	80	40	40	320	160	160	320	80
A/Lyon/2633/2012		2012-12-10	MDCK2/SIAT1	40	80	160	320	160	160	640	640	640	1280	320
A/Bremen/18/2012		2012-12-10	C2/SIAT1	80	80	320	160	320	160	640	160	640	1280	320
A/England/668/2012	3C	2012-12-17	MDCK1/SIAT1	<	40	80	160	80	80	320	320	320	640	320
A/Lyon/2743/2012		2012-12-17	MDCK2/SIAT1	<	40	80	160	80	80	320	160	640	640	320
A/Bremen/19/2012		2012-12-17	C2/SIAT1	80	80	320	320	320	160	640	160	640	1280	320
A/Schleswig Holstein/7/2012	3C	2012-12-17	C2/SIAT1	40	80	80	320	160	160	640	160	640	640	320
A/Mecklenburg Vorpommern/10/2012	5	2012-12-17	C2/SIAT1	40	40	160	160	320	160	640	160	320	640	320
A/England/670/2012	3C	2012-12-18	MDCK1/SIAT1	<	40	40	160	40	40	320	160	320	640	160
A/Lyon/2763/2012	3C	2012-12-19	MDCK2/SIAT1	<	40	40	160	80	80	320	160	320	320	160
A/Niedersachsen/33/2012		2012-12-19	C2/SIAT1	40	80	80	320	160	160	640	160	640	640	320
A/Berlin/170/2012		2012-12-20	C2/SIAT1	40	80	160	320	160	160	640	160	640	1280	320
A/Berlin/171/2012		2012-12-20	C2/SIAT1	80	80	160	320	160	160	640	160	640	1280	320
A/Thüringen/41/2012		2012-12-20	C2/SIAT1	40	80	80	320	160	160	640	160	640	640	320
A/Annecy/2809/2013		2012-12-21	MDCK2/SIAT1	<	40	40	160	80	80	320	320	320	640	160
A/Brandenburg/3/2012		2012-12-21	C2/SIAT1	40	80	160	320	160	160	640	160	640	1280	320
A/Annecy/2806/2012	3C	2012-12-22	MDCK2/SIAT1	<	40	80	160	80	80	640	320	640	640	320
A/Clermont-Ferrand/14/2013	3C	2012-12-23	MDCK2/SIAT1	<	40	40	160	40	80	320	320	320	640	160
A/Lyon-CHU/52.123/2012		2012-12-24	MDCK2/SIAT1	<	80	80	320	160	160	320	160	640	640	320
A/Lyon-CHU/52.131/2012	3C	2012-12-24	MDCK2/SIAT1	<	<	40	80	40	80	160	80	320	320	160
A/Mecklenburg Vorpommern/13/2012		2012-12-27	C2/SIAT1	160	160	320	320	640	320	640	320	1280	1280	640
A/Mecklenburg Vorpommern/12/2012	5	2012-12-27	C2/SIAT1	40	80	320	160	320	160	640	160	320	640	320
A/England/676/2012	3C	2012-12-29	SIAT1/SIAT1	<	40	40	160	40	80	320	320	320	640	160
A/Nordrhein-Westfalen/1/2013	3C	2013-01-03	C1/SIAT1	<	40	80	160	160	80	640	80	640	640	320
Sequences in phylogenetic trees									Vaccine			Vaccine		

1. < = <40

Table 16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-01-25

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret sera										
				A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F35/12	A/Berlin 93/11 T/C F11/12	A/Vic 361/11 T/C F34/12	A/Athens 112/12 F16/12	A/Texas 50/12 F36/12	A/Hawaii 22/12 F37/12
						group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	640	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	5120	1280	1280	2560	2560	2560	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT3	40	<	80	80	80	80	160	160	160	320	80
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT4	40	80	80	320	80	80	320	160	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	1280	1280	1280	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	320	160	640	2560	640	320	320	5120	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT3	160	80	160	320	320	160	640	640	640	1280	640
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	80	80	80	160	160	80	320	320	320	640	320
A/Athens/112/2012	3B	2012-02-01	S5	80	80	160	320	160	160	320	320	640	640	640
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	640	320	1280	1280	640	1280	640	1280	5120	2560
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	640	640	1280	640	1280	640	1280	5120	5120
TEST VIRUSES														
A/Dakar/04/2012	3C	2012-10-03	MDCK1/SIAT1	40	80	80	160	160	80	640	320	640	1280	320
A/Dakar/10/2012		2012-10-04	MDCK1/SIAT1	40	80	160	320	160	160	320	320	640	640	320
A/Dakar/11/2012		2012-10-04	MDCK2/SIAT1	40	80	160	160	160	160	320	320	640	640	320
A/Dakar/12/2012		2012-10-04	MDCK2/SIAT1	<	40	40	80	40	80	320	320	320	640	160
A/Dakar/16/2012	3C	2012-10-04	MDCK1/SIAT1	<	<	<	80	<	40	160	160	160	160	80
A/Dakar/09/2012		2012-10-05	MDCK1/SIAT1	80	80	80	160	160	160	640	320	320	640	320
A/Dakar/05/2012		2012-10-06	MDCK1/SIAT1	40	80	80	320	160	80	320	320	640	640	320
A/Dakar/06/2012		2012-10-06	MDCK1/SIAT1	<	40	40	160	80	80	320	160	320	640	320
A/Dakar/07/2012		2012-10-06	MDCK1/SIAT1	80	160	160	320	160	160	640	320	640	1280	640
A/Dakar/08/2012	3C	2012-10-06	MDCK1/SIAT1	80	80	160	320	160	160	640	640	640	1280	640
A/Dakar/14/2012	3C	2012-10-08	MDCK1/SIAT1	40	40	160	320	160	160	640	320	640	640	320
A/Dakar/15/2012	3C	2012-10-11	MDCK1/SIAT1	40	80	160	320	160	160	640	320	640	640	320
A/Cairo/5/2012	3B	2012-11-18	C1/SIAT1	160	160	320	320	320	320	640	640	640	1280	320
A/Cairo/47/2013	3B	2012-11-29	C1/SIAT1	<	40	80	80	160	80	160	160	160	320	80
A/Cairo/55/2013		2012-12-06	C1/SIAT1	80	80	320	320	320	160	640	640	640	1280	320
A/Cairo/56/2013		2012-12-06	C1/SIAT1	640	160	320	640	320	320	640	640	1280	2560	640
A/Cairo/59/2012	3B	2012-12-06	C1/SIAT1	640	160	320	320	320	320	640	640	1280	2560	640
A/Cairo/63/2012		2012-12-09	C1/SIAT1	320	160	320	320	320	320	640	640	1280	1280	640
A/Cairo/66/2012		2012-12-09	C1/SIAT1	160	160	320	320	640	320	640	640	640	1280	320
A/Cairo/81/2013	3B	2012-12-11	C1/SIAT1	640	160	320	640	640	640	1280	640	1280	2560	640
A/Cairo/135/2012		2012-12-27	C1/SIAT1	160	160	320	640	320	320	640	320	640	1280	640
A/Cairo/136/2012	3B	2012-12-27	C1/SIAT1	1280	160	320	640	320	640	1280	1280	1280	2560	640
A/Cairo/138/2012		2012-12-27	C1/SIAT1	320	160	320	320	320	320	640	640	640	1280	320
A/Cairo/140/2012	3B	2012-12-27	C1/SIAT1	320	160	320	640	320	320	1280	640	640	2560	640
A/Cairo/144/2012		2012-12-27	C1/SIAT1	160	160	320	640	640	320	1280	1280	1280	2560	640
A/Sachsen/2/2013	3C	2013-01-07	C1/SIAT2	160	80	160	320	160	160	640	640	640	1280	640

Sequences in phylogenetic trees

Vaccine

Vaccine

1. < = <40

Table 17. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-01-29

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret sera										
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12
				F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12
						group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	1280	2560	1280	2560	2560	2560	2560	1280	2560	5120	5120
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT3	40	<	80	80	80	80	160	160	160	320	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	80	320	160	80	320	320	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	640	640	1280	1280	640	1280	1280	1280	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	160	640	320	160	640	1280	640	320	160	2560	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT3	80	40	160	160	160	160	640	320	320	1280	320
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	80	80	160	160	160	160	320	320	640	1280	320
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	80	160	160	160	160	320	160	640	640	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	640	640	1280	1280	640	1280	640	1280	5120	1280
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	320	640	1280	640	1280	640	1280	5120	5120
TEST VIRUSES														
A/Ukraine/5491/2012		unknown	Cx/SIAT1	160	160	320	320	320	320	640	640	640	1280	320
A/Ukraine/5462/2012		2012-04-11	Cx/SIAT1	80	160	320	320	320	160	640	640	640	1280	320
A/Texas/50/2012	3C	2012-04-15	MDCK1/C2/SIAT1	160	160	160	320	160	160	640	320	640	1280	640
A/Hawaii/22/2012	3C	2012-07-09	C3/SIAT1	80	80	80	320	160	160	640	320	320	640	640
A/Dakar/25/12		2012-09-24	SIAT2	80	160	320	640	320	320	640	640	640	1280	640
A/Dakar/13/12		2012-10-08	MDCK1/SIAT2	80	80	160	320	160	160	640	320	640	1280	320
A/Dakar/24/12		2012-10-10	SIAT2	<	80	40	160	80	80	160	160	160	320	160
A/Paris/1654/2012		2012-10-26	MDCK2/SIAT1	80	160	160	320	160	160	640	320	320	640	320
A/Sapporo/125/2012		2012-11-01	MDCK2/SIAT1	80	160	160	320	160	160	640	640	640	640	320
A/Paris/1912/2012	5	2012-12-07	MDCK2/SIAT1	40	40	80	160	80	80	320	160	160	320	160
A/Ukraine/551/2012	3B	2012-12-11	SIAT3/SIAT1	320	160	160	320	160	320	640	320	640	1280	640
A/Paris/1966/2012		2012-12-12	MDCK1/SIAT1	40	80	80	320	80	160	320	320	320	640	320
A/Paris/1988/2012		2012-12-18	MDCK1/SIAT1	80	80	160	320	160	160	640	640	320	1280	320
A/Nord Pas de Calais/2000/2012	3C	2012-12-19	MDCK1/SIAT1	40	80	40	320	160	160	320	320	640	640	320
A/Haute Normandie/2022/2012	3C	2012-12-19	MDCK1/SIAT1	<	80	80	160	80	160	320	160	320	640	160
A/Athens GR/14/2013	3C	2013-01-08	SIAT2	<	80	40	160	80	80	320	160	320	640	160

Sequences in phylogenetic trees

Vaccine

Vaccine

1. < = <40

Table 18. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-02-01

Haemagglutination inhibition titre ¹														
Viruses	Collection Date	Passage History	Post infection ferret sera											
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12	
			F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12	
Genetic group	Genetic group				group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C	
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	1280	80	320	160	320	320	320	640	320	160	
A/Victoria/208/2009		2009-06-02	E3/E2	1280	2560	1280	1280	2560	1280	2560	5120	5120		
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT3	<	<	80	80	80	80	160	160	40		
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	160	160	640	320	160	640	640	320		
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	1280	1280	2560	1280		
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	320	160	640	2560	640	320	2560	1280	
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT3	160	160	320	320	320	320	640	640	1280	640	
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	160	160	320	320	320	320	640	640	1280	640	
A/Athens/112/2012	3B	2012-02-01	SIAT7	160	160	320	320	320	320	640	640	1280	640	
A/Texas/50/2012	3C	2012-04-15	E5/E1	640	1280	640	1280	1280	640	1280	640	5120	2560	
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	1280	320	640	1280	640	1280	2560	5120		
TEST VIRUSES														
A/Maryland/02/2012 (IVR-166)	3B	2012-01-18	E5/D6/E1	80	640	160	640	320	640	640	320	2560	1280	
A/Ukraine/5515/2012		2012-03-04	C1/SIAT1	40	80	160	160	160	160	320	320	1280	320	
A/Ukraine/5504/2012		2012-04-02	C1/SIAT1	160	160	320	320	640	320	1280	640	2560	320	
A/Ukraine/5513/2012		2012-04-03	C1/SIAT1	320	320	640	640	640	640	1280	1280	2560	640	
A/Ukraine/5512/2012		2012-04-04	C1/SIAT1	160	160	320	320	640	320	640	640	1280	320	
A/Ukraine/5503/2012		2012-04-09	C1/SIAT1	80	160	320	320	320	320	640	640	1280	320	
A/Ukraine/5506/2012		2012-04-09	C1/SIAT1	80	80	160	160	320	160	640	320	640	320	
A/Ukraine/5502/2012		2012-04-10	C1/SIAT1	40	80	80	160	160	160	320	320	640	320	
A/Ukraine/5507/2012		2012-04-10	C1/SIAT1	160	160	320	640	640	320	640	640	2560	320	
A/Ukraine/5508/2012		2012-04-10	C1/SIAT1	40	80	160	160	160	160	320	320	320	320	
A/Ukraine/5501/2012		2012-04-12	C1/SIAT1	160	160	320	640	640	160	640	640	1280	320	
A/Ukraine/5509/2012		2012-04-13	C1/SIAT1	40	80	160	160	160	160	320	320	1280	320	
A/Ukraine/5510/2012		2012-04-16	C1/SIAT1	80	160	160	320	320	320	320	640	1280	320	
A/Ukraine/5496/2012		2012-04-19	C1/SIAT2	80	160	160	320	320	160	320	320	640	160	
A/Ukraine/5498/2012		2012-04-19	C1/SIAT2	80	80	160	320	320	320	320	640	640	160	
A/Singapore/22/2012	3C	2012-06-06	E4/E1	160	640	160	640	640	1280	640	640	1280	2560	
A/Ukraine/5511/2012		2012-07-22	C1/SIAT1	160	160	160	640	320	320	1280	640	640	320	
A/Sydney/195/2012	3A	2012-08-06	E4/E1	640	2560	640	5120	5120	2560	2560	1280	5120	5120	
A/Yamaguchi/30/2012	3C	2012-10-10	MDCK3/SIAT2	<	40	40	80	40	80	160	160	320	160	
A/Norway/2102/2012		2012-10-15	SIAT1/SIAT1	80	160	160	320	160	160	640	640	640	320	
A/Norway/2103/2012		2012-10-22	SIAT1/SIAT1	80	160	160	320	160	160	640	640	640	320	
A/Paris/1651/2012	3C	2012-10-29	MDCK2/SIAT2	80	160	80	320	160	160	640	640	1280	320	
A/Acores/SU43/2012	5	2012-11-06	SIAT2	160	160	320	320	320	320	640	640	1280	320	
A/South Carolina/16/2012	5	2012-11-07	MDCK1/C1/SIAT1	80	160	160	320	320	320	640	640	1280	320	
A/Delaware/15/2012	3C	2012-11-12	C2/SIAT1	40	80	80	320	160	160	640	320	640	320	
A/Lyon/2496/2012	3C	2012-11-19	MDCK2/SIAT3	80	160	160	160	160	160	320	640	640	160	
A/Norway/2423/2012		2012-11-26	MDCK1/SIAT1	80	160	160	320	160	160	640	320	640	640	
A/Switzerland/7587197/2012		2012-12-14	SIAT2	<	40	40	80	40	80	320	160	320	40	
A/Ireland/00462/2012		2012-12-20	MDCK1/SIAT1	80	80	160	320	320	320	640	320	640	320	
A/Belgium/S0304/2012		2012-12-21	SIAT2	640	320	320	640	640	320	1280	640	1280	640	
A/Belgium/G1010/2012		2012-12-24	SIAT2	40	160	160	320	320	160	640	320	640	320	
A/Belgium/S0325/2012		2012-12-26	SIAT2	40	80	80	160	160	160	320	320	640	160	
A/Lisboa/SU15/2012	3C	2013-01-04	SIAT2	40	80	80	320	160	80	640	320	640	320	
A/Latvia/1-32593/2013		2013-01-06	MDCK1/SIAT1	40	160	160	320	160	320	640	320	640	320	
A/Madeira/MS3/2012		2013-01-07	SIAT2	40	80	80	320	160	80	640	320	320	320	
A/Belgium/G0044 /2013		2013-01-07	SIAT2	40	80	160	320	160	160	640	320	640	320	
A/Latvia/1-37023/2013		2013-01-13	MDCK1/SIAT1	40	160	160	320	160	160	640	320	1280	640	

Sequences in phylogenetic trees

Vaccine

Vaccine

1. < = <40

Table 19. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-02-05

Haemagglutination inhibition titre ¹														
Viruses	Collection Date	Passage History	Post infection ferret sera											
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12	
			F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12	
Genetic group			group 5			group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C	
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	80	160	160	320	160	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	1280	2560	2560	2560	5120	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	160	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	80	320	160	80	640	320	640	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	1280	1280	2560	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	160	80	640	2560	1280	320	160	2560	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT6	80	80	160	320	320	320	640	320	640	2560	320
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	80	80	160	320	320	160	640	640	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT8	40	80	160	160	160	160	320	320	640	640	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	1280	320	1280	1280	1280	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	320	640	1280	640	1280	640	1280	5120	2560
TEST VIRUSES														
A/Hawaii/22/2012 (NIB-80A)	3C	2012-07-09	E7	160	320	160	320	320	320	640	320	320	1280	1280
A/Hawaii/22/2012 (NIB-80B)	3C	2012-07-09	E7	80	320	160	320	320	320	640	320	640	1280	1280
A/Hawaii/22/2012 (NIB-80C)	3C	2012-07-09	E7	80	320	160	320	320	320	640	320	320	1280	1280
A/Hawaii/22/2012 (NIB-80F)	3C	2012-07-09	E7	160	320	160	320	320	320	640	320	640	1280	1280
A/Texas/50/2012 (NIB-81) ²	3C	2012-04-15	E6	160	320	160	320	320	320	640	320	640	1280	640
A/Galicia/15/2012		2012-02-07	SIAT2/SIAT1	80	80	160	320	320	160	640	320	640	1280	320
A/Galicia/RR9908/2012		2012-03-09	SIAT2/SIAT1	80	80	320	160	320	320	640	320	320	1280	320
A/Ukraine/5500/2012		2012-04-11	1/SIAT2	<	40	40	80	80	80	160	80	160	160	80
A/Ghana/DILI-0405/2012		2012-08-02	SIAT1/SIAT3	40	40	80	160	160	80	320	160	320	320	160
A/Israel/10/2012		2012-10-30	P1/SIAT1	<	<	<	80	80	80	160	40	160	160	40
A/Madrid/323/2012		2012-11-30	SIAT1/SIAT1	40	80	80	320	320	160	640	320	640	640	320
A/Turkey/193/2012		2012-12-03	C0/SIAT1	40	80	80	320	160	160	1280	320	640	1280	320
A/Lisboa/SU91/2012		2012-12-11	SIAT3	40	40	160	160	160	80	320	160	320	320	160
A/Switzerland/7586983/2012		2012-12-14	SIAT3	<	40	40	160	80	80	320	160	320	320	160
A/Turkey/426/2012		2012-12-27	C0/SIAT1	80	80	160	320	320	320	1280	320	640	1280	640
A/Roma/02/2013		2012-12-31	Cx/SIAT1	<	<	<	80	40	40	320	80	160	320	80
A/Roma/03/2013		2012-12-31	Cx/SIAT1	<	40	40	160	80	80	320	160	320	640	80
A/Roma/04/2013		2012-12-31	Cx/SIAT1	<	40	<	160	160	80	320	320	320	640	160
A/Israel/15/2013		2013-01-01	P2/SIAT1	40	40	80	160	80	160	640	320	320	640	320
A/Israel/17/2013		2013-01-01	P2/SIAT1	<	<	40	80	80	40	320	160	160	320	160
A/Ireland/00367/2013		2013-01-02	MDCK2/SIAT2	160	160	160	320	320	320	1280	640	640	1280	640
A/Denmark/60/2013		2013-01-07	SIAT2/SIAT1	<	40	40	160	160	160	640	320	640	640	320
A/Israel/20/2013		2013-01-08	P1/SIAT1	<	<	40	80	40	40	160	80	160	320	80
A/Denmark/20/2013		2013-01-10	SIAT2/SIAT1	<	80	80	320	160	160	640	320	320	640	160
A/Israel/24/2013		2013-01-14	P1/SIAT1	40	80	80	160	80	160	640	320	320	640	320
A/Finland/311/2013		2013-01-23	SIAT1/SIAT1	40	80	80	320	160	160	640	320	640	640	320
A/Finland/296/2012	3C	2013-01-24	IAT1	40	80	80	160	160	80	640	320	320	320	320
A/Finland/298/2012	3C	2013-01-24	IAT1	40	80	40	160	80	80	640	320	640	640	160

Sequences in phylogenetic trees

Vaccine

Vaccine

1. < = <40; 2. H3N1 reassortant

Results on one-way tests for potential vaccine candidates

Table 20. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-02-08

			Haemagglutination inhibition titre ¹										
Viruses	Collection Date	Passage History	Post infection ferret sera										
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12
			F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12
Genetic group				group 5	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C	
REFERENCE VIRUSES													
A/Perth/16/2009	2009-07-04	E3/E2	640	40	80	80	160	160	320	160	320	320	80
A/Victoria/208/2009	2009-06-02	E3/E2	640	2560	1280	2560	2560	2560	5120	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	160	160	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	80	320	160	640	320	640	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	1280	1280	2560	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	160	80	640	1280	320	160	2560	1280
A/Berlin/93/2011	3C	2011-12-07	NVD3/SIAT6	80	80	160	320	320	320	640	320	640	320
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT2	80	80	160	320	320	640	640	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT8	40	80	160	160	160	320	320	640	640	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	320	1280	320	1280	1280	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C	2012-07-09	E4/E1	320	640	320	640	1280	1280	640	1280	5120	2560
TEST VIRUSES													
A/Galicia/RR9907/2012	2012-03-09	SIAT1/SIAT2	160	160	320	320	320	320	1280	320	640	1280	640
A/Galicia/RR9911/2012	2012-04-18	SIAT2/SIAT2	640	640	640	1280	1280	640	2560	1280	1280	2560	1280
A/Arges/126697/2012	2012-10-29	MDCK3/SIAT1	<	80	40	160	160	80	640	320	320	640	320
A/Algiers/107/2012	2012-11-07	C1/SIAT1	40	80	80	320	160	160	640	320	320	640	640
A/Algiers/120/2012	2012-11-17	C1/SIAT1	40	80	80	320	160	160	640	320	640	640	640
A/Trieste/65/2012	2012-11-26	Cx/SIAT1	80	160	160	320	320	320	640	640	640	1280	640
A/Latvia/1-30021/2013	2013-01-01	MDCK1/SIAT3	80	160	160	160	160	160	640	320	640	640	640
A/Trieste/01/2013	2013-01-04	Cx/SIAT1	80	160	160	640	320	320	1280	320	640	1280	640
A/Israel/19/2013	2013-01-15	Cx/SIAT1	80	160	160	320	320	160	1280	320	1280	1280	640

1. < = <40

Vaccine

Vaccine

Table 21. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation² (MDCK-SIAT) 2013-01-23

					Neutralisation titre ¹				
Viruses	Genetic group	Collection Date	Passage History	Plaque Type	Post infection ferret sera				
					A/Stockholm 18/11 T/C F28/11 group 3A	A/Athens 112/12 T/C F16/12 group 3B	A/Vic 361/11 Egg F35/12 group 3C	A/Vic 361/11 T/C F14/12 group 3C	A/Texas 50/12 Egg F36/12 group 3C
REFERENCE VIRUSES									
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT4	mix single/very small	160	160	40	40	ND
A/Athens/112/2012	3B	2012-02-01	SIAT4	single cell	160	320	40	160	ND
A/Victoria/361/2011	3C	2011-10-24	E3/E2	big	80	160	5120	320	ND
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT5	single cell	40	160	40	320	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	big	320	640	320	640	2560
TEST VIRUSES									
A/Stockholm/30/2012	5	2012-07-12	MDCK1/SIAT1	mix big and small	40	160	40	160	160
A/Stockholm/29/2012	3C	2012-07-27	MDCK1/SIAT1	mix big and small	40	80	40	160	160
A/Denmark/68/2012	3C	2012-09-25	Cx/SIAT1	mix big and small	80	320	80	320	320
A/Denmark/67/2012	3C	2012-09-27	MDCK3/SIAT1	mix big and small	80	320	80	320	640
A/Stockholm/31/2012	5	2012-10-17	MDCK2/SIAT1	mix big and small	160	640	160	640	1280
A/Denmark/71/2012	3C	2012-11-08	MDCK4/SIAT1	mix big and small	80	160	80	320	320
A/Stockholm/39/2012	3A	2012-11-08	MDCK1/SIAT1	mix big and small	160	320	160	320	640
A/Stockholm/40/2012	5	2012-11-14	MDCK1/SIAT1	mix big and small	80	320	80	320	320
A/Stockholm/37/2012	5	2012-11-15	MDCK0/SIAT1	mix big and small	160	320	160	640	640
A/Stockholm/36/2012	3C	2012-11-16	MDCK1/SIAT1	mix big and small	80	160	160	320	640
Sequences in phylogenetic trees									
1. <=<40 2. 80% reduction									
Vaccine									

Table 22. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation² (MDCK-SIAT) 2013-02-06

Viruses	Genetic group	Collection Date	Passage History	plaques	Neutralisation titre ¹				
					Post infection ferret sera				
					A/Stockholm 18/11 T/C F28/11 group 3A	A/Athens 112/12 T/C F16/12 group 3B	A/Vic 361/11 Egg F35/12 group 3C	A/Vic 361/11 T/C F14/12 group 3C	A/Texas 50/12 E F36/12 group 3C
REFERENCE VIRUSES									
A/Stockholm/18/2011	3A	2011-03-28	MDCK2/SIAT4	mix single/very small	160	160	40	40	ND
A/Athens/112/2012	3B	2012-02-01	SIAT4	single cell	160	320	40	160	ND
A/Victoria/361/2011	3C	2011-10-24	E3/E2	big	80	160	5120	320	ND
A/Victoria/361/2011	3C	2011-10-24	MDCK2/SIAT5	single cell	40	160	40	320	320
A/Texas/50/2012	3C	2012-04-15	E5/E1	big	320	640	320	640	2560
TEST VIRUSES									
A/Stockholm/12-16700/2012	3C		MDCK2/SIAT1	mix big and small	40	80	40	160	160
A/England/565/2012	3B	2012-05-31	SIAT2/SIAT6	single cell	320	640	320	1280	1280
A/Dakar/04/2012	3C (128, 142)	2012-10-03	MDCK1/SIAT1	mix big and small	80	320	80	320	320
A/Dakar/11/2012		2012-10-04	MDCK2/SIAT1	mix big and small	40	160	40	160	160
A/Dakar/09/2012		2012-10-05	MDCK1/SIAT1	mix big and small	160	320	160	640	640
A/Dakar/08/2012	3C (128, 142)	2012-10-06	MDCK1/SIAT1	mix big and small	160	320	80	640	320
A/Dakar/14/2012	3C (128, 142)	2012-10-08	MDCK1/SIAT1	mix big and small	40	80	40	160	160
A/England/587/2012	3C (128, 142)	2012-10-31	SIAT1/SIAT1	single cell	320	640	160	640	1280
A/England/586/2012	3C (128, 142)	2012-11-02	SIAT1/SIAT1	single cell	320	640	320	640	1280
A/Niedersachsen/29/2012	3C (128, 142)	2012-11-05	C1/SIAT1	single cell	320	640	160	640	640
A/Stockholm/38/2012	5	2012-11-05	MDCK1/SIAT1	mix big and small	80	160	80	320	320
A/Austria/705367/2012	5	2012-11-06	SIAT1/SIAT1	big	160	640	160	640	640
A/England/593/2012	3C (128, 142)	2012-11-12	SIAT1/SIAT1	single cell	320	640	160	640	640
A/Thuringen/33/12	3C	2012-11-12	C2/SIAT1	single cell	320	640	320	1280	1280
A/Cairo/5/2012	3C	2012-11-18	C1/SIAT1	big	160	640	160	640	640
A/Cairo/47/2013	3C	2012-11-29	C1/SIAT1	mix big and small	80	160	80	320	320
A/Austria/709778/2012	3C	2012-12-06	SIAT1/SIAT1	mix big and small	80	320	80	320	640
A/Cairo/55/2013		2012-12-06	C1/SIAT1	big	160	320	160	640	640
A/Cairo/56/2013		2012-12-06	C1/SIAT1	big	160	320	160	640	640
A/Cairo/59/2012	3B	2012-12-06	C1/SIAT1	mix big and small	160	320	160	320	640
A/Cairo/63/2012		2012-12-09	C1/SIAT1	mix big and small	160	320	160	640	640
A/Cairo/66/2012		2012-12-09	C1/SIAT1	big	160	640	160	640	640
A/Norway/2496/2012	3C	2012-12-10	SIAT1	mix big and small	160	320	160	320	640
A/Austria/710638/2012	5	2012-12-11	SIAT1/SIAT1	mix big and small	160	320	160	640	640
A/Cairo/81/2013	3B	2012-12-11	C1/SIAT1	big	80	320	160	320	640
A/Cairo/135/2012		2012-12-27	C1/SIAT1	big	160	160	160	320	640
A/Cairo/136/2012	3B	2012-12-27	C1/SIAT1	big	160	320	160	320	640
A/Cairo/138/2012		2012-12-27	C1/SIAT1	big	160	320	160	320	640
A/Cairo/140/2012	3B	2012-12-27	C1/SIAT1	mix big and small	160	320	160	320	640
A/Cairo/144/2012		2012-12-27	C1/SIAT1	big	160	640	160	640	640

Sequences in phylogenetic trees

1. <=40 2. 80% reduction 3C (128, 142) = 3C subgroup T128A, R142G

Vaccine

Figure 11. Phylogenetic comparison of influenza A(H3N2) HA genes

Vaccine virus

Reference viruses

Collection date

Oct 2012

Nov 2012

Dec 2012

Jan 2013

See HI Tables
for antigenic
characterisation

HA2 numbering

proposed serology
antigens

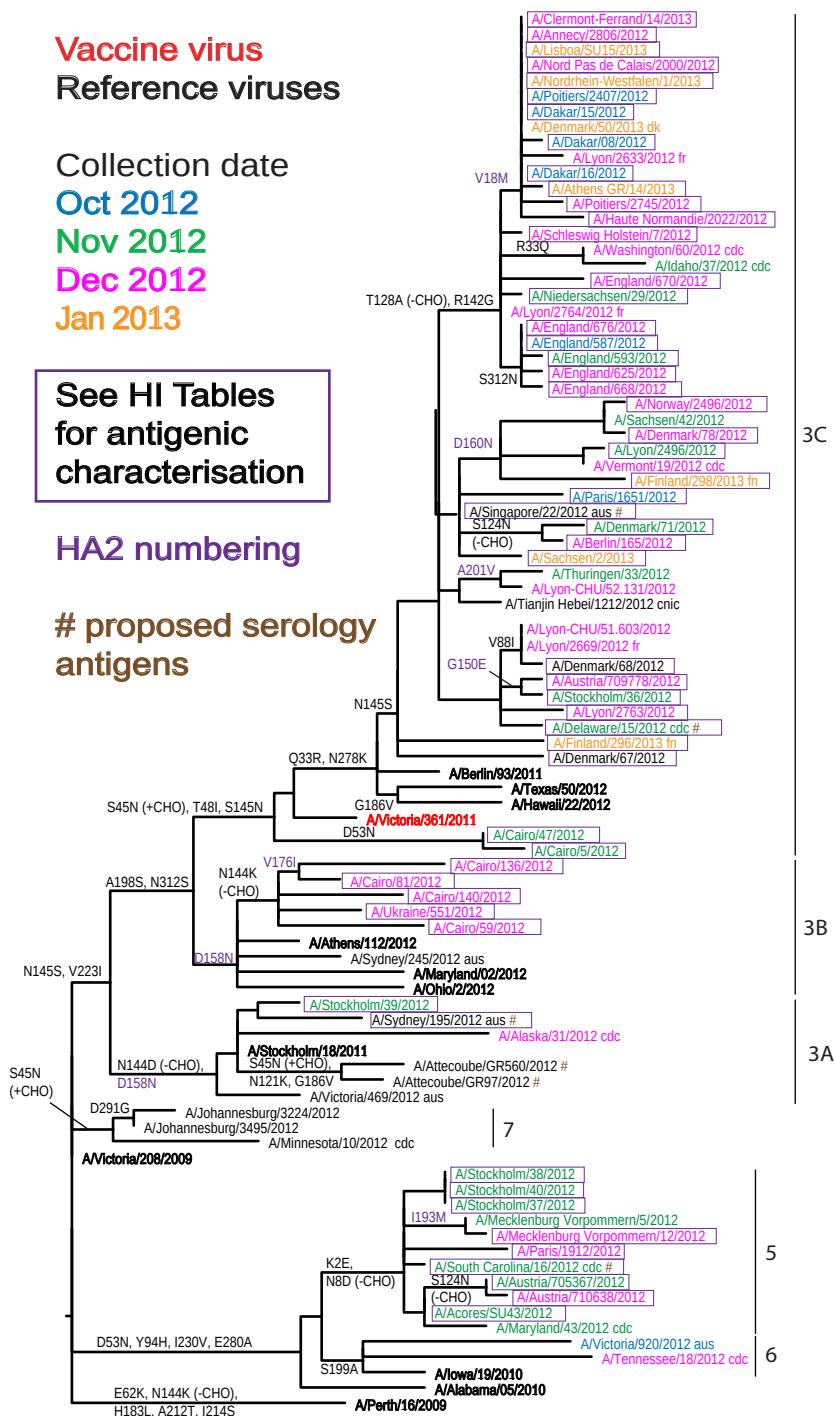


Figure 12. HA protein alignment for A(H3N2) viruses.

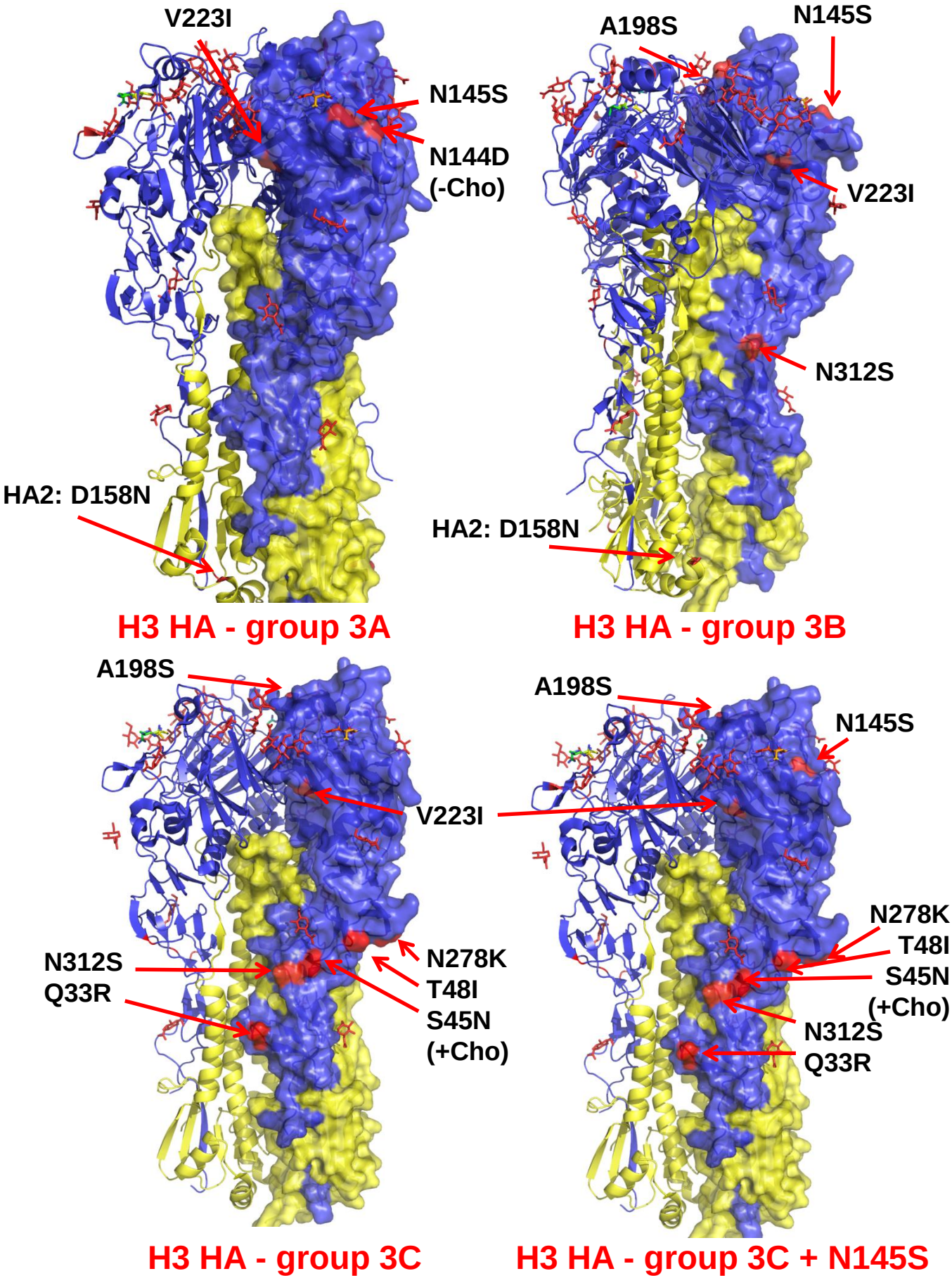
Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

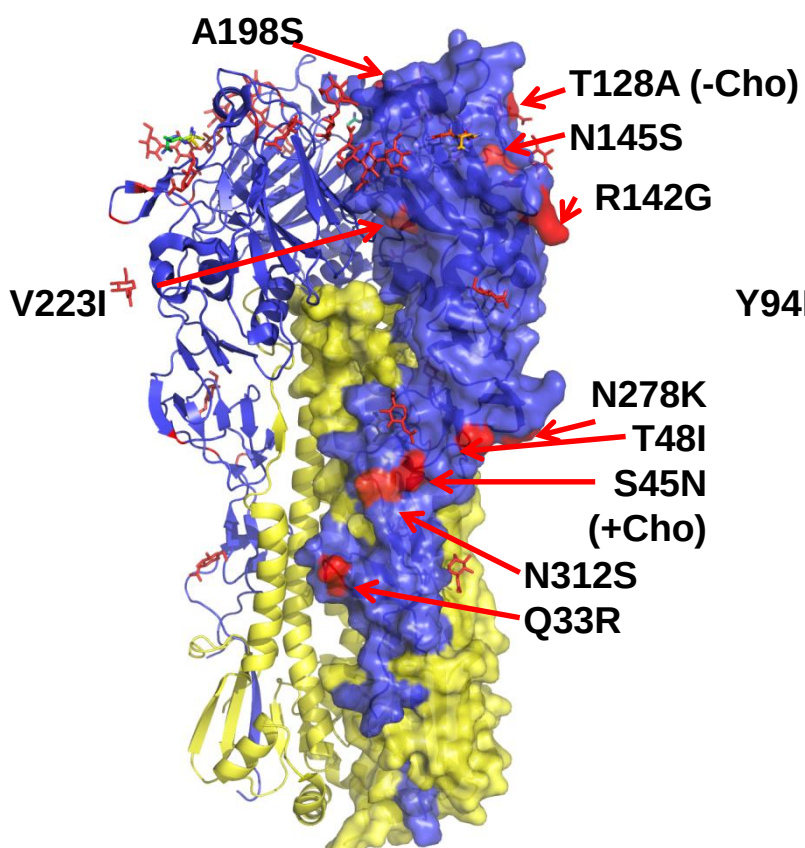
	10	20	30	40	50	60	70	80	90	100
A/Perth/16/2009	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Alabama/05/2010	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Iowa/19/2010	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Tennessee/18/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Austria/710638/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Paris/1912/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Victoria/208/2009	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Johannesburg/3495/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Attacoube/GR560/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Stockholm/18/2011	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Stockholm/39/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Ohio/2/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Maryland/02/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Athens/112/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Ukraine/551/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Cairo/140/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Cairo/47/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Victoria/361/2011	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Hawaii/22/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Texas/50/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Berlin/93/2011	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Finland/296/2013 fn	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Stockholm/36/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Austria/709778/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Sachsen/2/2013	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Denmark/78/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/England/668/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Niedersachsen/29/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Athens GR/14/2013	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
A/Annecy/2806/2012	QKLP	GNDS	STAT	LC	LGH	AV	P	N	G	T
	110	120	130	140	150	160	170	180	190	200
A/Perth/16/2009	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Alabama/05/2010	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Iowa/19/2010	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Tennessee/18/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Austria/710638/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Paris/1912/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Victoria/208/2009	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Johannesburg/3495/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Attacoube/GR560/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Stockholm/18/2011	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Stockholm/39/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Ohio/2/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Maryland/02/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Athens/112/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Ukraine/551/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Cairo/140/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Cairo/47/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Victoria/361/2011	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Hawaii/22/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Texas/50/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Berlin/93/2011	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Finland/296/2013 fn	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Stockholm/36/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Austria/709778/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Sachsen/2/2013	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Denmark/78/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/England/668/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Niedersachsen/29/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Athens GR/14/2013	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
A/Annecy/2806/2012	DVPD	YASL	RSIV	ASSG	TLEF	NNE	SF	NWT	GV	T
	210	220	230	240	250	260	270	280	290	300
A/Perth/16/2009	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Alabama/05/2010	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Iowa/19/2010	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Tennessee/18/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Austria/710638/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Paris/1912/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Victoria/208/2009	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Johannesburg/3495/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Attacoube/GR560/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Stockholm/18/2011	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Stockholm/39/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Ohio/2/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Maryland/02/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Athens/112/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Ukraine/551/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Cairo/140/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Cairo/47/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Victoria/361/2011	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Hawaii/22/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Texas/50/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Berlin/93/2011	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Finland/296/2013 fn	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Stockholm/36/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Austria/709778/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Sachsen/2/2013	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Denmark/78/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/England/668/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Niedersachsen/29/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Athens GR/14/2013	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS
A/Annecy/2806/2012	RITV	TKRS	QQT	VSP	NIG	SR	PR	VR	NI	PS

	310	320	330	340	350	360	370	380	390	400
A/Perth/16/2009	TYGACPRVYKQNTLKLATGMRNVPEKQ	TGIFGAIAGFTIENGWEGMVDGWYGFRH	QNSEGRGQAADLKSTQAAIDQINGKLNRL	IGKTNEKFPHQIEKEFS						
A/Alabama/05/2010										
A/Iowa/19/2010										
A/Tennessee/18/2012										
A/Austria/710638/2012										
A/Paris/1912/2012										
A/Victoria/208/2009										
A/Johannesburg/3495/2012										
A/Attacoube/GR560/2012										
A/Stockholm/18/2011										
A/Stockholm/39/2012										
A/Ohio/2/2012		S								
A/Maryland/02/2012		S								
A/Athens/112/2012		S								
A/Ukraine/551/2012		S								
A/Cairo/140/2012		S								
A/Cairo/47/2012		S								
A/Victoria/361/2011		S								
A/Hawaii/22/2012		S			M					
A/Texas/50/2012		S								
A/Berlin/93/2011		S								
A/Finland/296/2013 fn		S								
A/Stockholm/36/2012		S								
A/Austria/709778/2012		S								
A/Sachsen/2/2013		S								
A/Denmark/78/2012		S								
A/England/668/2012		S								
A/Niedersachsen/29/2012		S								
A/Athens GR/14/2013		S			M					
A/Annecy/2806/2012		S			M					
	410	420	430	440	450	460	470	480	490	500
A/Perth/16/2009	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALE	QHTIDLTDSSEMNKLFETKKQLRENAEDMG	NGCFKIYHKCDNACIGSIRNGTYDHDVYR	DEALNNRF						
A/Alabama/05/2010										
A/Iowa/19/2010										
A/Tennessee/18/2012										
A/Austria/710638/2012		X								
A/Paris/1912/2012										
A/Victoria/208/2009										
A/Johannesburg/3495/2012										
A/Attacoube/GR560/2012										
A/Stockholm/18/2011										
A/Stockholm/39/2012										
A/Ohio/2/2012										
A/Maryland/02/2012							M			
A/Athens/112/2012										
A/Ukraine/551/2012										
A/Cairo/140/2012										
A/Cairo/47/2012										
A/Victoria/361/2011										
A/Hawaii/22/2012										
A/Texas/50/2012										
A/Berlin/93/2011										
A/Finland/296/2013 fn										
A/Stockholm/36/2012							E			
A/Austria/709778/2012							E			
A/Sachsen/2/2013										
A/Denmark/78/2012										
A/England/668/2012										
A/Niedersachsen/29/2012										
A/Athens GR/14/2013										
A/Annecy/2806/2012										
	510	520	530	540	550					
A/Perth/16/2009	QIKGVELKSGYKDWILWISFAISCFLLCVALLG	FIMWACQKGNIRCNICI								
A/Alabama/05/2010										
A/Iowa/19/2010										
A/Tennessee/18/2012			T							
A/Austria/710638/2012										
A/Paris/1912/2012										
A/Victoria/208/2009										
A/Johannesburg/3495/2012										
A/Attacoube/GR560/2012										
A/Stockholm/18/2011										
A/Stockholm/39/2012										
A/Ohio/2/2012										
A/Maryland/02/2012										
A/Athens/112/2012										
A/Ukraine/551/2012										
A/Cairo/140/2012										
A/Cairo/47/2012										
A/Victoria/361/2011										
A/Hawaii/22/2012										
A/Texas/50/2012										
A/Berlin/93/2011				T						
A/Finland/296/2013 fn										
A/Stockholm/36/2012										
A/Austria/709778/2012										
A/Sachsen/2/2013										
A/Denmark/78/2012										
A/England/668/2012										
A/Niedersachsen/29/2012										
A/Athens GR/14/2013										
A/Annecy/2806/2012										

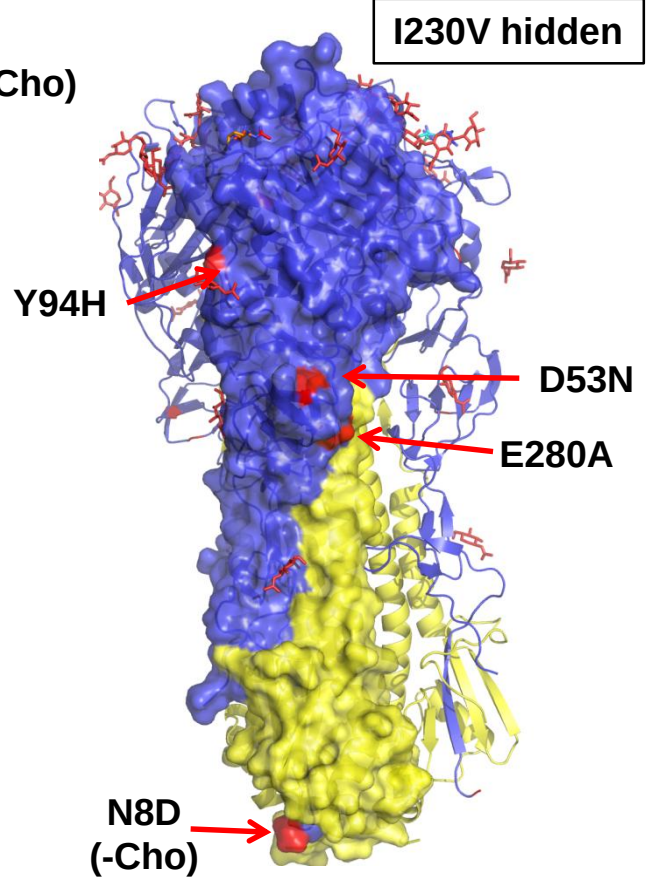
Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 13. H3-HA locations of genetic group defining amino acid substitutions

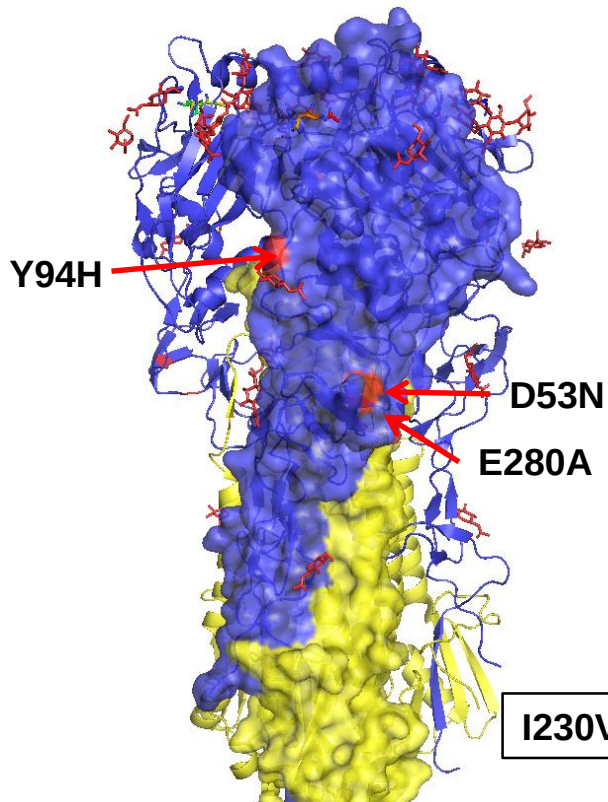




**H3 HA - group 3C + N145S,
T128A, R142G**



H3 HA - group 5



H3 HA - group 6

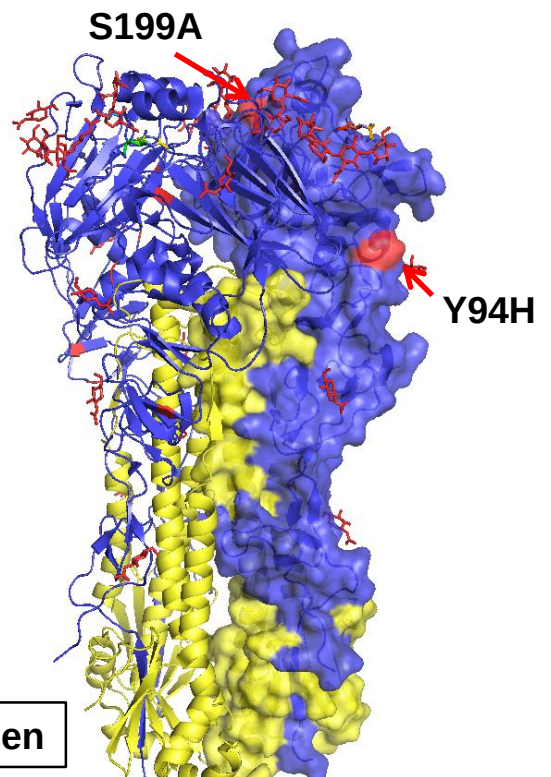


Figure 14. Phylogenetic comparison of influenza A(H3N2) NA genes

Vaccine virus
Reference viruses

Collection date

Oct 2012

Nov 2012

Dec 2012

Jan 2013

See HI Tables
for antigenic
characterisation

proposed serology
antigens

Substitutions in
the 150 loop
(148, 151)

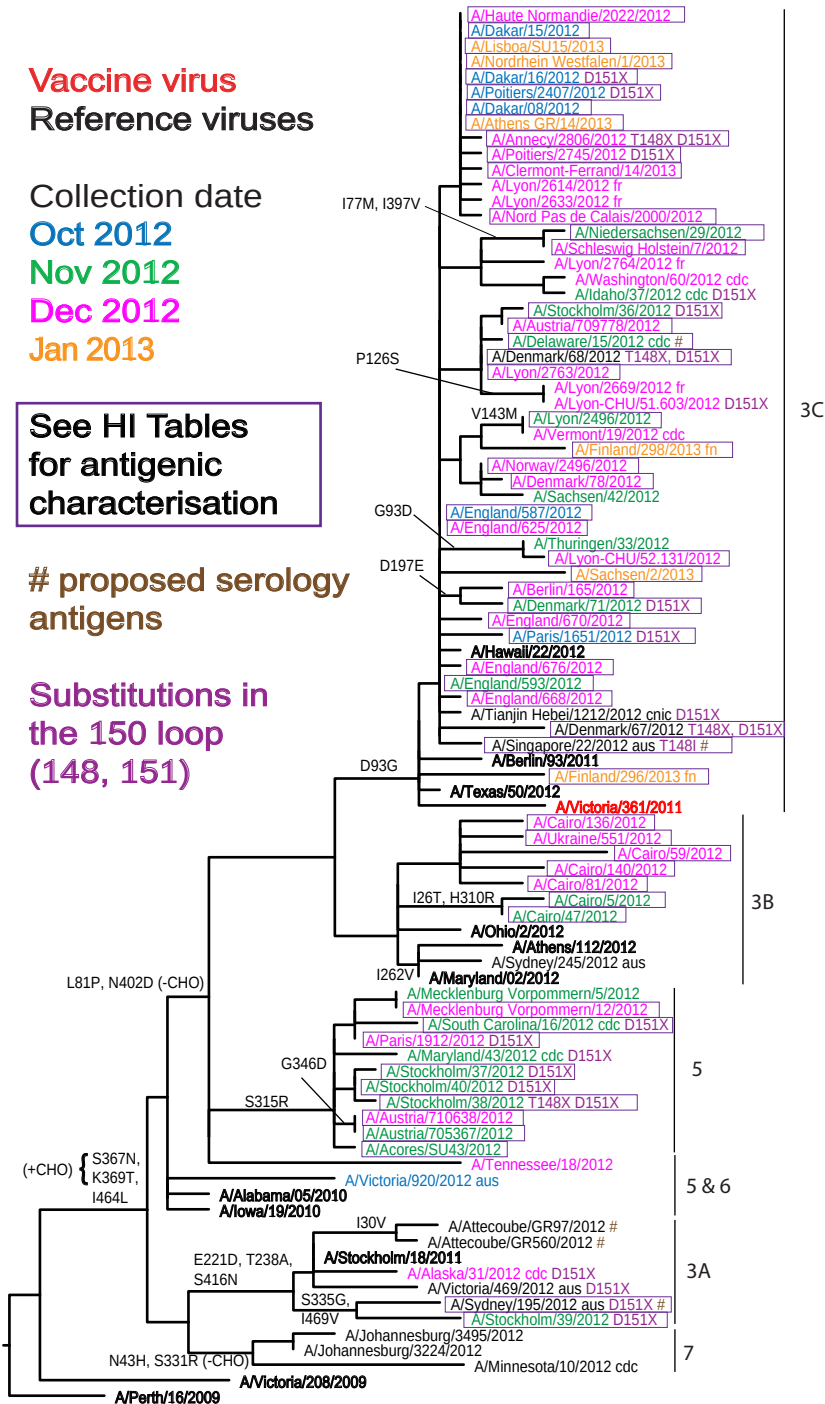
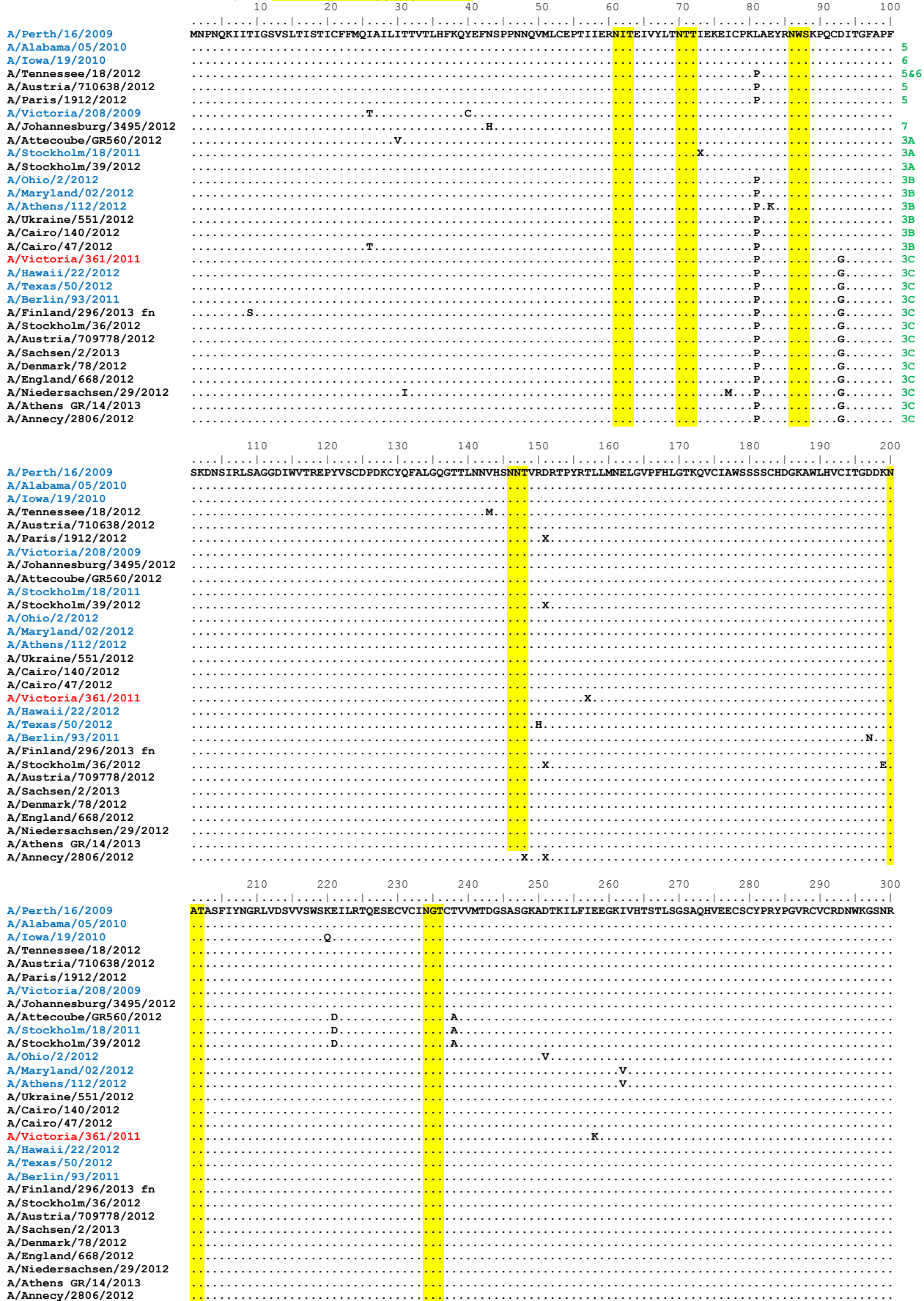


Figure 15. NA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

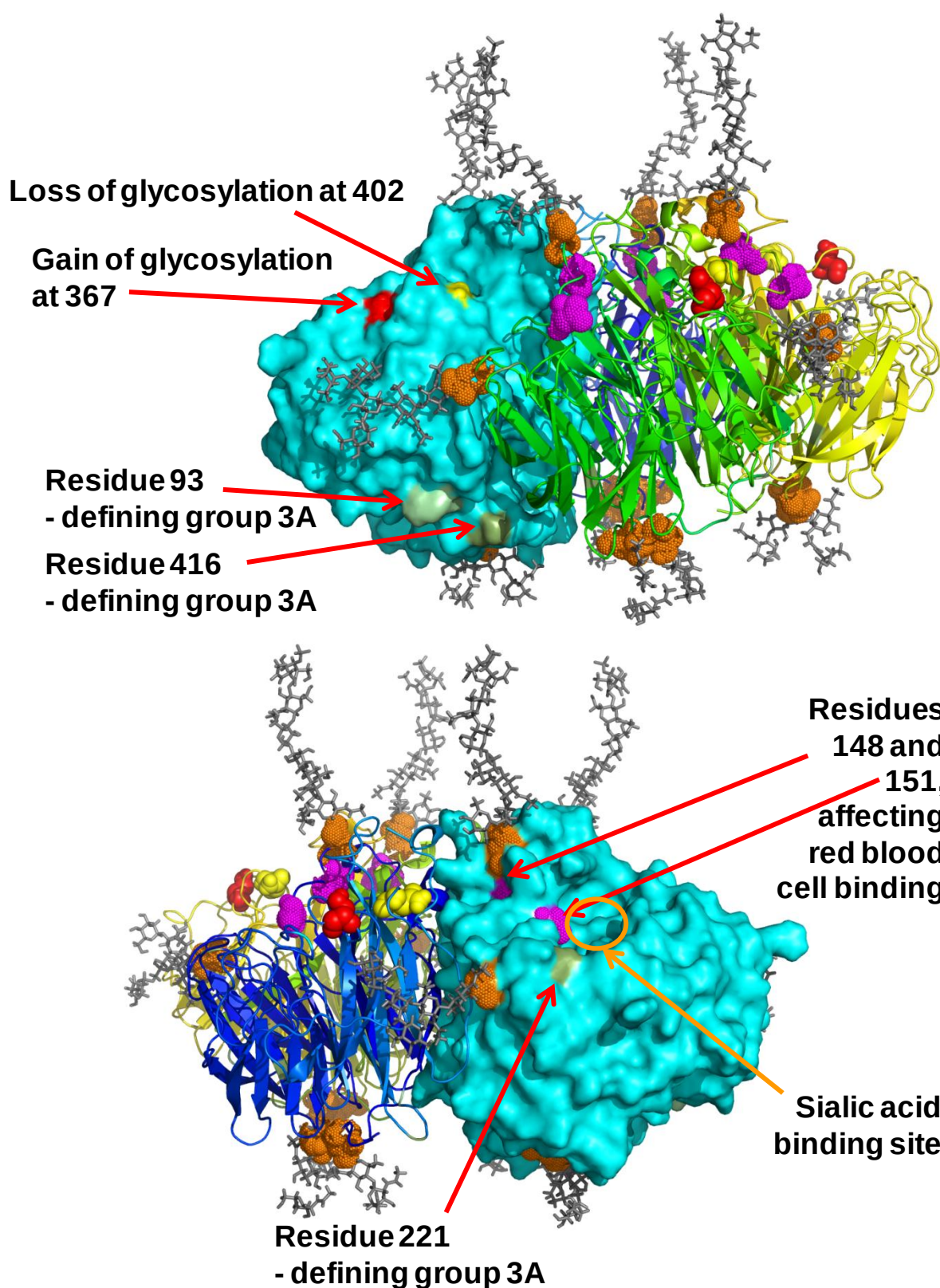


	310	320	330	340	350	360	370	380	390	400
A/Perth/16/2009	P	I	V	D	I	N	K	D	H	S
A/Alabama/05/2010	I	V	S	S	I	V	C	S	G	L
A/Iowa/19/2010	V	G	D	T	P	R	K	N	S	S
A/Tennessee/18/2012	S	S	S	S	H	C	L	D	P	N
A/Austria/710638/2012	N	E	E	G	G	H	G	V	K	G
A/Paris/1912/2012	W	A	F	D	D	G	N	D	V	M
A/Victoria/208/2009	G	R	T	I	S	E	K	S	R	L
A/Johannesburg/3495/2012	G	Y	E	T	F	K	V	I	E	G
A/Attecoube/GR560/2012	S	N	P	K	S	K	L	Q	I	N
A/Stockholm/18/2011	R	Q	V	I	D	R				
A/Stockholm/39/2012										
A/Ohio/2/2012										
A/Maryland/02/2012										
A/Athens/112/2012										
A/Ukraine/551/2012										
A/Cairo/140/2012										
A/Cairo/47/2012										
A/Victoria/361/2011										
A/Hawaii/22/2012										
A/Texas/50/2012										
A/Berlin/93/2011										
A/Finland/296/2013 fn										
A/Stockholm/36/2012										
A/Austria/709778/2012										
A/Sachsen/2/2013										
A/Denmark/78/2012										
A/England/668/2012										
A/Niedersachsen/29/2012										
A/Athens GR/14/2013										
A/Annecy/2806/2012										

	410	420	430	440	450	460
A/Perth/16/2009	G	N	R	S	G	Y
A/Alabama/05/2010	S	G	I	F	S	V
A/Iowa/19/2010	E	G	K	S	C	I
A/Tennessee/18/2012	N	R	C	F	Y	V
A/Austria/710638/2012	L	I	R	G	R	K
A/Paris/1912/2012	E	T	V	L	W	T
A/Victoria/208/2009	S	N	S	I	V	F
A/Johannesburg/3495/2012	C	G	T	S	G	T
A/Attecoube/GR560/2012	Y	G	T	G	S	W
A/Stockholm/18/2011	P	D	G	A	D	I
A/Stockholm/39/2012	N	L	M	P	I	
A/Ohio/2/2012						
A/Maryland/02/2012						
A/Athens/112/2012						
A/Ukraine/551/2012						
A/Cairo/140/2012						
A/Cairo/47/2012						
A/Victoria/361/2011						
A/Hawaii/22/2012						
A/Texas/50/2012						
A/Berlin/93/2011						
A/Finland/296/2013 fn						
A/Stockholm/36/2012						
A/Austria/709778/2012						
A/Sachsen/2/2013						
A/Denmark/78/2012						
A/England/668/2012						
A/Niedersachsen/29/2012						
A/Athens GR/14/2013						
A/Annecy/2806/2012						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 16. N2-NA locations of genetic group defining & N-linked glycosylation site modifying amino acid substitutions



N2 NA, conserved (gold) and variable (loss: yellow, gain: red) glycosylation sites, residues affecting red blood cell binding (magenta) & group-defining substitutions.

Table 23. Summary of influenza B (Victoria-lineage) samples analysed, collected between 2012-09-01 and 2013-01-31

MONTH	B Victoria lineage				
Country	Number received	Number propagated¹	Fold reduction in HI titre		
			≤2 fold²	4 fold²	≥8 fold²
OCTOBER					
France	2	in process			
Norway	1	1	1		
United Kingdom	2	2	1	1	
NOVEMBER					
Algeria	1	in process			
Belgium	1	1	1		
Denmark	2	2	2		
Finland	1	1	1		
Hong Kong	2	in process			
Ireland	1	in process			
Jordan	1	1			1
Norway	2	in process			
Spain	1	1	1		
United Kingdom	1	1		1	
DECEMBER					
France	6	6	5	1	
Germany	1	1	1		
Jordan	4	4	4		
Portugal	1	1	1		
Spain	6	in process	4		
United Kingdom	1	1	1		
JANUARY					
Finland	1	1	1		
Germany	1	1		1	
Hong Kong	1	in process			
Ireland	1	1	1		
Italy	1	1	1		
	42	27	26 84%	4 13%	1 3%

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 24. Antigenic analyses of influenza B viruses (Victoria lineage) 2012-10-09

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹							
				Post infection ferret sera							
				B/Bris ²	B/Mal	B/Eng	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
				60/08 Sh 523	2506/04 F28/05	393/08 F05/11	60/08 F22/12	1762/08 F17/11	514/09 F13/10	3886/10 F19/11	636714/11 F33/11
				1A		1A	1A	1A	1B	1B	1A
REFERENCE VIRUSES											
B/Malaysia/2506/2004		2004-12-06	E3/E5	1280	640	40	160	<	<	10	160
B/England/393/2008	1A	2008-08-29	E1/E1	2560	80	320	640	40	40	80	640
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	320	640	40	40	80	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK1	5120	10	20	80	80	80	0.8	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK3	5120	<	20	80	80	80	320	20
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK1	2560	<	20	40	80	80	160	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	640	40	40	80	640
TEST VIRUSES											
B/Buenos Aires/R146/2012	1A	unknown	MDCK1/MDCK1	5120	10	40	80	160	160	160	40
B/Santa Fe/335651/2012	1A	unknown	MDCK2 /MDCK1	2560	<	40	80	80	80	80	40
B/Buenos Aires/R335/2012	1A	unknown	MDCK1/MDCK1	2560	10	40	80	80	80	80	40
B/Tucumán/A423/2012	1A	unknown	Cx/MDCK2	2560	<	40	80	80	80	80	40
B/Mauritius/571/2012	1A	2012-06-29	MDCK3/MDCK1	2560	40	160	320	40	40	40	160
B/Buenos Aires/770/2012	1A	2012-07-06	MDCK2 /MDCK1	5120	<	20	80	160	80	160	40
B/Mauritius/614/2012	1A	2012-07-11	MDCK1/MDCK1	5120	10	40	80	80	80	80	40
B/Mauritius/678/2012	1A	2012-07-21	MDCK1/MDCK1	2560	160	160	320	40	40	40	160
B/Mauritius/718/2012	1A	2012-07-31	MDCK1/MDCK1	2560	<	20	80	80	80	80	40
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK1	5120	<	40	80	160	80	80	40
B/Tucumán/I2010/2012		2012-08-15	MDCK1/MDCK1	2560	10	40	80	80	80	80	40
B/Santa Fe/915530/2012	1A	2012-08-21	MDCK1/MDCK1	2560	<	20	80	80	80	80	20
B/Tucumán/A458/2012	1A	2012-08-28	MDCK1/MDCK1	2560	<	20	80	80	80	80	20

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

T199A>T (-CHO)

T199I (-CHO)

Table 25. Antigenic analyses of influenza B viruses (Victoria lineage) 2012-12-27

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹							
				Post infection ferret sera							
				B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/Eng 393/08 F05/11	B/Bris 60/08 F22/12	B/Paris 1762/08 F17/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11
				1A		1A	1A	1A	1B	1B	1A
REFERENCE VIRUSES											
B/Malaysia/2506/2004		2004-12-06	E3/E6	640	320	10	40	<	<	<	40
B/England/393/2008	1A	2008-08-29	E1/E1	2560	80	320	640	40	40	40	640
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	5120	160	320	640	40	40	80	640
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	20	20	40	40	40	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	<	10	20	80	80	160	20
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK1	5120	<	10	20	80	80	160	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	160	320	640	40	40	40	640
TEST VIRUSES											
B/Norway/2090/2012	1B	2012-10-02	LLC-MK2/ MDCK1/MDCK1	2560	<	10	10	80	80	80	20
B/England/577/2012	1A	2012-10-22	SIAT1/MDCK1	2560	10	10	20	40	40	40	10
B/England/579/2012	1A	2012-10-30	SIAT1/MDCK1	2560	<	10	20	80	80	80	20
B/Denmark/23/2012	1A	2012-11-05	MDCK3/MDCK1	2560	<	10	20	80	40	80	20
B/Denmark/22/2012	1A	2012-11-07	MDCK2/MDCK1	2560	<	10	20	80	40	80	40
B/England/595/2012	1A	2012-11-09	SIAT2/MDCK1	2560	<	10	20	80	40	40	20

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

Table 26. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-01-22

				Haemagglutination inhibition titre ¹							
Viruses	Genetic group	Collection date	Passage History	Post infection ferret sera							
				B/Bris ²	B/Mal	B/Eng	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
				60/08	2506/04	393/08	60/08	1762/08	514/09	3886/10	636714/11
				Sh 523	F28/05	F05/11	F22/12	F17/11	F13/10	F19/11	F33/11
				1A		1A	1A	1A	1B	1B	1A
REFERENCE VIRUSES											
B/Malaysia/2506/2004		2004-12-06	E3/E6	640	640	40	80	<	<	<	80
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	160	320	40	20	40	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	160	320	40	20	40	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	20	40	40	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	<	10	20	80	80	160	10
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	2560	<	20	20	80	160	160	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	160	320	40	20	40	320
TEST VIRUSES											
B/England/626/2012	1A	2012-12-06	SIAT1/MDCK1	2560	20	20	40	80	80	80	20
B/Lyon/2696/2012		2012-12-06	MDCK3/MDCK1	2560	10	20	40	80	80	80	40
B/Lyon/2664/2012		2012-12-11	MDCK2/MDCK1	2560	20	20	40	80	80	80	20
B/Poitiers/2821/2012	1A	2012-12-12	MDCK1/MDCK1	2560	<	20	40	80	80	80	40
B/Lyon/2673/2012		2012-12-13	MDCK2/MDCK1	2560	20	80	80	40	40	40	80
B/Lyon-CHU/50.569/2012		2012-12-13	MDCK2/MDCK1	2560	<	20	40	80	80	80	20
B/Lyon/2760/2012	1A	2012-12-17	MDCK2/MDCK1	1280	<	20	40	80	40	80	20
B/Valladolid/96/2012	1A	2012-12-26	MDCK1/MDCK1	2560	10	10	20	80	40	80	20
B/Bayern/1/2013	1A	2013-01-03	C2/MDCK1	1280	<	10	20	40	40	40	10

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

Table 27. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-01-25

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹							
				Post infection ferret sera							
				B/Bris ²	B/Mal	B/Eng	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
				60/08	2506/04	393/08	60/08	1762/08	514/09	3886/10	636714/11
				Sh 523	F28/05	F05/11	F22/12	F17/11	F13/10	F19/11	F33/11
				1A		1A	1A	1A	1B	1B	1A
REFERENCE VIRUSES											
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	40	80	<	<	<	80
B/England/393/2008	1A	2008-08-29	E1/E1	1280	80	320	320	40	40	40	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	640	40	40	40	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	20	20	40	80	80	80	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	<	10	20	80	80	160	20
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	2560	<	20	40	80	40	80	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	160	320	640	40	40	80	320
TEST VIRUSES											
B/Sachsen/26/2012	1A	2012-12-18	C2/MDCK1	2560	20	40	40	80	160	80	40
B/LisboaPT/8/2012	1A	2012-12-18	SIAT1/MDCK1	2560	80	40	80	80	160	160	80

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

Table 28. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-01-29

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret sera									
				B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/Eng 393/08 F05/11 1A	B/Bris 60/08 F22/12 1A	B/Paris 1762/08 F17/11 1A	B/HK 514/09 F13/10 1B	B/Odessa 3886/10 F19/11 1B	B/Malta 636714/11 F33/11 1A	B/Jhb 3964/12 F01/13 1A	B/For V2367/12 F04/13 1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	1280	80	160	<	<	<	160	320	160
B/England/393/2008	1A	2008-08-29	E1/E2	2560	160	320	640	40	40	80	320	640	640
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	160	320	640	40	40	40	320	320	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	1280	10	20	40	40	80	80	20	80	320
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4/MDCK1	5120	<	20	20	80	80	160	20	80	160
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK2	1280	40	80	80	20	40	40	80	160	320
B/Malta/636714/2011	1A	2011-03-07	E4	2560	160	320	320	40	40	40	320	640	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E1	2560	320	320	640	40	40	80	320	640	640
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	20	80	80	40	80	40	80	160	320
TEST VIRUSES													
B/Jordan/30465/2012		2012-11-27	C1/MDCK1	1280	10	10	20	<	40	10	20	10	80
B/Jordan/20437/2012		2012-12-09	C1/MDCK1	2560	10	20	40	40	40	80	20	80	160
B/Jordan/20440/2012	1A	2012-12-10	C1/MDCK1	5120	80	160	320	40	40	80	320	320	320
B/Jordan/30511/2012		2012-12-16	C1/MDCK1	2560	10	20	40	40	80	80	40	80	160
B/Jordan/30014/2012	1A	2012-12-30	C1/MDCK1	5120	40	40	40	80	80	80	40	160	320

T199N=T (-CHO)

T199N=T (-CHO)

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

Table 29. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-02-05

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹																
				Post infection ferret sera																
				B/Bris ²	B/Mal	B/Eng	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta	B/Jhb	B/For							
				60/08 Sh 523	2506/05 F28/05	393/08 F05/11	60/08 F22/12	1762/08 F17/11	514/09 F13/10	3886/10 F19/11	636714/11 F33/11	3964/12 F01/13	V2367/12 F04/13							
												1A	1A	1A	1A	1B	1B	1A	1A	1A
REFERENCE VIRUSES																				
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	1280	40	80	<	<	<	80	160	80							
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	160	160	20	20	20	160	160	160							
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	160	320	40	40	40	160	320	320							
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	1280	10	10	20	20	40	40	20	40	80							
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	<	10	40	40	80	80	20	40	80							
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	20	40	80	20	40	40	80	80	160							
B/Malta/636714/2011	1A	2011-03-07	E4	1280	80	160	320	40	40	40	320	640	640							
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E1	1280	320	320	640	40	40	40	320	640	640							
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	10	20	80	40	80	40	40	160	160							
TEST VIRUSES																				
B/Belgium/G886/2012		2012-11-12	MDCK2	2560	<	20	40	40	40	40	40	80	160							
B/Burgos/95/2012	1A	2012-11-14	MDCK2	2560	10	10	20	40	40	40	20	40	80							
B/Finland/293/2012	1A	2012-11-20	MDCK2/MDCK1	2560	<	20	20	40	40	40	40	80	80							
B/Andalucia/359/2012		2012-12-03	SIAT1/MDCK1	2560	<	20	20	40	80	40	40	80	80							
B/Andalucia/358/2012		2012-12-04	SIAT1/MDCK1	2560	<	10	20	40	80	40	20	80	80							
B/Valladolid/99/2012	1A	2012-12-20	MDCK2	2560	10	20	40	40	40	40	40	80	80							
B/Ireland/00125/2013		2013-01-03	MDCK1/MDCK1	2560	<	10	20	40	40	40	40	80	80							
B/Roma/01/2013		2013-01-07	Cx/MDCK1	2560	10	20	40	40	80	40	40	80	160							
B/Finland/310/2013		2013-01-23	MDCK1/MDCK1	2560	<	20	40	40	40	40	40	80	80							

Sequences in phylogenetic trees

1. < = <10; 2. hyperimmune sheep serum

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Oct 2012
Nov 2012
Dec 2012
Jan 2013

See HI Tables
for antigenic
characterisation

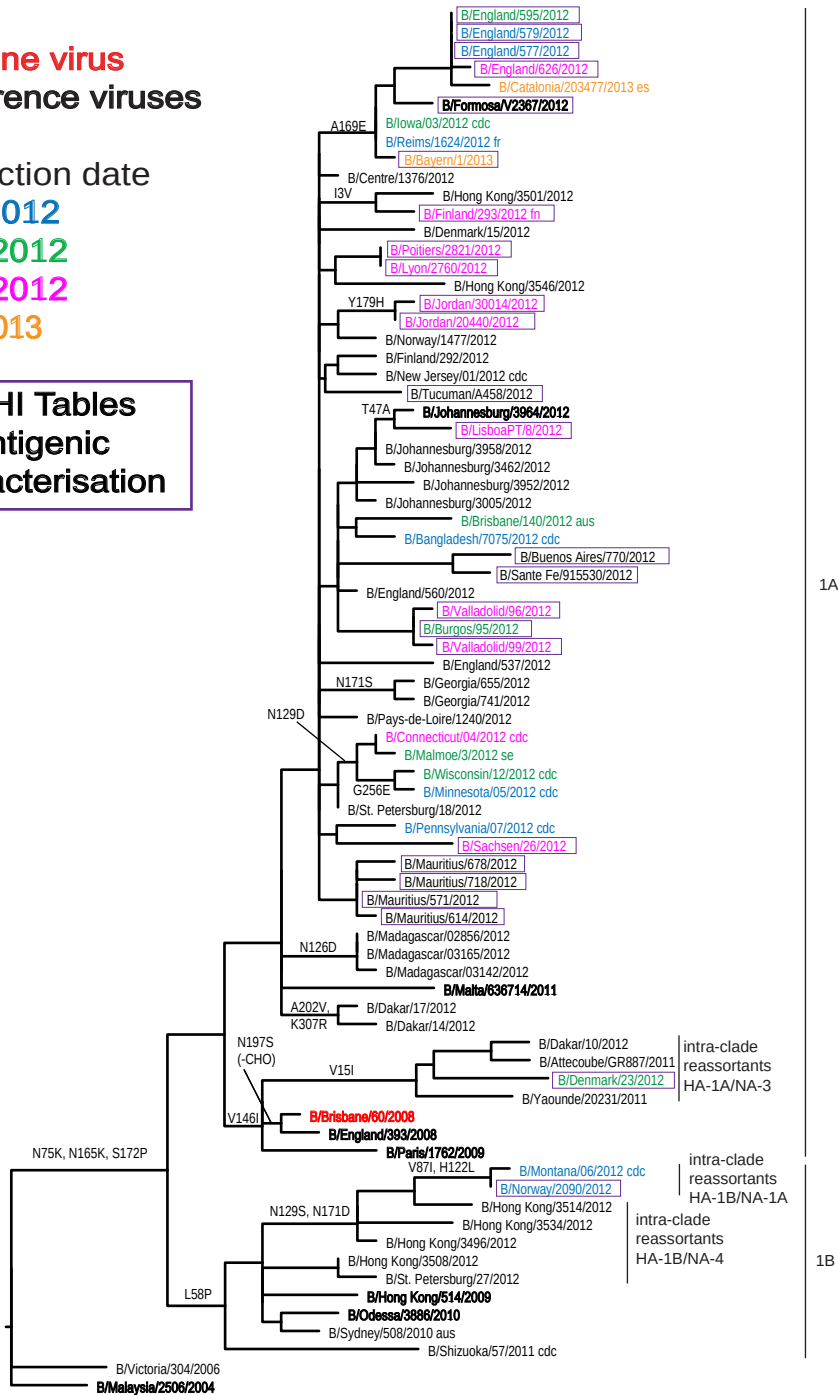


Figure 18. HA protein alignment for B (Victoria-lineage) viruses.

Last used Vaccine virus: **Reference virus:** **Genetic clade:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**

	10	20	30	40	50	60	70	80	90	100
B/Malaysia/2506/2004	DRICTGTSSNSPHVKTATQGEVNVTGVIPLTTPTKSHFANKGTETRGKLCPCCLNCTDLDVALGRPKCTGNIP SARVSILHEVRPVTSGCFFIMHD									4
B/Odessa/3886/2010	.	.	.	P.	.	K.	.	.	.	1E
B/Hong Kong/514/2009	.	.	.	P.	.	K.	.	.	.	1E
B/Hong Kong/3534/2012	.	.	.	P.	.	K.	.	.	.	1E
B/Norway/2090/2012	.	.	.	P.	.	K.	.	I.	.	1E
B/Paris/1762/2009	.	.	.	.	.	K.	.	.	.	1A
B/England/393/2008	.	.	.	.	.	K.	.	.	.	1A
B/Brisbane/60/2008	.	.	.	.	.	K.	.	.	.	1A
B/Denmark/23/2012	.	I.	.	.	.	M.KV.	.	.	.	1A
B/Dakar/10/2012	.	I.	.	.	.	K.	.	.	.	1A
B/Dakar/17/2012	.	.	.	.	.	K.	.	.	.	1A
B/Malta/636714/2011	.	.	.	.	.	K.	.	.	.	1A
B/Madagascar/03165/2012	.	.	.	.	.	K.	.	.	.	1A
B/Mauritius/718/2012	.	.	.	.	.	K.K.	.	.	.	1A
B/Sachsen/26/2012	.	.	.	.	.	K.	.	A.	.	1A
B/Wisconsin/12/2012 cdc	.	.	.	.	.	K.	.	.	.	1A
B/Malmoe/3/2012 se	.	.	.	.	.	K.	.	.	.	1A
B/Georgia/741/2012	.	.	.	.	.	K.	.	.	.	1A
B/Valladolid/96/2012	.	.	.	.	.	K.	.	.	.	1A
B/Sante Fe/915530/2012	.	.	.	.	.	K.	.	.	.	1A
B/LisboaPT/8/2012	.	.	.	A.	.	K.	.	.	.	1A
B/Johannesburg/3964/2012	.	.	.	A.	.	K.	.	.	.	1A
B/Jordan/30014/2012	.	.	.	.	.	K.	.	.	.	1A
B/Poitiers/2821/2012	.	.	.	.	.	K.	.	.	.	1A
B/Finland/293/2012 fn	V.	.	.	.	F.	K.	.	.	.	1A
B/Hong Kong/3501/2012	V.	.	.	.	.	K.	.	.	.	1A
B/Bayern/1/2013	.	.	.	.	.	K.	.	.	.	1A
B/Formosa/V2367/2012	.	.	A.	.	.	K.	.	.	.	1A
B/Catalonia/203477/2013 es	.	.	.	.	.	K.	.	.	.	1A
B/England/626/2012	.	.	.	.	.	K.	.	.	.	1A

	110	120	130	140	150	160	170	180	190	200
B/Malaysia/2506/2004	RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCP	NVT	NGNGFFATMAWAVPKNDN	NK	TATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ					
B/Odessa/3886/2010						K	P			T
B/Hong Kong/514/2009						K	P			T
B/Hong Kong/3534/2012			S			K	DP			T
B/Norway/2090/2012		L	S			K	DP			T
B/Paris/1762/2009				I		K	P		I	T
B/England/393/2008				I		K	P			S.T
B/Brisbane/60/2008				I		K	P			S.T
B/Denmark/23/2012				I		K	P			T
B/Dakar/10/2012				I		K	P			T
B/Dakar/17/2012						K	P			T
B/Malta/636714/2011						K	P			S.T
B/Madagascar/03165/2012			D			K	P			T
B/Mauritius/718/2012						K	P			X
B/Sachsen/26/2012						K	P			T
B/Wisconsin/12/2012 cdc			D			K	P			T
B/Malmö/3/2012 se			D		F	K	P			T
B/Georgia/741/2012						K	SP		I	T
B/Valladolid/96/2012						K	P			T
B/Sante Fe/915530/2012						K	P			T
B/LisboaPT/8/2012			I			K	P			T
B/Johannesburg/3964/2012						K	P			X.T
B/Jordan/30014/2012						K	P		H	T
B/Poitiers/2821/2012						K	P			T
B/Finland/293/2012 fn			V			K	P			T
B/Hong Kong/3501/2012						K	P			X
B/Bayern/1/2013						K	E. P			T
B/Formosa/V2367/2012						K	E. P			T
B/Catalonia/203477/2013 es						K	E. P			T
B/England/626/2012						K	E. P			T

[illegible]

	310	320	330	340	350	360	370	380	390	400	
B/Malaysia/2506/2004	GGLN	KSPYYT	GEHAKA	IGNCPI	WVKTP	PLKLAN	GT	KYRPP	KARLLKE	RGFF	GAIAGF
B/Odessa/3886/2010											
B/Hong Kong/514/2009											
B/Hong Kong/3534/2012											
B/Norway/2090/2012											
B/Paris/1762/2009											
B/England/393/2008											
B/Brisbane/60/2008											
B/Denmark/23/2012											
B/Dakar/10/2012											
B/Dakar/17/2012											
B/Malta/636714/2011											
B/Madagascar/03165/2012											
B/Mauritius/718/2012											
B/Sachsen/26/2012											
B/Wisconsin/12/2012 cdc											
B/Malmoe/3/2012 se											
B/Georgia/741/2012											
B/Valladolid/96/2012											
B/Sante Fe/915530/2012											
B/LisboaPT/8/2012											
B/Johannesburg/3964/2012											
B/Jordan/30014/2012											
B/Poitiers/2821/2012											
B/Finland/293/2012 fn											
B/Hong Kong/3501/2012											
B/Bayern/1/2013											
B/Formosa/V2367/2012											
B/Catalonia/203477/2013 es											
B/England/626/2012											

	410	420	430	440	450	460	470	480	490	500
B/Malaysia/2506/2004	SLSLE	VKNLQ	RLSGA	MDLH	NEILE	DEKVD	DLRAD	TISSQ	IELAV	LLSNEG
B/Odessa/3886/2010										
B/Hong Kong/514/2009										
B/Hong Kong/3534/2012										
B/Norway/2090/2012										
B/Paris/1762/2009										
B/England/393/2008										
B/Brisbane/60/2008										
B/Denmark/23/2012										
B/Dakar/10/2012										
B/Dakar/17/2012										
B/Malta/636714/2011										
B/Madagascar/03165/2012										
B/Mauritius/718/2012										
B/Sachsen/26/2012										
B/Wisconsin/12/2012 cdc										
B/Malmoe/3/2012 se										
B/Georgia/741/2012										
B/Valladolid/96/2012										
B/Sante Fe/915530/2012										
B/LisboaPT/8/2012										
B/Johannesburg/3964/2012										
B/Jordan/30014/2012										
B/Poitiers/2821/2012										
B/Finland/293/2012 fn										
B/Hong Kong/3501/2012										
B/Bayern/1/2013										
B/Formosa/V2367/2012										
B/Catalonia/203477/2013 es										
B/England/626/2012										

	510	520	530	540	550	560	570
B/Malaysia/2506/2004	AGTFD	AGESL	PTFDS	LNIT	TAASL	NDDGL	NHET
B/Odessa/3886/2010							
B/Hong Kong/514/2009							
B/Hong Kong/3534/2012							
B/Norway/2090/2012							
B/Paris/1762/2009							
B/England/393/2008							
B/Brisbane/60/2008							
B/Denmark/23/2012							
B/Dakar/10/2012							
B/Dakar/17/2012							
B/Malta/636714/2011							
B/Madagascar/03165/2012							
B/Mauritius/718/2012							
B/Sachsen/26/2012							
B/Wisconsin/12/2012 cdc							
B/Malmoe/3/2012 se							
B/Georgia/741/2012							
B/Valladolid/96/2012							
B/Sante Fe/915530/2012							
B/LisboaPT/8/2012							
B/Johannesburg/3964/2012							
B/Jordan/30014/2012							
B/Poitiers/2821/2012							
B/Finland/293/2012 fn							
B/Hong Kong/3501/2012							
B/Bayern/1/2013							
B/Formosa/V2367/2012							
B/Catalonia/203477/2013 es							
B/England/626/2012							

Last used Vaccine virus: Reference virus: Genetic clade: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 19. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus
Reference viruses

Collection date

Oct 2012

Nov 2012

Dec 2012

Jan 2013

See HI Tables
for antigenic
characterisation

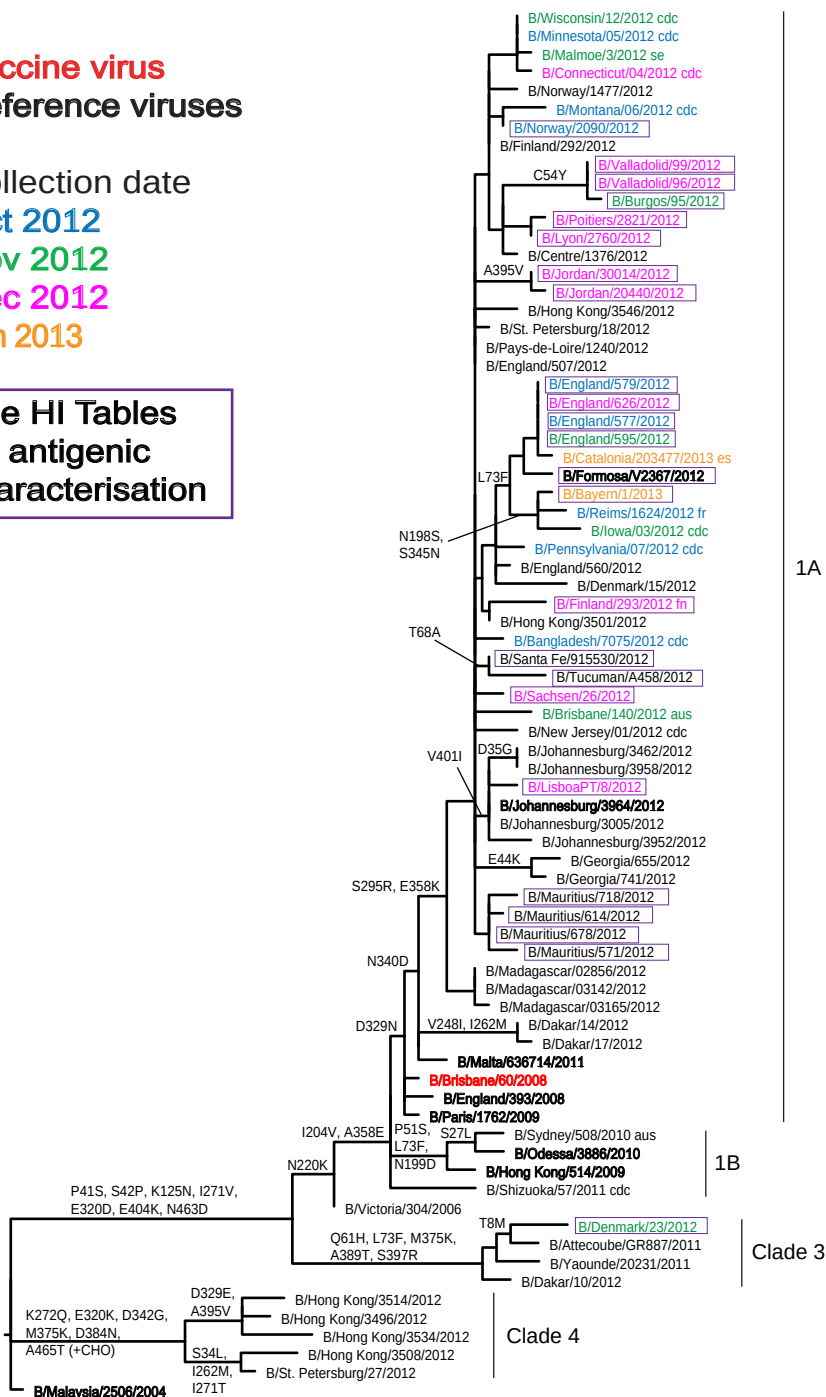


Figure 20. NA protein alignment for B (Victoria-lineage) viruses.

Last used Vaccine virus: Reference virus: Genetic clade: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100	
B/Malaysia/2506/2004	M	L	P	S	T	I	Q	T	L	T	4
B/Odessa/3886/2010											1B
B/Hong Kong/514/2009											1B
B/Hong Kong/3534/2012											4
B/Norway/2090/2012											1A
B/Paris/1762/2009											1A
B/England/393/2008											1A
B/Brisbane/60/2008											1A
B/Denmark/23/2012											3
B/Dakar/10/2012											3
B/Dakar/17/2012											1A
B/Malta/636714/2011											1A
B/Madagascar/03165/2012											1A
B/Mauritius/718/2012											1A
B/Sachsen/26/2012											1A
B/Wisconsin/12/2012 cdc											1A
B/Malmoe/3/2012 se											1A
B/Georgia/741/2012											1A
B/Valladolid/96/2012											1A
B/Santa Fe/915530/2012											1A
B/LisboaPT/8/2012											1A
B/Johannesburg/3964/2012											1A
B/Jordan/30014/2012											1A
B/Poitiers/2821/2012											1A
B/Finland/293/2012 fn											1A
B/Hong Kong/3501/2012											1A
B/Bayern/1/2013											1A
B/Formosa/V2367/2012											1A
B/Catalonia/203477/2013 es											1A
B/England/626/2012											1A

	110	120	130	140	150	160	170	180	190	200	
B/Malaysia/2506/2004	H	R	F	G	E	T	K	G	N	S	
B/Odessa/3886/2010											
B/Hong Kong/514/2009											
B/Hong Kong/3534/2012											
B/Norway/2090/2012											
B/Paris/1762/2009											
B/England/393/2008											
B/Brisbane/60/2008											
B/Denmark/23/2012											
B/Dakar/10/2012											
B/Dakar/17/2012											
B/Malta/636714/2011											
B/Madagascar/03165/2012											
B/Mauritius/718/2012											
B/Sachsen/26/2012											
B/Wisconsin/12/2012 cdc											
B/Malmoe/3/2012 se											
B/Georgia/741/2012											
B/Valladolid/96/2012											
B/Santa Fe/915530/2012											
B/LisboaPT/8/2012											
B/Johannesburg/3964/2012											
B/Jordan/30014/2012											
B/Poitiers/2821/2012											
B/Finland/293/2012 fn											
B/Hong Kong/3501/2012											
B/Bayern/1/2013											
B/Formosa/V2367/2012											
B/Catalonia/203477/2013 es											
B/England/626/2012											

	210	220	230	240	250	260	270	280	290	300	
B/Malaysia/2506/2004	L	L	K	I	K	Y	G	E	A	T	
B/Odessa/3886/2010											
B/Hong Kong/514/2009											
B/Hong Kong/3534/2012											
B/Norway/2090/2012											
B/Paris/1762/2009											
B/England/393/2008											
B/Brisbane/60/2008											
B/Denmark/23/2012											
B/Dakar/10/2012											
B/Dakar/17/2012											
B/Malta/636714/2011											
B/Madagascar/03165/2012											
B/Mauritius/718/2012											
B/Sachsen/26/2012											
B/Wisconsin/12/2012 cdc											
B/Malmoe/3/2012 se											
B/Georgia/741/2012											
B/Valladolid/96/2012											
B/Santa Fe/915530/2012											
B/LisboaPT/8/2012											
B/Johannesburg/3964/2012											
B/Jordan/30014/2012											
B/Poitiers/2821/2012											
B/Finland/293/2012 fn											
B/Hong Kong/3501/2012											
B/Bayern/1/2013											
B/Formosa/V2367/2012											
B/Catalonia/203477/2013 es											
B/England/626/2012											

	310	320	330	340	350	360	370	380	390	400																			
B/Malaysia/2506/2004	PFVKLN	VE	TD	TAEIR	LMCT	ET	YLD	TPRP	DDGS	ITGP	CES	NGDK	SGSG	GKGG	FVHQ	RMA	SKIG	RWYS	SR	TMSK	TKRM	GMGL	YV	YDGP	WADSD	ALAF	S	G	V
B/Odessa/3886/2010		D													E						E								
B/Hong Kong/514/2009		D													E						E								
B/Hong Kong/3534/2012		K		E				G							E	K		N										V	
B/Norway/2090/2012		D		N			D								K						E								
B/Paris/1762/2009		D		N											E						E								
B/England/393/2008		D		N											E						E								
B/Brisbane/60/2008		D		N											E						E								
B/Denmark/23/2012		D						G													E	K					T		R
B/Dakar/10/2012		D																			E	K					TN		R
B/Dakar/17/2012		D		N			D								E						E								
B/Malta/636714/2011		D		N			D								E						E								
B/Madagascar/03165/2012		D		N			D								K						E								
B/Mauritius/718/2012		D		N			D								K						E								
B/Sachsen/26/2012		D		N			D								K						E								
B/Wisconsin/12/2012 cdc		D		N			D								K						E								
B/Malmoe/3/2012 se		D		N			D								K						E								
B/Georgia/741/2012		D		N			D								K						E								
B/Valladolid/96/2012		D		N			D								K						E								
B/Santa Fe/915530/2012		D		N			D								K						E								
B/LisboaPT/8/2012		D		N			D		I						K						E								
B/Johannesburg/3964/2012		D		N			D								K						E								
B/Jordan/30014/2012		D		N			D								K						E							V	
B/Poitiers/2821/2012		D		N			D								K						E								
B/Finland/293/2012 fn		D		N			D								K						E						T		E
B/Hong Kong/3501/2012		D		N			D								K						E								
B/Bayern/1/2013		D		N			D		N						K						E								
B/Formosa/V2367/2012		D		N			D								K						E								
B/Catalonia/203477/2013 es		D		N			D								K						E								
B/England/626/2012		D		N			D								K						E								

	410	420	430	440	450	460
B/Malaysia/2506/2004	VSMEEP	GWYS	FGFE	IKDK	KCDV	PCIG
B/Odessa/3886/2010		K				D
B/Hong Kong/514/2009		K				D
B/Hong Kong/3534/2012						T
B/Norway/2090/2012		K				D
B/Paris/1762/2009		K				D
B/England/393/2008		K				D
B/Brisbane/60/2008		K				D
B/Denmark/23/2012	I	K				D
B/Dakar/10/2012		K				D
B/Dakar/17/2012		K				D
B/Malta/636714/2011		K				D
B/Madagascar/03165/2012		K				D
B/Mauritius/718/2012		K				D
B/Sachsen/26/2012		K				D
B/Wisconsin/12/2012 cdc		K				D
B/Malmoe/3/2012 se		K				D
B/Georgia/741/2012		K				D
B/Valladolid/96/2012		K				D
B/Santa Fe/915530/2012		K				D
B/LisboaPT/8/2012	I	K				D
B/Johannesburg/3964/2012	I	K				D
B/Jordan/30014/2012		K				D
B/Poitiers/2821/2012		K				D
B/Finland/293/2012 fn		K				D
B/Hong Kong/3501/2012		K				D
B/Bayern/1/2013		K				D
B/Formosa/V2367/2012		K				D
B/Catalonia/203477/2013 es		K				D
B/England/626/2012		K				D

Last used Vaccine virus: Reference virus: Genetic clade: Potential N-linked glycosylation site

Table 30. Summary of influenza B (Yamagata-lineage) samples analysed, collected between 2012-09-01 and 2013-01-31

MONTH	B Yamagata lineage								
			Fold reduction in HI titre						
			B/Stockholm/12/2011 antisera Clade 3				B/Estonia/55669/2011 antisera Clade 2		
Country	Number received	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²		≤2 fold ³	4 fold ³	≥8 fold ³
SEPTEMBER									
France	1	1	1				1		
Ghana	1	1	1					1	
Norway	2	2	1	1			1		1
OCTOBER									
France	1	in process							
Germany	1	1	1				1		
Ghana	1	1	1						1
Norway	5	4	1		3		4		
United Kingdom	3	3		1	2		3		
NOVEMBER									
Algeria	1	in process							
Belgium	1	1	1						1
Denmark	3	3			3		3		
France	5	in process	1	1			1		1
Germany	3	3	1	1	1		1		2
Hong Kong	1	in process							
Ireland	5	5		2	3		5		
Italy	2	2	2						2
Netherlands	1	in process							
Norway	11	in process	3	4	1		6		2
Portugal	1	1	1						1
Spain	6	5	3	2			5		
Sweden	1	1		1					1
DECEMBER									
Belgium	23	in process		2			2		
Denmark	1	1			1		1		
France	5	5			5		5		
Germany	4	4	2		2		1		3
Ghana	1	in process							
Ireland	5	in process							
Italy	11	in process	5	4			7		2
Latvia	1	1	1					1	
Portugal	6	6	5	1			2		4
Spain	2	in process	1					1	
Switzerland	5	in process	2						2
United Kingdom	4	4		1	3		4		
JANUARY									
Belgium	9	in process							
Estonia	1	1	1					1	
Germany	4	4		1	3		3		1
Hong Kong	2	in process							
Ireland	2	2		1	1		2		
Israel	1	1	1				1		
Latvia	1	1		1			1		
Netherlands	1	in process							
Portugal	2	2	2						2
Ukraine	1	1		1			1		
	148	67	38	25 91	28		61	4 91	26
Total sequenced		36	11	8	17		22	0	14
Clade 2		19	0	4 (50%)	15 (88.2%)		18 (81.8%)	0	1 (7.1%)
Clade 3		16	11 (100%)	4 (50%)	1 (5.9%)		3 (13.6%)	0	13 (92.9%)
Clade 1		1	0	0	1 (5.9%)		1 (4.6%)	0	0

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antiserum raised against tissue-culture grown B/Stockholm/12/2011 (Clade 3)

3. Compared to homologous titres for ferret antiserum raised against tissue-culture grown B/Estonia/55669/2011 (Clade 2)

Table 31. Antigenic analyses of influenza B viruses (Yamagata lineage) 2012-10-09

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre											
				Post infection ferret sera											
				B/Fi ³	B/Eg ¹	B/Fi ¹	B/Bris ²	B/Eng ²	B/Bang ²	B/Wis ²	B/Stock ²	B/Serbia ²	B/Stock ²	B/Estonia ²	
				4/06 SH479	144/05 F3/07	4/06 F01/10	3/07 F21/12	145/08 F12/11	3333/07 F25/08	1/10 F7/12	12/11 F34/11	1894/11 F25/11	12/11 T/C F8/12	55669/11 F26/11	
				Group 1	Group 1	Group 1	Group 2	Group 3	Group 3	Group 3	Group 3	Group 3	Group 3	Group 2	
REFERENCE VIRUSES															
B/Egypt/144/2005	1	2005-05-01	E3/E5	5120	640	1280	1280	320	640	320	640	20	640	160	
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	320	2560	1280	640	640	320	640	20	640	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	160	640	1280	160	320	160	640	10	320	160	
B/England/145/2008	3		Ex/E4	640	<	160	80	320	40	40	80	<	80	<	
B/Bangladesh/3333/2007	3	2007-08-07	E4/E4	5120	80	640	320	80	320	80	640	20	640	<	
B/Wisconsin/1/2010	3	2010-02-20	E3/E1	1280	80	320	160	160	160	160	160	20	320	10	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	5120	160	640	320	160	320	160	640	40	640	10	
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	5120	160	320	160	320	160	80	320	320	320	160	
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK4	5120	80	320	160	160	160	80	640	320	320	160	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK2	2560	80	320	160	80	20	20	160	80	40	640	
TEST VIRUSES															
B/Buenos Aires/8833/2012	2	2012-08-26	MDCK1/MDCK2	1280	<	80	40	20	10	10	80	20	20	320	
B/Tucumán/A285/2012	2	2012-08-14	MDCK1/MDCK1	2560	40	160	80	40	10	20	160	20	20	640	

Sequences in phylogenetic trees

Vaccine

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Table 32. Antigenic analyses of influenza B viruses (Yamagata lineage) 2012-10-30

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre										
				Post infection ferret sera										
				B/Fi ³ 4/06 SH479	B/Eg ¹ 144/05 F3/07	B/Fi ¹ 4/06 F01/10	B/Bris ² 3/07 F21/12	B/Wis ² 1/10 F23/10	B/Stock ² 12/11 F34/11	B/Serbia ² 1894/11 F25/11	B/Stock ² 12/11 T/C F8/12	B/Novo ² 1/12 F31/12	B/Estonia ² 55669/11 F26/11	B/HK ² 3577/12 F33/12
				Group 1	Group 1	Group 1	Group 2	Group 3	Group 3	Group 3	Group 3	Group 3	Group 2	Group 2
REFERENCE VIRUSES														
B/Egypt/144/2005	1	2005-05-01	E3/E5	5120	320	2560	1280	320	640	40	640	160	320	1280
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	160	2560	1280	320	640	20	640	80	320	1280
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	160	1280	1280	160	320	10	640	80	320	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E1	5120	160	640	320	320	160	80	640	320	40	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	5120	80	640	320	160	1280	40	640	160	20	160
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	5120	40	160	160	80	320	320	320	640	160	640
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK3	5120	40	320	160	80	320	320	320	320	160	1280
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	2560	40	160	160	40	160	40	320	640	160	640
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK2	5120	40	160	160	20	160	160	80	640	1280	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	2560	<	320	160	20	320	160	80	160	1280	1280
TEST VIRUSES														
B/Norway/1864/2012	3	2012-09-16	MDCK2/MDCK1	5120	40	160	160	40	320	160	160	640	160	640
B/Norway/1494/2012	2	2012-06-12	MDCK1/MDCK1	5120	<	160	160	20	160	40	40	80	640	1280
B/Norway/1836/2012	2	2012-09-07	MDCK2/MDCK1	5120	<	160	160	20	160	80	80	320	1280	1280

Sequences in phylogenetic trees

Vaccine

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Table 33. Antigenic analyses of influenza B viruses (Yamagata lineage) 2012-12-11

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post infection ferret sera									
				B/Fi ³ 4/06 SH479	B/Fi ¹ 4/06 F01/10	B/Bris ² 3/07 F21/12	B/Wis ² 1/10 F26/10	B/Stock ² 12/11 F34/11	B/Serbia ² 1894/11 F25/11	B/Stock ² 12/11 T/C F8/12	B/Novo ² 1/12 F31/12	B/Estonia ² 55669/11 F32/12	B/HK ² 3577/12 F33/12
				Group 1	Group 1	Group 2	Group 3	Group 3	Group 3	Group 3	Group 3	Group 2	Group 2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	640	320	320	20	640	80	160	640
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	160	160	20	320	40	80	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	640	320	320	320	40	320	80	40	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	320	160	320	40	640	80	20	80
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	2560	320	160	160	320	320	320	320	320	640
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	1280	160	80	80	320	80	160	160	80	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	320	160	80	320	80	160	320	80	320
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	320	160	40	160	80	80	160	320	1280
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	2560	320	160	80	320	40	160	160	640	1280
TEST VIRUSES													
B/Norway/2290/2012	3	2012-11-11	MDCK2	320	160	80	20	160	<	40	20	10	40
B/Norway/2157/2012	2	2012-11-04	MDCK1/MDCK1	1280	160	80	10	80	40	20	40	160	640
B/Norway/2117/2012	2	2012-10-30	MDCK1/MDCK1	1280	160	80	10	160	40	20	40	160	1280
B/Baden Württemberg/39/2012	3	2012-10-25	C1/MDCK1	2560	320	160	160	320	320	320	320	320	640
B/Norway/1995/2012	3	2012-10-11	MDCK1/MDCK1	1280	320	160	160	320	160	320	320	320	640
B/Baden Württemberg/40/2012	2	2012-11-05	C1/MDCK1	1280	160	160	40	160	40	40	40	320	640
B/Norway/2114/2012	2	2012-10-29	MDCK1/MDCK1	1280	160	80	20	160	40	20	40	320	640
B/Norway/2026/2012	2	2012-10-09	MDCK1/MDCK1	1280	160	80	20	160	20	20	80	320	640

Sequences in phylogenetic trees

Vaccine

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Table 34. Antigenic analyses of influenza B viruses (Yamagata lineage) 2012-12-27

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post infection ferret sera									
				B/FI ³	B/FI ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Serbia ²	B/Stock ²	B/Novo ²	B/Estonia ²	B/HK ²
				4/06 SH479	4/06 F01/10	3/07 F21/12	1/10 F26/10	12/11 F12/12	1894/11 F25/11	12/11 T/C F8/12	1/12 F31/12	55669/11 F32/12	3577/12 F33/12
				Group 1	Group 1	Group 2	Group 3	Group 3	Group 3	Group 3	Group 3	Group 2	Group 2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	1280	320	1280	10	640	80	160	640
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	2560	1280	640	1280	10	1280	80	320	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	1280	1280	640	1280	40	640	160	80	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	640	160	320	640	10	320	40	20	40
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	2560	320	160	80	320	160	160	320	320	320
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	1280	160	80	40	160	80	160	160	80	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	160	80	80	160	40	80	160	80	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	80	20	80	20	40	80	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	5120	160	80	10	80	10	20	40	320	640
TEST VIRUSES													
B/Stockholm/14/2012	3	2012-11-04	MDCK1/MDCK1	320	160	80	40	320	10	40	40	20	40
B/Denmark/24/2012	2	2012-11-15	MDCK3/MDCK1	1280	80	80	10	40	10	20	40	160	640
B/Denmark/26/2012	2	2012-11-26	MDCK2/MDCK1	640	80	40	10	40	10	10	20	160	640
B/Denmark/31/2012	2	2012-12-06	MDCK1/MDCK1	1280	160	80	10	40	10	10	20	160	640
B/England/594/2012	2	2012-10-30	SIAT1/MDCK1	640	80	40	10	20	10	10	20	160	640
B/England/581/2012	1	2012-10-30	SIAT2/MDCK1	2560	160	160	20	80	20	20	40	320	640
B/England/580/2012	2	2012-10-23	SIAT1/MDCK1	2560	160	160	40	160	40	40	40	320	1280
B/Denmark/27/2012	2	2012-11-28	MDCK1/MDCK1	1280	160	80	20	80	10	20	40	320	1280

Sequences in phylogenetic trees

Vaccine

1. <= <40; 2. <= <10 ; 3. hyperimmune sheep serum

Table 35. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-01-22

Viruses	Genetic group	Collection date	Passage History	Haemagglutination Inhibition Titre								
				B/FI ³ 4/06 SH479	Post infection ferret sera							
					B/FI ¹ 4/06 F21/07 Group 1	B/Bris ² 3/07 F21/12 Group 2	B/Wis ² 1/10 F26/10 Group 3	B/Stock ² 12/11 F12/12 Group 3	B/Stock ² 12/11 T/C F8/12 Group 3	B/Novo ² 1/12 F31/12 Group 3	B/Estonia ² 55669/11 F26/11 Group 2	B/HK ² 3577/12 F33/12 Group 2
REFERENCE VIRUSES												
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	640	640	320	640	640	40	160	640
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	320	640	640	40	160	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	160	320	320	640	640	80	40	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	40	80	80	320	160	40	10	40
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	640	<	80	40	80	160	160	40	160
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	<	80	80	160	80	160	40	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	40	80	40	40	20	80	640	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	1280	40	80	20	80	20	40	640	640
TEST VIRUSES												
B/Baden Württemberg/41/2012	3	2012-11-26	C2/MDCK1	320	<	40	10	20	20	40	10	40
B/England/708/2012		2012-12-10	SIAT1/MDCK1	640	<	20	10	20	20	40	10	40
B/England/709/2012	3	2012-12-20	MDCK1/MDCK1	640	<	40	20	40	40	40	10	80
B/Rheinland-Pfalz/4/2013	3	2013-01-04	C2/MDCK1	640	<	40	20	40	40	40	20	80
B/Nordrhein-Westfalen/5/2012		2012-12-20	C2/MDCK1	320	<	40	20	40	20	40	20	80
B/England/706/2012	2	2012-12-17	MDCK1/MDCK1	640	<	40	10	20	10	20	40	80
B/Thüringen/1/2013	2	2013-01-07	C1/MDCK1	640	<	40	10	40	10	20	320	320
B/Rheinland-Pfalz/2/2013		2013-01-07	C1/MDCK1	640	<	40	10	40	10	20	320	320
B/Lyon/2771/2012	2	2012-12-17	MDCK2/MDCK1	640	40	40	10	20	10	40	320	320
B/Lyon/2749/2012		2012-12-17	MDCK2/MDCK1	320	<	40	10	20	10	20	320	320
B/LyonCHU/51.410/2012		2012-12-19	MDCK2/MDCK1	640	40	40	10	20	10	20	320	320
B/Lyon/2791/2012		2012-12-20	MDCK2/MDCK1	640	<	40	10	20	10	20	320	320
B/LyonCHU/51.559/2012	2	2012-12-21	MDCK2/MDCK1	640	40	40	10	20	10	20	320	320
B/Rheinland-Pfalz/3/2013		2013-01-03	C2/MDCK1	640	<	40	10	20	10	20	320	320
B/England/707/2012	2	2012-12-29	SIAT1/MDCK1	640	40	80	10	40	20	40	320	640

Sequences in phylogenetic trees

Vaccine

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Table 36. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-01-25

Viruses	Genetic group	Collection date	Passage History	Haemagglutination Inhibition Titre								
				Post infection ferret sera								
				B/FI ³	B/FI ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Serbia ²	B/Stock ²	B/Estonia ²	B/HK ²
				4/06 SH479	4/06 F21/07	3/07 F24/07	1/10 F26/10	12/11 F12/12	1894/11 F25/11	12/11 T/C F8/12	55669/11 F26/11	3577/12 F33/12
REFERENCE VIRUSES												
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	640	640	640	40	640	320	640
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	320	640	10	640	160	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	160	320	640	640	20	640	20	80
B/Stockholm/12/2011	3	2011-03-28	E4/E2	2560	80	80	160	320	10	320	10	80
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	2560	40	80	80	160	80	160	80	320
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	2560	<	80	80	80	40	160	40	320
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	80	40	40	80	40	40	640	1280
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	2560	80	40	20	80	20	40	640	640
TEST VIRUSES												
B/Baden Württemberg/42/2012		2012-12-03	C2/MDCK1	640	<	40	20	40	20	80	10	80
B/Brandenburg/2/2012		2012-12-19	C2/MDCK1	640	<	40	20	20	40	80	10	80
B/Bayern/57/2012		2012-11-28	C2/MDCK1	640	<	40	40	40	40	80	20	80
B/LisboaPT/7/2012	3	2012-11-26	SIAT1/MDCK1	1280	<	80	80	160	40	160	40	160
B/BragaPT/11/2012		2012-12-21	SIAT1/MDCK1	2560	160	40	80	160	40	160	40	160
B/BragaPT/10/2012	3	2012-12-23	SIAT1/MDCK1	2560	40	40	80	160	40	160	40	160
B/BragaPT/13/2012		2012-12-27	SIAT1/MDCK1	2560	80	40	80	160	40	160	40	160
B/PortoPT/14/2012	3	2012-12-31	SIAT1/MDCK1	2560	80	40	80	160	40	160	40	160
B/BragaPT/12/2012	3	2012-12-21	SIAT1/MDCK1	2560	80	40	80	160	80	160	320	320
B/LisboaPT/9/2012	2	2012-12-27	SIAT1/MDCK1	2560	80	40	40	160	40	40	640	640
B/Saarland/2/2012	2	2012-12-17	C2/MDCK1	1280	<	40	10	160	10	20	640	640

Sequences in phylogenetic trees

Vaccine

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Table 37. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-01-30

Viruses	Genetic group	Collection date	Passage History	Haemagglutination Inhibition Titre											
				B/FI ³ 4/06 SH479	Post infection ferret sera										
					B/FI ¹ 4/06 F21/07 Group 1	B/Bris ² 3/07 F24/07 Group 2	B/Wis ² 1/10 F26/10 Group 3	B/Stock ² 12/11 F12/12 Group 3	B/Serbia ² 1894/11 F25/11 Group 3	B/Stock ² 12/11 T/C F8/12 Group 3	B/Novo ² 1/12 F31/12 Group 3	B/Estonia ² 55669/11 F26/11 Group 2	B/HK ² 3577/12 F33/12 Group 2	B/Mass ² 2/12 Egg F02/13 Group 2	B/Mass ² 2/12 T/C F03/13 Group 2
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	1280	640	1280	80	1280	160	320	1280	1280	640
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	320	640	10	320	80	160	640	1280	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	320	80	640	640	40	640	80	40	160	640	40
B/Stockholm/12/2011	3	2011-03-28	E4/E2	1280	80	40	160	640	10	320	40	10	40	320	20
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	2560	80	80	80	160	160	160	320	160	320	160	320
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	2560	160	80	160	320	20	320	320	320	640	320	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	<	80	80	160	80	160	320	80	320	160	80
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	80	80	160	80	160	160	640	1280	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	1280	160	40	40	80	20	80	80	640	640	160	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	1280	320	160	160	320	20	320	20	80	320	640	40
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	2560	160	80	40	160	20	40	40	160	640	640	160
TEST VIRUSES															
B/Caen/1808/2012		2012-11-16	MDCK1/MDCK1	2560	40	40	80	320	40	160	80	20	160	160	80
B/Ghana/DILI-0362/2012		2012-07-25	SIAT3/MDCK1	1280	80	20	40	80	80	160	160	40	80	80	80
B/Ghana/DILI-0309/2012		2012-07-18	SIAT2/MDCK1	2560	40	40	80	320	80	160	160	40	160	160	80
B/Hyogo/4002/2012	3	2012-12-11	MDCK2/MDCK1	1280	80	40	80	160	40	160	160	40	160	320	80
B/Paris/1443/2012		2012-04-16	MDCK2/MDCK1	1280	80	40	160	320	80	320	320	40	320	320	80
B/Ghana/DILI-0859/2012	3	2012-10-19	SIAT2/MDCK1	2560	80	40	80	320	80	320	320	40	320	320	80
B/Ghana/DILI-0369/2012		2012-07-25	SIAT2/MDCK1	2560	40	40	80	320	80	160	320	80	160	160	80
B/Ghana/ARI 1648/2012	3	2012-08-30	SIAT3/MDCK1	2560	80	40	80	160	80	160	320	80	160	160	80
B/Ghana/DILI-0409/2012		2012-08-02	SIAT1/MDCK1	2560	80	40	160	320	80	320	320	80	320	160	160
B/Ghana/FS-968/2012		2012-07-27	SIAT2/MDCK1	2560	40	20	80	320	80	320	320	80	320	320	80
B/Ghana/DILI-0396/2012	3	2012-08-02	SIAT1/MDCK1	2560	80	40	80	320	80	320	320	80	320	320	80
B/Ghana/FS-1120/2012		2012-09-09	SIAT3/MDCK1	2560	80	40	80	160	80	320	320	160	320	160	160
B/Yokohama/82/2012	2	2012-12-07	MDCK2/MDCK1	1280	80	80	40	80	40	80	80	320	320	160	160
B/Paris/1466/2012		2012-09-24	MDCK2/MDCK1	2560	160	80	80	160	40	160	80	640	640	320	160
B/New Mexico/04/2012	2	2012-11-26	MDCK1/C1/MDCK1	2560	160	80	40	160	40	40	40	640	640	320	160
B/Paris/1448/2012		2012-07-13	MDCK2/MDCK1	2560	160	80	40	160	20	80	40	640	640	320	160
B/Pays de Loire/1791/2012		2012-11-19	MDCK1/MDCK1	2560	160	160	80	320	80	80	80	640	640	320	320
B/Ukraine/2/2013		2013-01-07	MDCK2/MDCK1	2560	320	80	80	320	40	80	80	640	1280	320	320

Sequences in phylogenetic trees

Vaccine

1. <= <40; 2. <= <10 ; 3. hyperimmune sheep serum

Table 38. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-02-05

Haemagglutination Inhibition Titre															
Viruses	Genetic group	Collection date	Passage History	Post infection ferret sera											
				*											
				B/FI ¹ 4/06 SH479	B/FI ¹ 4/06 F21/07 Group 1	B/Bris ² 3/07 F24/07 Group 2	B/Wis ¹ 1/10 F26/10 Group 3	B/Stock ² 12/11 F12/12 Group 3	B/Serbia ² 1894/11 F25/11 Group 3	B/Stock ² 12/11 T/C F8/12 Group 3	B/Novo ² 1/12 F31/12 Group 3	B/Estonia ² 55669/11 F26/11 Group 2	B/HK ² 3577/12 F33/12 Group 2	B/Mass ² 2/12 Egg F02/13 Group 2	B/Mass ² 2/12 T/C F03/13 Group 2
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	1280	160	640	20	640	80	160	640	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	320	80	320	<	320	40	160	320	640	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	160	80	80	640	20	640	80	10	80	640	40
B/Stockholm/12/2011	3	2011-03-28	E4/E2	1280	40	40	<	320	10	320	40	<	40	160	20
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	1280	<	40	40	80	80	160	160	80	320	160	160
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	2560	80	40	40	160	160	160	320	160	320	320	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	<	40	40	80	40	80	160	40	320	160	80
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	40	40	80	40	20	40	80	320	1280	160	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	1280	80	40	40	40	40	40	80	320	640	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	2560	320	160	40	320	<	320	20	40	320	320	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	1280	40	40	<	40	<	40	40	160	320	160	160
TEST VIRUSES															
B/Porto/EuroEva38/2012	3	2013-01-04	MDCK2	2560	<	20	<	160	40	160	80	10	80	160	40
B/Belgium/G901/2012		2012-11-28	MDCK3	1280	<	20	<	160	40	160	80	20	80	160	40
B/Parma/03/2012		2012-12-19	MDCK2/MDCK1	1280	<	<	<	160	40	80	80	20	80	160	40
B/Braga/SU17/2012		2013-01-02	MDCK2	1280	<	20	40	160	40	160	80	20	80	160	80
B/Ghana/FS-0899/2012		2012-07-16	SIAT2/MDCK2	2560	40	20	40	160	40	160	80	20	160	160	80
A/Switzerland/7628394/2012		2012-12-17	MDCK2	2560	40	40	40	320	40	160	80	40	160	160	80
B/Milano/36/2012		2012-12-04	MDCK1/MDCK1	1280	40	20	40	160	80	160	160	40	160	160	80
A/Switzerland/7587143/2012		2012-12-10	MDCK2	1280	40	20	40	160	40	80	160	40	160	160	80
B/Milano/38/2012		2012-11-26	MDCK1/MDCK1	1280	<	20	<	80	80	160	160	40	160	160	80
B/Norway/2276/2012		2012-11-19	MDCK1/MDCK1	1280	40	40	40	320	80	160	160	40	320	160	80
B/Milano/37/2012		2012-11-26	MDCK1/MDCK1	2560	80	40	80	320	80	320	160	40	320	320	80
B/Victoria/524/2012	2	2012-07-20	E4/E1	640	40	80	<	40	<	10	10	80	160	160	40
B/Latvia/12-32427p/2012		2012-12-03	MDCK2/MDCK1	1280	80	40	40	160	80	160	160	80	160	160	160
B/Estonia/74419/2013		2013-01-10	MDCK1/MDCK1	1280	40	40	40	160	80	160	160	80	160	160	160
B/Pais Vasco/13005/2012		2012-12-31	SIAT1/MDCK1	2560	40	40	40	160	320	160	160	80	320	160	160
B/Ireland/02490/2013		2013-01-08	MDCK1/MDCK1	640	40	40	<	40	<	20	20	160	160	80	80
B/Roma/02/2012		2012-12-10	Cx/MDCK1	2560	160	80	160	320	320	320	320	160	320	160	320
B/Pais Vasco/284/2012		2012-11-07	SIAT1/MDCK1	2560	80	80	40	160	160	160	320	160	320	160	320
B/Roma/03/2012		2012-12-10	Cx/MDCK1	2560	160	80	80	640	320	640	320	320	160	320	160
B/Norway/2250/2012		2012-11-09	MDCK2/MDCK1	1280	80	40	40	80	10	40	40	320	320	160	160
B/Belgium/G922/2012		2012-12-07	MDCK3	1280	40	40	<	80	10	40	20	320	320	160	160
B/Ireland/89450/2012		2012-11-22	MDCK1/MDCK1	2560	80	40	<	80	10	40	40	320	640	160	80
B/Ireland/91035/2012		2012-11-28	MDCK1/MDCK1	1280	40	40	<	80	<	20	40	320	640	160	160
B/Ireland/90800/2012		2012-11-26	MDCK1/MDCK1	1280	40	40	<	80	<	20	20	320	640	160	160
B/Norway/2349/2012		2012-11-26	MDCK1/MDCK1	1280	40	40	<	80	10	80	40	320	640	160	160
B/Norway/2182/2012		2012-11-07	MDCK1/MDCK1	1280	40	40	40	80	10	40	40	320	640	160	160
B/Belgium/G908/2012		2012-12-04	MDCK2	2560	80	40	<	80	10	40	40	320	640	160	160
B/Parma/02/2012		2012-12-14	MDCK1/MDCK1	1280	80	40	<	80	10	40	40	320	640	160	160
B/Torino/06/2012		2012-12-31	Cx/MDCK1	2560	80	40	<	80	10	40	40	320	640	160	160
B/Ireland/00329/2013		2013-01-02	MDCK1/MDCK1	1280	80	40	<	80	10	40	40	320	640	160	160
B/Torino/01/2012		2012-12-19	MDCK1/MDCK1	1280	80	40	<	80	10	40	20	320	640	160	160
B/Latvia/1-40275p/2013		2013-01-15	MDCK1/MDCK1	1280	40	40	<	80	10	40	20	320	640	160	160
B/Ireland/88486/2012		2012-11-21	MDCK1/MDCK1	1280	40	40	<	80	10	20	40	320	640	160	160
B/Milano/44/2012		2012-12-17	MDCK1/MDCK1	2560	80	40	<	80	20	40	40	320	640	320	160
B/Ireland/90999/2012		2012-11-25	MDCK1/MDCK1	2560	80	40	<	80	10	40	40	320	640	320	160
B/Norway/2256/2012		2012-11-01	MDCK2/MDCK1	1280	80	40	40	80	<	40	40	320	1280	160	160
B/Wellington/3/2012	2	2012-07-16	E2/E1	1280	<	40	<	20	10	10	20	640	640	160	80
B/Navarra/303/2012		2012-11-05	SIAT1/MDCK1	1280	80	40	<	40	20	80	80	640	640	160	160
B/Navarra/299/2012		2012-11-20	SIAT1/MDCK1	1280	80	40	<	40	20	40	80	640	640	160	160
B/Navarra/300/2012		2012-11-16	SIAT1/MDCK1	2560	80	40	<	80	40	80	160	640	640	160	320
B/Navarra/304/2012		2012-11-12	SIAT1/MDCK1	1280	40	40	<	40	20	40	80	640	640	160	320
B/Norway/2281/2012		2012-11-14	MDCK1/MDCK1	2560	80	40	80	160	20	80	80	640	640	320	160
B/Milano/42/2012		2012-12-11	MDCK1/MDCK1	2560	80	80	<	160	20	80	40	640	640	320	160
B/Israel/1/2013		2013-01-15	P1/MDCK1	2560	80	40	<	40	20	80	80	640	640	320	320
B/Parma/04/2012		2012-12-27	MDCK1/MDCK1	2560	80	80	80	160	10	40	40	640	1280	320	160

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

* Note the higher dilution of this antiserum compared with the other B-Yamagata lineage HI tables presented

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Oct 2012

Nov 2012

Dec 2012

Jan 2013

See HI Tables
for antigenic
characterisation

HA2 numbering

proposed serology
antigens

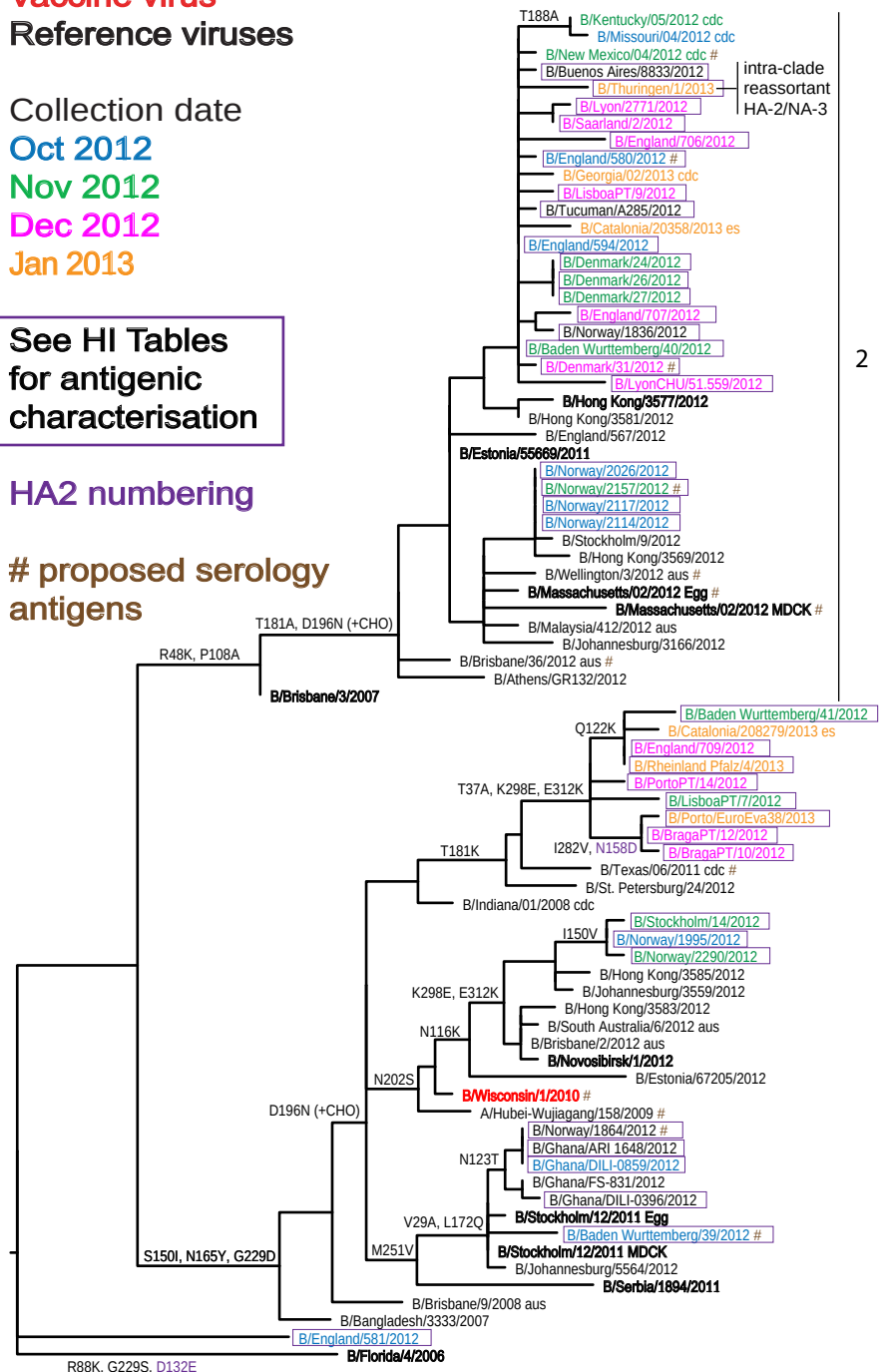


Figure 22. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic clade: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Florida/4/2006	DRIC	TG	IT	SS	NS	PH	VV	K	T	A
B/Bangladesh/3333/2007									R	
B/Serbia/1894/2011									R	
B/Stockholm/12/2011 MDCK			A						R	
B/Stockholm/12/2011 Egg			A						R	
B/Baden Wurttemberg/39/2012			A						R	
B/Ghana/0859/2012			A						R	
B/Wisconsin/1/2010									R	
B/Novosibirsk/1/2012									R	
B/Johannesburg/3559/2012									R	
B/Norway/2290/2012									R	
B/Stockholm/14/2012									R	
B/BragaPT/12/2012				A					R	
B/Porto/EuroEva38/2013				A					R	
B/England/709/2012				A					R	
B/Catalonia/208279/2013 es				A					R	
B/Brisbane/3/2007					K				R	
B/Massachusetts/02/2012 Egg					K				R	
B/Massachusetts/02/2012 MDCK					K				R	
B/Hong Kong/3569/2012					K				R	
B/Norway/2157/2012					K				R	
B/Estonia/55669/2011					K				R	
B/Hong Kong/3577/2012					K				R	
B/Denmark/31/2012					K				R	
B/Tucuman/A285/2012					K				R	
B/Georgia/02/2013 cdc					K				R	
B/England/580/2012					K				R	
B/Saarland/2/2012					K				R	
B/Lyon/2771/2012					K				R	
B/Kentucky/05/2012 cdc					K				R	
	110	120	130	140	150	160	170	180	190	200
B/Florida/4/2006	R	K	I	R	Q	L	P	N	L	R
B/Bangladesh/3333/2007										
B/Serbia/1894/2011										
B/Stockholm/12/2011 MDCK										
B/Stockholm/12/2011 Egg										
B/Baden Wurttemberg/39/2012										
B/Ghana/0859/2012			T							
B/Wisconsin/1/2010										
B/Novosibirsk/1/2012			K							
B/Johannesburg/3559/2012			K							
B/Norway/2290/2012			K							
B/Stockholm/14/2012			K							
B/BragaPT/12/2012										
B/Porto/EuroEva38/2013										
B/England/709/2012										
B/Catalonia/208279/2013 es										
B/Brisbane/3/2007										
B/Massachusetts/02/2012 Egg										
B/Massachusetts/02/2012 MDCK										
B/Hong Kong/3569/2012										
B/Norway/2157/2012										
B/Estonia/55669/2011										
B/Hong Kong/3577/2012										
B/Denmark/31/2012										
B/Tucuman/A285/2012										
B/Georgia/02/2013 cdc										
B/England/580/2012										
B/Saarland/2/2012										
B/Lyon/2771/2012										
B/Kentucky/05/2012 cdc										
	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006	K	N	L	Y	G	S	N	P	K	F
B/Bangladesh/3333/2007										
B/Serbia/1894/2011										
B/Stockholm/12/2011 MDCK										
B/Stockholm/12/2011 Egg										
B/Baden Wurttemberg/39/2012										
B/Ghana/0859/2012										
B/Wisconsin/1/2010										
B/Novosibirsk/1/2012										
B/Johannesburg/3559/2012										
B/Norway/2290/2012										
B/Stockholm/14/2012										
B/BragaPT/12/2012										
B/Porto/EuroEva38/2013										
B/England/709/2012										
B/Catalonia/208279/2013 es										
B/Brisbane/3/2007										
B/Massachusetts/02/2012 Egg										
B/Massachusetts/02/2012 MDCK										
B/Hong Kong/3569/2012										
B/Norway/2157/2012										
B/Estonia/55669/2011										
B/Hong Kong/3577/2012										
B/Denmark/31/2012										
B/Tucuman/A285/2012										
B/Georgia/02/2013 cdc										
B/England/580/2012										
B/Saarland/2/2012										
B/Lyon/2771/2012										
B/Kentucky/05/2012 cdc										

	310	320	330	340	350	360	370	380	390	400														
B/Florida/4/2006	GLN	KSK	PYYT	GEHAK	IGNCPI	WVKTP	PLK	LANG	NT	KRYR	PPAKLLKE	RG	FFG	AIAG	FLEGG	WEGM	IAGWH	GYTSH	GAHGV	AAADL	KSTQ	EAIN	KITK	NLS
B/Bangladesh/3333/2007																								
B/Serbia/1894/2011																								
B/Stockholm/12/2011 MDCK																								
B/Stockholm/12/2011 Egg																								
B/Baden Wurttemberg/39/2012																								
B/Ghana/0859/2012																								
B/Wisconsin/1/2010																								
B/Novosibirsk/1/2012																								
B/Johannesburg/3559/2012																								
B/Norway/2290/2012																								
B/Stockholm/14/2012																								
B/BragaPT/12/2012																								
B/Porto/EuroEva38/2013																								
B/England/709/2012																								
B/Catalonia/208279/2013 es																								
B/Brisbane/3/2007																								
B/Massachusetts/02/2012 Egg																								
B/Massachusetts/02/2012 MDCK																								
B/Hong Kong/3569/2012																								
B/Norway/2157/2012																								
B/Estonia/55669/2011																								
B/Hong Kong/3577/2012																								
B/Denmark/31/2012																								
B/Tucuman/A285/2012																								
B/Georgia/02/2013 cdc																								
B/England/580/2012																								
B/Saarland/2/2012																								
B/Lyon/2771/2012																								
B/Kentucky/05/2012 cdc																								

	410	420	430	440	450	460	470	480	490	500																												
B/Florida/4/2006	LSE	LEV	KNL	QRL	SGAM	DEL	HN	EILE	LDK	VDDL	LRAD	TI	SSQ	IEL	AVLL	SN	EG	IIN	SEDE	HLL	AL	ER	KL	KK	ML	GP	SA	VE	IGN	GC	FF	ET	KHK	CN	QT	CL	DR	IAA
B/Bangladesh/3333/2007																																						
B/Serbia/1894/2011																																						
B/Stockholm/12/2011 MDCK																																						
B/Stockholm/12/2011 Egg																																						
B/Baden Wurttemberg/39/2012																																						
B/Ghana/0859/2012																																						
B/Wisconsin/1/2010																																						
B/Novosibirsk/1/2012																																						
B/Johannesburg/3559/2012																																						
B/Norway/2290/2012																																						
B/Stockholm/14/2012																																						
B/BragaPT/12/2012																																						
B/Porto/EuroEva38/2013																																						
B/England/709/2012																																						
B/Catalonia/208279/2013 es																																						
B/Brisbane/3/2007																																						
B/Massachusetts/02/2012 Egg																																						
B/Massachusetts/02/2012 MDCK																																						
B/Hong Kong/3569/2012																																						
B/Norway/2157/2012																																						
B/Estonia/55669/2011																																						
B/Hong Kong/3577/2012																																						
B/Denmark/31/2012																																						
B/Tucuman/A285/2012																																						
B/Georgia/02/2013 cdc																																						
B/England/580/2012																																						
B/Saarland/2/2012																																						
B/Lyon/2771/2012																																						
B/Kentucky/05/2012 cdc																																						

	510	520	530	540	550	560
B/Florida/4/2006	GTF	NAGE	FSL	PTF	DSL	NIT
B/Bangladesh/3333/2007						
B/Serbia/1894/2011						
B/Stockholm/12/2011 MDCK						
B/Stockholm/12/2011 Egg						
B/Baden Wurttemberg/39/2012						
B/Ghana/0859/2012						
B/Wisconsin/1/2010						
B/Novosibirsk/1/2012						
B/Johannesburg/3559/2012						
B/Norway/2290/2012						
B/Stockholm/14/2012						
B/BragaPT/12/2012						
B/Porto/EuroEva38/2013						
B/England/709/2012						
B/Catalonia/208279/2013 es						
B/Brisbane/3/2007						
B/Massachusetts/02/2012 Egg						
B/Massachusetts/02/2012 MDCK						
B/Hong Kong/3569/2012						
B/Norway/2157/2012						
B/Estonia/55669/2011						
B/Hong Kong/3577/2012						
B/Denmark/31/2012						
B/Tucuman/A285/2012						
B/Georgia/02/2013 cdc						
B/England/580/2012						
B/Saarland/2/2012						
B/Lyon/2771/2012						
B/Kentucky/05/2012 cdc						

Vaccine virus: Reference virus: Genetic clade: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 24. NA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic clade: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100	
B/Florida/4/2006	M	L	P	S	T	I	Q	T	L	F	L
B/Bangladesh/3333/2007	.	.	.	.	R	.	T	.	.	.	.
B/Serbia/1894/2011	.	.	.	.	R	.	T	.	.	.	.
B/Stockholm/12/2011 MDCK	.	.	.	.	R	.	T	.	P	.	.
B/Stockholm/12/2011 Egg	.	.	.	.	R	.	T	.	P	.	.
B/Baden Wurttemberg/39/2012	.	.	.	.	R	.	T	.	P	.	.
B/Ghana/0859/2012	.	.	.	.	R	.	T	.	P	.	.
B/Wisconsin/1/2010	.	.	.	.	R	.	T	.	.	.	.
B/Novosibirsk/1/2012	.	.	.	.	R	.	T	.	.	.	.
B/Johannesburg/3559/2012	.	.	.	.	R	.	T	.	.	.	.
B/Norway/2290/2012	.	.	.	.	R	.	T	.	.	.	.
B/Stockholm/14/2012	.	.	.	.	R	.	T	.	.	.	.
B/BragaPT/12/2012	.	.	.	.	R	.	T	.	A	.	.
B/Porto/EuroEva38/2013	.	.	.	.	R	.	T	.	A	.	.
B/England/709/2012	.	.	.	.	R	K	.	T	.	A	.
B/Catalonia/208279/2013 es	.	.	.	.	R	.	T	.	A	.	.
B/Brisbane/3/2007	.	.	.	.	K	.	.	.	.	.	.
B/Massachusetts/02/2012 Egg	.	.	.	.	.	.	H	.	.	.	.
B/Massachusetts/02/2012 MDCK	.	.	.	.	.	.	H	.	.	.	.
B/Hong Kong/3569/2012	.	.	.	.	.	.	H	.	.	.	.
B/Norway/2157/2012	.	.	.	.	.	.	H	.	.	.	.
B/Estonia/55669/2011	.	.	.	.	.	.	.	.	.	.	.
B/Hong Kong/3577/2012	.	.	.	.	.	.	.	.	.	.	.
B/Denmark/31/2012	.	.	.	.	.	.	.	.	.	.	.
B/Tucuman/A285/2012	.	.	.	.	.	T	.	.	.	.	.
B/Georgia/02/2013 cdc	.	.	.	.	.	T	.	.	.	.	.
B/England/580/2012	.	.	.	.	.	.	.	.	.	.	.
B/Saarland/2/2012	.	.	.	.	.	.	.	.	.	.	.
B/Lyon/2771/2012	.	.	.	.	.	.	.	.	.	.	.
B/Kentucky/05/2012 cdc	.	.	.	.	T	.	.	.	.	.	.
	110	120	130	140	150	160	170	180	190	200	
B/Florida/4/2006	H	R	F	G	E	T	K	G	N	S	A
B/Bangladesh/3333/2007	.	.	.	.	.	.	.	.	.	.	.
B/Serbia/1894/2011	.	.	.	.	.	.	R	.	.	R	.
B/Stockholm/12/2011 MDCK	.	.	.	.	.	.	.	.	.	R	.
B/Stockholm/12/2011 Egg	.	.	.	.	.	.	.	.	.	R	.
B/Baden Wurttemberg/39/2012	.	.	.	.	.	.	.	.	.	R	.
B/Ghana/0859/2012	.	.	.	.	.	.	.	.	.	R	.
B/Wisconsin/1/2010	.	.	.	.	.	.	.	.	.	R	.
B/Novosibirsk/1/2012	.	.	.	.	.	.	.	.	.	R	.
B/Johannesburg/3559/2012	.	.	.	.	.	.	.	.	.	R	.
B/Norway/2290/2012	.	.	.	.	.	.	.	.	.	R	I
B/Stockholm/14/2012	.	.	.	.	.	.	.	.	R	I	.
B/BragaPT/12/2012	.	.	.	.	.	.	.	.	R	.	.
B/Porto/EuroEva38/2013	.	.	.	.	.	.	.	.	R	.	.
B/England/709/2012	.	.	.	.	.	.	.	.	R	.	N
B/Catalonia/208279/2013 es	.	.	.	.	.	.	.	.	R	.	N
B/Brisbane/3/2007	.	.	.	.	.	.	.	.	.	.	.
B/Massachusetts/02/2012 Egg	.	I	.	.	.	.	.	.	.	.	.
B/Massachusetts/02/2012 MDCK	.	I	.	.	.	.	.	.	.	.	.
B/Hong Kong/3569/2012	.	I	.	.	.	.	.	.	.	.	.
B/Norway/2157/2012	.	I	.	.	.	.	.	.	.	.	.
B/Estonia/55669/2011	.	I	.	.	.	.	.	.	.	.	.
B/Hong Kong/3577/2012	.	I	.	.	.	.	.	.	.	.	.
B/Denmark/31/2012	.	I	.	.	.	.	.	.	.	.	.
B/Tucuman/A285/2012	.	I	.	.	.	.	.	.	.	.	.
B/Georgia/02/2013 cdc	.	I	.	.	.	.	.	.	.	.	.
B/England/580/2012	.	I	.	.	.	.	.	.	.	.	.
B/Saarland/2/2012	.	I	.	.	.	.	.	.	.	.	.
B/Lyon/2771/2012	.	I	.	.	.	.	.	.	.	.	.
B/Kentucky/05/2012 cdc	.	I	.	.	.	.	.	.	.	.	.
	210	220	230	240	250	260	270	280	290	300	
B/Florida/4/2006	L	L	K	I	K	Y	E	A	T	D	T
B/Bangladesh/3333/2007	.	.	.	.	.	.	.	.	.	.	.
B/Serbia/1894/2011	.	.	.	.	.	.	.	.	.	.	.
B/Stockholm/12/2011 MDCK	.	.	.	.	.	.	.	.	.	.	.
B/Stockholm/12/2011 Egg	.	.	.	.	.	.	.	.	.	.	.
B/Baden Wurttemberg/39/2012	.	.	.	.	.	.	.	.	.	.	.
B/Ghana/0859/2012	.	.	.	.	.	.	.	.	.	.	.
B/Wisconsin/1/2010	.	.	.	.	.	.	.	.	.	.	.
B/Novosibirsk/1/2012	.	.	.	.	.	.	.	.	.	.	.
B/Johannesburg/3559/2012	.	.	.	.	.	.	.	.	.	.	.
B/Norway/2290/2012	.	.	.	.	.	.	.	.	.	.	.
B/Stockholm/14/2012	.	.	.	.	.	.	.	.	.	.	.
B/BragaPT/12/2012	.	.	.	.	.	.	.	.	.	.	.
B/Porto/EuroEva38/2013	.	.	.	.	.	.	.	.	.	.	.
B/England/709/2012	.	.	.	.	.	.	.	.	.	.	.
B/Catalonia/208279/2013 es	.	.	.	.	.	.	.	.	.	.	.
B/Brisbane/3/2007	.	.	.	.	.	.	.	.	.	.	.
B/Massachusetts/02/2012 Egg	.	.	.	.	.	.	.	.	.	.	.
B/Massachusetts/02/2012 MDCK	.	.	.	.	.	.	.	.	.	.	.
B/Hong Kong/3569/2012	.	.	.	.	.	.	.	.	.	.	.
B/Norway/2157/2012	.	.	.	.	.	.	.	.	.	.	.
B/Estonia/55669/2011	.	.	.	.	.	.	.	.	.	.	.
B/Hong Kong/3577/2012	.	.	.	.	.	.	.	.	.	.	.
B/Denmark/31/2012	.	.	.	.	.	.	.	.	.	.	.
B/Tucuman/A285/2012	.	.	.	.	.	.	.	.	.	.	.
B/Georgia/02/2013 cdc	.	.	.	.	.	.	.	.	.	.	.
B/England/580/2012	.	.	.	.	.	.	.	.	.	.	.
B/Saarland/2/2012	.	.	.	.	.	.	.	.	.	.	.
B/Lyon/2771/2012	.	.	.	.	.	.	.	.	.	.	.
B/Kentucky/05/2012 cdc	.	.	.	.	.	.	.	.	.	.	.

	310	320	330	340	350	360	370	380	390	400
B/Florida/4/2006	PFVKLN	VETDTAEIRLMCTET	YLDTPRPNDG	SITGP	CESDGD	KGSGG	IKGGFV	HQRMASK	IGRWYSRT	MSKTKRMGMGLYVKYDGPWTDSEALALSGVM
B/Bangladesh/3333/2007						N				
B/Serbia/1894/2011						N				
B/Stockholm/12/2011 MDCK						N	E			
B/Stockholm/12/2011 Egg						N	E			
B/Baden Wurttemberg/39/2012						N	E			
B/Ghana/0859/2012						N	NE			T
B/Wisconsin/1/2010						N				
B/Novosibirsk/1/2012						N				
B/Johannesburg/3559/2012						N				
B/Norway/2290/2012						N				
B/Stockholm/14/2012						N				
B/BragaPT/12/2012						N				
B/Porto/EuroEva38/2013						N				
B/England/709/2012						N				
B/Catalonia/208279/2013 es						N				
B/Brisbane/3/2007										
B/Massachusetts/02/2012 Egg										
B/Massachusetts/02/2012 MDCK										
B/Hong Kong/3569/2012										
B/Norway/2157/2012										
B/Estonia/55669/2011										
B/Hong Kong/3577/2012										
B/Denmark/31/2012										
B/Tucuman/A285/2012										
B/Georgia/02/2013 cdc										
B/England/580/2012										
B/Saarland/2/2012										
B/Lyon/2771/2012										
B/Kentucky/05/2012 cdc										
	410	420	430	440	450	460				
B/Florida/4/2006	VSME	EPGWYSFGFEIKDKKCDVPC	IGIEMVHDG	GKTTWHS	AAATAYCLMGSGQLLWD	TVTG	VD	MA	L	
B/Bangladesh/3333/2007										
B/Serbia/1894/2011								N	T	
B/Stockholm/12/2011 MDCK								N	T	
B/Stockholm/12/2011 Egg								N	T	
B/Baden Wurttemberg/39/2012								N	T	
B/Ghana/0859/2012								N	T	
B/Wisconsin/1/2010								N	T	
B/Novosibirsk/1/2012								N	T	
B/Johannesburg/3559/2012								N	T	
B/Norway/2290/2012								N	T	
B/Stockholm/14/2012							A	N	T	
B/BragaPT/12/2012								N	T	
B/Porto/EuroEva38/2013								N	T	
B/England/709/2012								N	T	
B/Catalonia/208279/2013 es								N	T	
B/Brisbane/3/2007										
B/Massachusetts/02/2012 Egg										
B/Massachusetts/02/2012 MDCK										
B/Hong Kong/3569/2012										
B/Norway/2157/2012							G			
B/Estonia/55669/2011										
B/Hong Kong/3577/2012										
B/Denmark/31/2012										
B/Tucuman/A285/2012										
B/Georgia/02/2013 cdc										
B/England/580/2012										
B/Saarland/2/2012										
B/Lyon/2771/2012										
B/Kentucky/05/2012 cdc										

Vaccine virus: Reference virus: Genetic clade: Potential N-linked glycosylation site

Table 39. Antiviral (NI) testing from 2012-09-01 to 2013-01-11

Year	Month of Collection	Type/Subtype of influenza						
		Flu B		A(H3N2)		A(H1N1)pdm09		Total Number Tested
		Number Tested	Reduced Inhibition	Number Tested	Reduced Inhibition	Number Tested	Highly Reduced Inhibition	
2010	March	1						1
2011	June	2						2
	July	2						2
	August	3						3
	November			1				1
2012	January			3				3
	February			4				4
	March	4		11		1		16
	April	8		24				32
	May	3		12				15
	June	9		14				23
	July	8		4		6	1 ¹	18
	August	9				16	2 ²	25
	September	4		3				7
	October	12		22		4		38
	November	17		17		12		46
	December	37		51		38		126
	2013	January	5		6		18	
Total collected from 2012-09-01		75		99		72		246
Unknown collection date		4		4		1		9
Total		128		176		96		400

All viruses were subjected to phenotypic testing against oseltamivir and zanamivir

¹A/Salta/1341/2012 gave an IC₅₀ of 187.89 nM against oseltamivir (highly reduced inhibition) - carried NA N295S substitution

²A/Netherlands/492/2012 and A/Netherlands/507/2012 gave IC₅₀s of 571.31nM and 312.2nM respectively against oseltamivir (highly reduced inhibition) - both carried NA H275Y substitutions

Figure 25. Phylogenetic comparison of influenza A(H1N1)pdm09 and A(H3N2) M genes

